ASSET INVENTORY

GUIDE - VERSION 4.2.0

INTRODUCTION

Put your Unity asset management workflow on steroids and say good-bye to the Package Manager as you know it! Asset Inventory is your ultimate asset companion: a lightning-fast search for assets for your current project. Find content in assets you purchased or downloaded without importing and bring single files in with just a click.

Eliminate the time-consuming task of finding a sound file, a texture or a model you know you purchased but which is hidden inside one of your many Asset Store purchases. The Asset Inventory provides a **complete list of all assets you own**, including their content.

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FEATURES

The tool aims to provide **what you always wanted the Asset Store & Package Manager to be**: The best possible Asset Management solution for Unity.

Find assets effectively. Search inside your purchases. Include custom packages you downloaded from Humble Bundle or other places. And packages from a registry. Manage everything through tags. Enhance it with AI. Perform bulk operations. Import multiple assets and packages in one go. Identify used assets inside your projects. Search using asset-specific criteria. And much more.

Powerful Search

Browse & find anything inside your purchased Asset Store packages without importing. Quickly preview audio files. Narrow your search using asset type, tags, image dimensions, audio length, vertex count, colors and more. Exclude items you don’t want to see. **Save and recall** searches.

Easy Setup

Works immediately out of the box. Lots of optional view & configuration options. Hassle-free indexing. Start, stop & resume at any time. Works with Unity 2021.3 and higher. Windows, Mac & Linux. Lightning-fast search.

Intelligent Updates

Handle updates from the Asset Store and from the Unity registry in one holistic UI. Define update strategies per package. Instantly see if there are any updates you should consider. Full package manager functionality for registry packages including add/update/remove and compatibility information.

Import & Export

Import only what you need instead of a whole package. Automatically determines asset dependencies to import complex prefabs and materials. Save space & reduce clutter in your project. **Bulk import** multiple packages at once. Automatically store imported assets in a specific sub-folder and keep the Assets root clean. **Export** assets easily for reuse in other contexts.

Many Sources

Your complete asset library: Automatically indexes Asset Store purchases. Triggers download of missing assets. Handles packages from registries. Integrate the Unity Cloud Asset Manager. Add custom folders to search through Unity packages downloaded from other locations. Indexes folders and zip/rar archives containing **arbitrary media files** like 3D models, audio libraries, textures and more. Automatically generates previews.

Organize

Automatically imports labels from the Asset Store. Use additional **tags** to group assets effectively. Assign tags to either packages or individual files inside packages. Assign colors and group by tags. Exclude unwanted items. Browse all sub-packages. Perform bulk operations. Import multiple assets and packages in one go. Builds on Package2Folder to allow importing packages into a custom sub-folder. **Backup** packages automatically to ensure you always have a working version to go back to.

Reverse Lookup

Quickly identify used assets and packages in your project. Ensure license compliance.

Constant Updates & Support

Receive regular updates, optimizations, and new features. Super-fast support. Discuss ideas in a great Discord community.

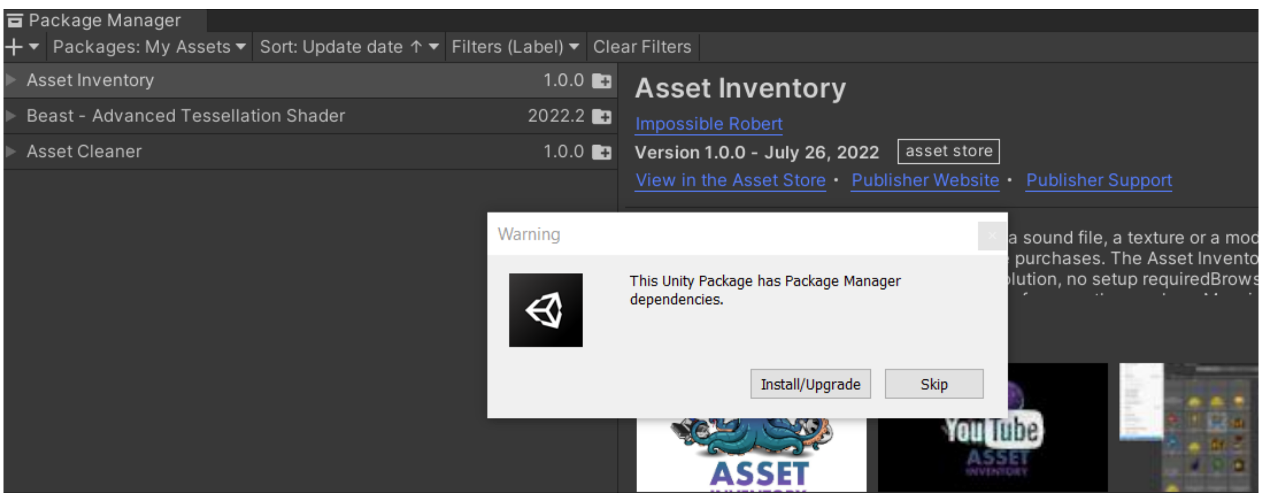
GETTING STARTED

INSTALLATION

It is strongly recommended to **remove any previous version** of the Asset Inventory before installing a new version. The best way to do so is to close Unity and afterwards delete the full *AssetInventory* folder since used libraries cannot be replaced if Unity is active.

Install the package through the Unity Package Manager. When asked if to install additional dependencies, select **Install/Upgrade**, otherwise the asset will not work. Installed package dependencies include *Newtonsoft Json*, *SharpLibZip* and *EditorCoroutines*.

If you have many Unity projects, an alternative way of installing and maintaining the tool for you might be a package-based approach as described later in the "Installing as a package" chapter.

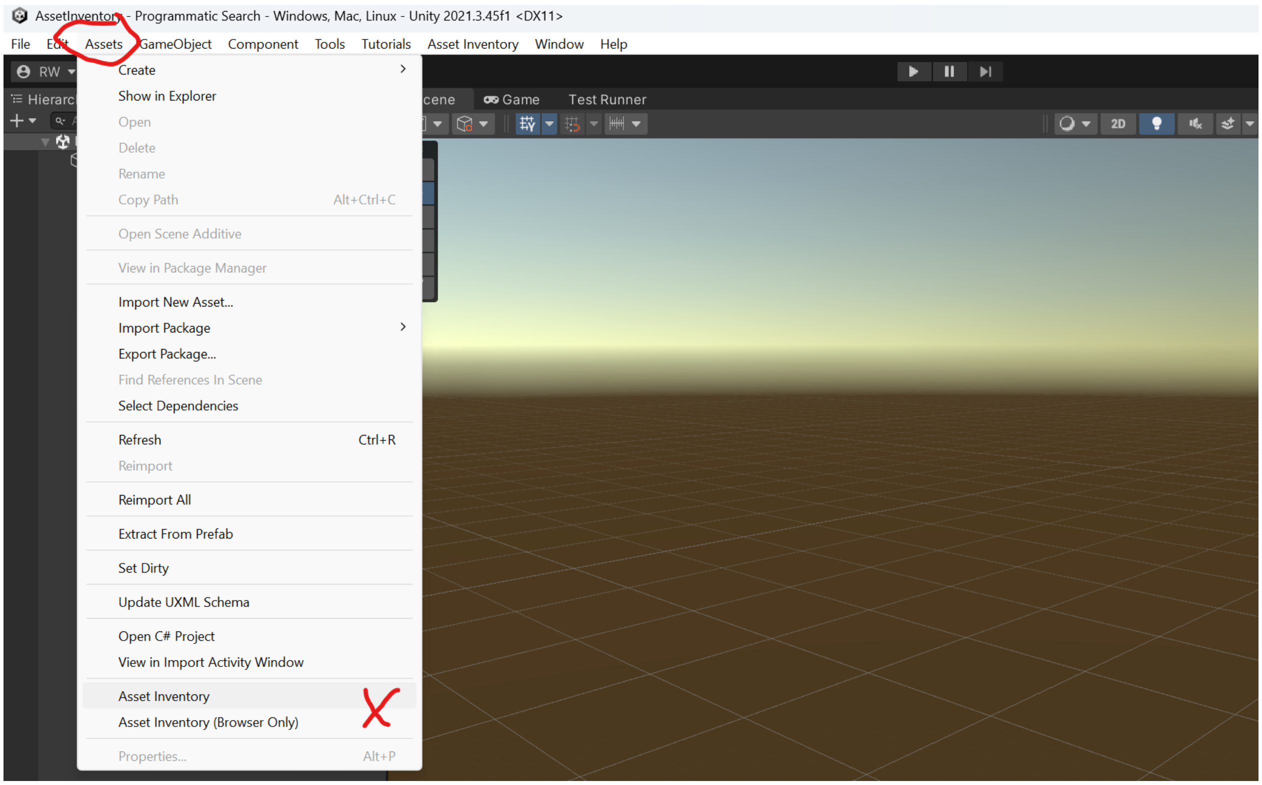


When indexing packages Unity might try to call third party applications like Blender, 3ds Max or Maya. It is recommended to at least **install Blender** (free) and the **Text Mesh Pro** package to make sure previews for *.blend* and UI prefab files are generated correctly.

Once installed you can access **integrated tutorials** which will guide you through the configuration and features. If the Unity Tutorials package is already installed, these will appear automatically. If not, you can install it at the launch screen of the tool, see next section.

USAGE

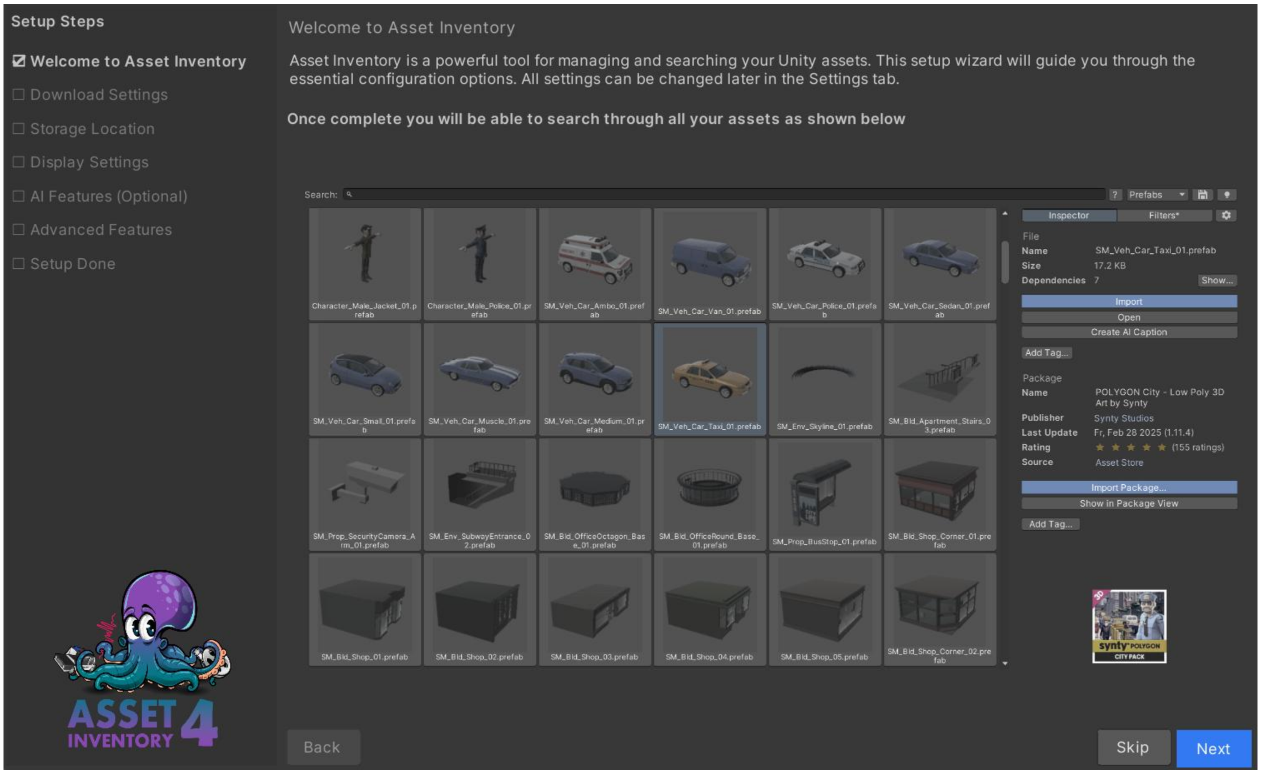
Open the Asset Inventory through the **Assets menu**.



The **Setup Wizard** will launch automatically. The default settings are usually good to go with and everything can be adjusted later, but some preferences make sense to check up-front. The main item to think about is where to store the database and caches since these require a fair amount of disk space for storing preview images and temporary files during extraction and browsing. You might also want to review bandwidth usage since downloading assets automatically might be taxing on your allowance.

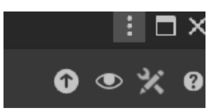
Once the wizard is done, the initial indexing will **start automatically**. Only a fraction of your assets will be indexed at the first to give you a chance to evaluate the tool and setup. If you are happy how everything works, press the *Run Actions* button under Settings to compile the full index. This might take a couple of hours and is best run over night.

When everything is done it can be beneficial to **delete your project's *Library* folder** which has accumulated lots of temporary unneeded data.



Whenever you purchase new packages or add new assets to your folders, make sure to press the *Run Actions* button under **Settings** once so that the index gets updated.

Tip: The UI displays only the core functionality (easy mode). You can access many **additional options** by holding down the **CTRL key** or pressing the *eye icon* at the top.



PERFORMANCE

Here are things to consider to get the most out of the tool:

- If you own an SSD, consider storing at least the database on it. If space is an issue, this is the order of priority for the data to be stored on SSD:
  - Database
  - Previews
  - Cache
  - Backups (no real need to store on SSD)
- Run the initial indexing or bigger updates in a **new empty Unity project** and use the newest available stable Unity release. The indexing results will be stored in a central location for reuse in any other project. This will give you the highest indexing performance since no other assets/scripts need to be refreshed and new versions of Unity typically bring big performance improvements for object and sound importers (in some cases 10x).
- Index in a **BIRP** project, as otherwise additional URP/HDRP dependencies need to be resolved or converted which takes additional time.
- Delete the **library folder** regularly as it can become very big and slow down Asset Database operations.
- **Optimize the database** after big operations (*Settings/Maintenance* menu).
- **Deactivate sorting** of search results for faster paging.
- **Deactivate *Search Without Input*** in the search settings for faster initial loading of the window.
- Less impactful and only do if you are handling large FBX of hundreds of MB: Deactivate **FBX dependency scanning**. In 95% of cases *fbx* files do not have dependencies listed inside.

ASSET SEARCH

OVERVIEW

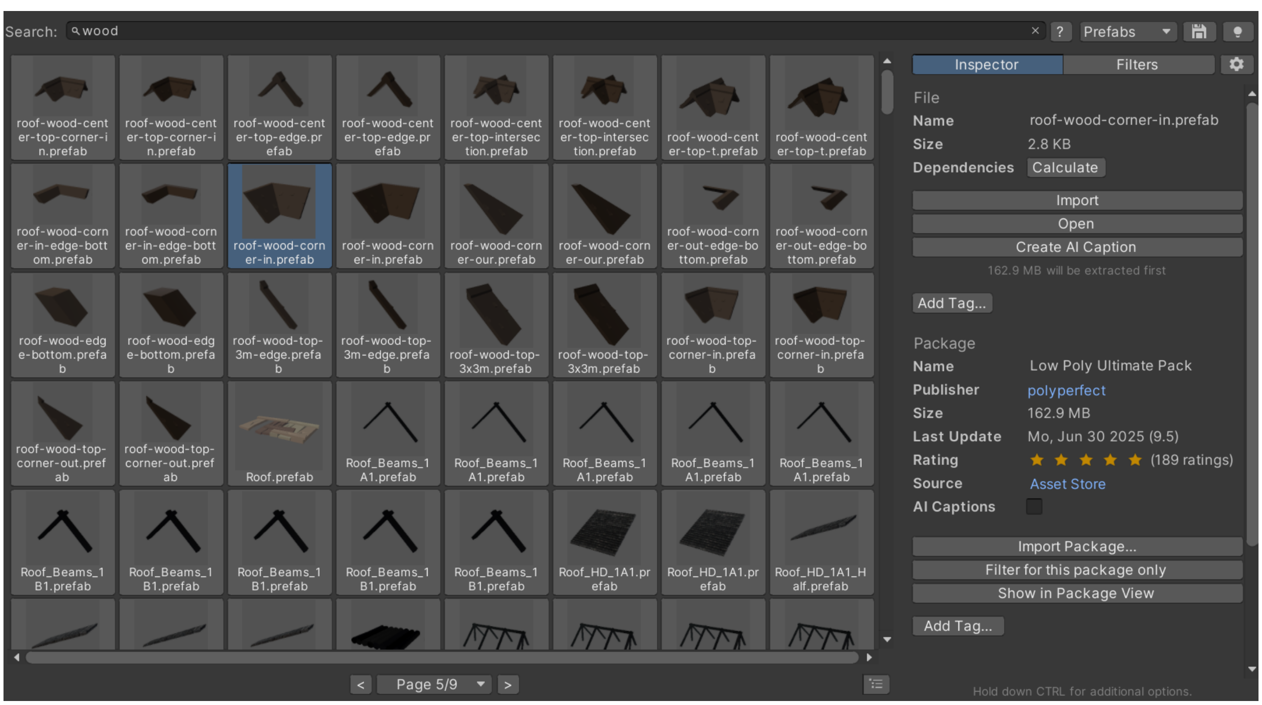
Once an asset is indexed, it can be found in the search. Searching will automatically start when typing. By default, every word entered needs to match the search result but not necessarily in the same order. If the exact phrase should be matched, prefix the search with ~.

Basic examples:

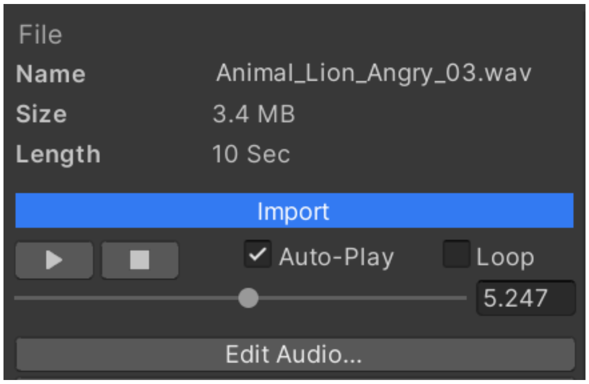
- *"car interior"* will return results like *"CarBlueInterior.fbx"* and *"InteriorCarDes.png"*
- *"~car interior"* will not return the above but only results like *"Car Interior.fbx"*
- *"car +interior -fbx"* will return results that match *"car"* and *"interior"* but not *"fbx"*

There is also an **expert search** available (search starting with "=") which is described later, where you have full database access.

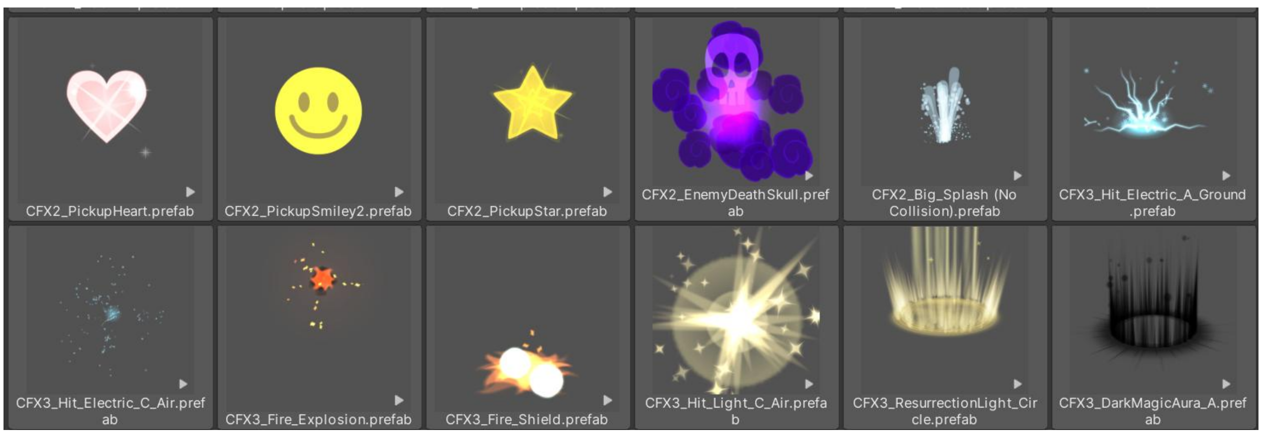
Additional filters can be selected to narrow down the results further. Once an asset is selected, details are shown on the right-hand side about the file and the package the file is contained in.



Selecting an audio file will **automatically play** it. This way it is easy to quickly preview audio files. Initial playback might take a while until the corresponding package is temporarily extracted.



Some results like animated FBX, videos, VFX and particle systems will be **animated**. A little play icon shows such previews. When selected they start playing automatically.

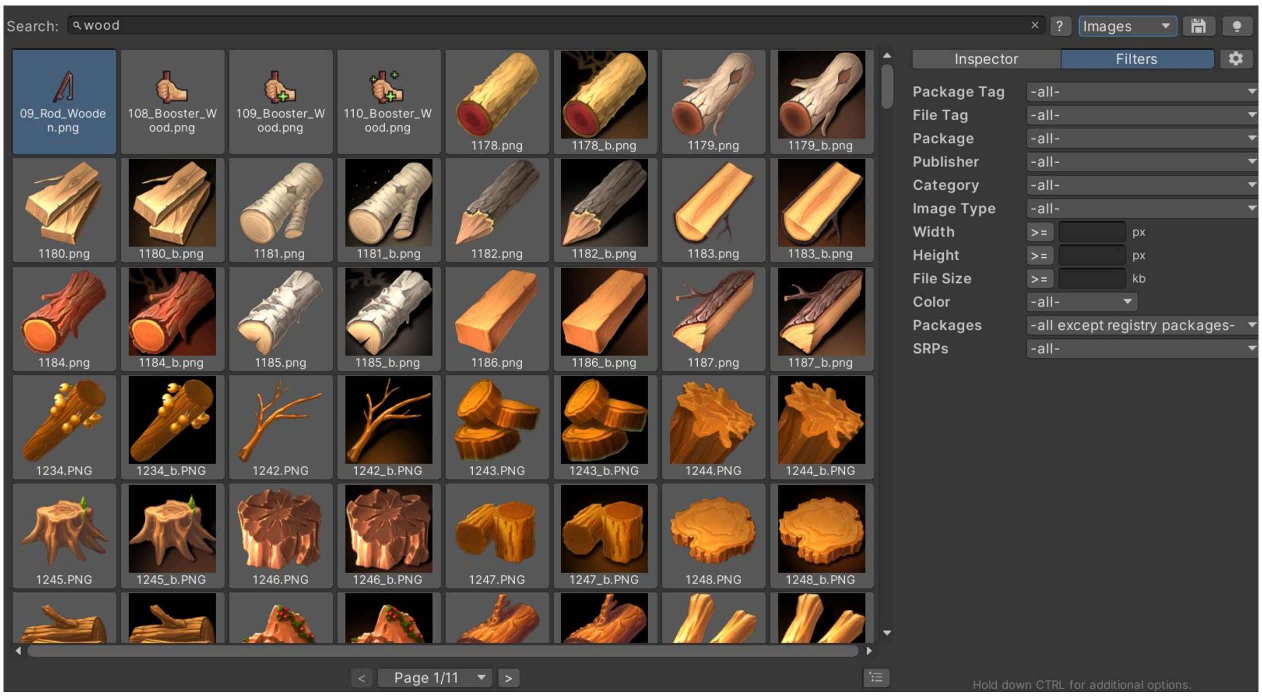


Per default one item plays at a time. To get a quicker overview you can press the play icon (using advanced visibility) in the lower left corner to automatically play all animations at once.

Multi-select is possible by holding down the *Shift* or *Ctrl* key. Selecting all results is done via *CTRL + a*.

ADVANCED FILTERS & SETTINGS

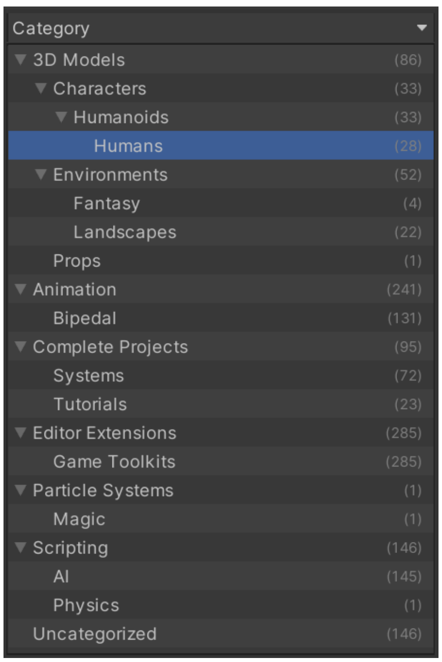
The **Filters** tab contains additional filters and media-specific properties to filter for specific image dimensions or audio length. Using the buttons in-front of each value toggles if results match that are bigger or smaller than entered. The filter dropdowns will only show values if respective assets are available. If the search type is limited (e.g. to Images), some filters might not show up (e.g. *length*) as they are only applicable to other types.



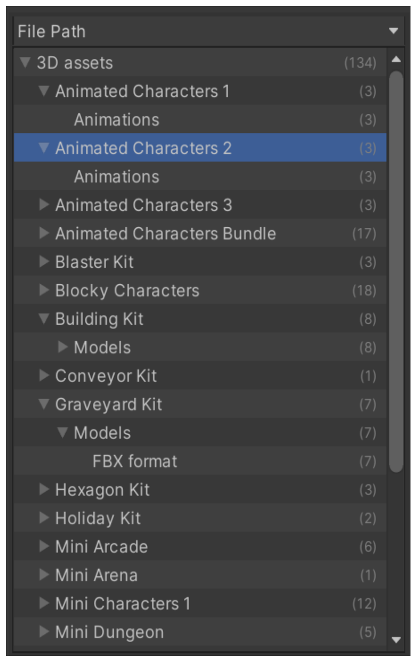
To get a better overview how the search results relate to different packages, categories, publishers, folders or file types, you can open the **hierarchy browser** on the left. It will show additional information for the currently shown search results and can be used to filter quickly to the files you need.

The browser is toggled with the little icon in the lower left corner.

Search:
fbx
Animations
File Path
3D assets (134)
Animated Characters 1 (3)
Animated Characters 2 (3)
Animations (3)
Animated Characters 3 (3)
Animated Characters Bundle (17)
Blaster Kit (3)
Blocky Characters (18)
Building Kit (8)
Conveyor Kit (1)
Graveyard Kit (7)
Models (7)
FBX format (7)
Hexagon Kit (3)
Holiday Kit (2)
Mini Arcade (6)
Mini Arena (1)
Mini Characters 1 (12)
Mini Dungeon (5)
Mini Market (3)
Mini Skate (2)
Modular Dungeon Kit (2)
Pirate Kit (1)
Platformer Kit (8)
Prototype Kit (25)
door.fbx
character-ghost.fbx
character-keeper.fbx
character-skeleton.fbx
character-vampire.fbx
character-zombie.fbx
crypt-door.fbx
crypt-large-door.fbx
building-mill.fbx
building-watermill.fbx
unit-mill.fbx
cabin-door-rotate.fbx
cabin-overhang-door-rotate.fbx
character-employee.fbx
character-gamer.fbx
claw-machine.fbx
prize-wheel.fbx
ticket-machine.fbx
wall-door-rotate.fbx
character-soldier.fbx
Page 1/10
Details
Filters
File
Name character-zombie.fbx
Size 0.7 MB
Animations 32
Materials 1
Vertices 1.072
Dependencies 1
Script Import -Never Import-
Import
Open
Add Tag...
Package
Name Kenney Game Assets All-in-1 3.3.0
Size 456.1 MB
Last Update Fr, Jan 9 2026
Source Archive
Import Package...
Filter for this package only
Show in Package View
Add Tag...
Hold down CTRL for additional options.



Category
3D Models (86)
Characters (33)
Humanoids (33)
Humans (28)
Environments (52)
Fantasy (4)
Landscapes (22)
Props (1)
Animation (241)
Bipedal (131)
Complete Projects (95)
Systems (72)
Tutorials (23)
Editor Extensions (285)
Game Toolkits (285)
Particle Systems (1)
Magic (1)
Scripting (146)
AI (145)
Physics (1)
Uncategorized (146)



File Path
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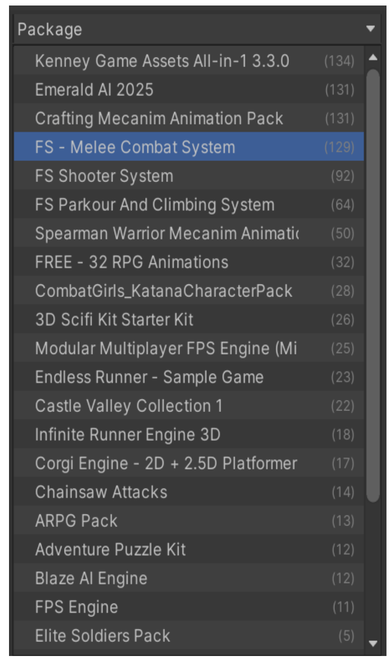
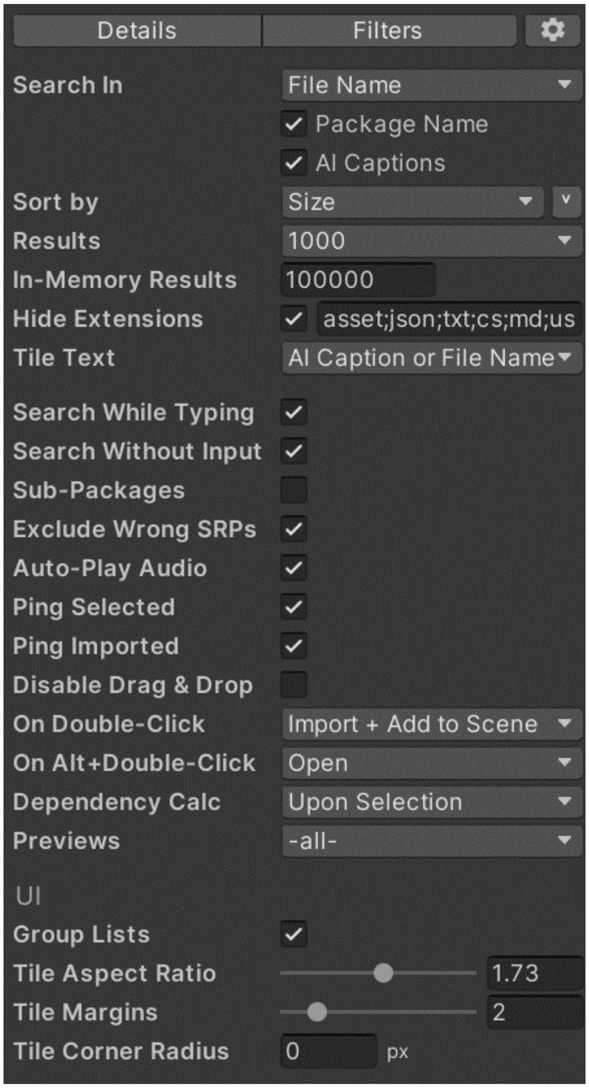
Mini Dungeon (5)



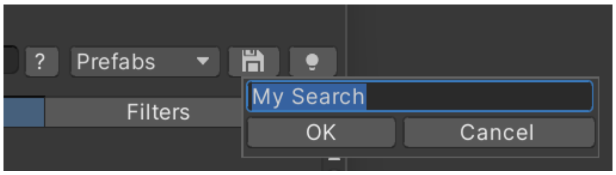
Package
Kenney Game Assets All-in-1 3.3.0 (134)
Emerald AI 2025 (131)
Crafting Mecanim Animation Pack (131)
FS - Melee Combat System (129)
FS Shooter System (92)
FS Parkour And Climbing System (64)
Spearman Warrior Mecanim Animati (50)
FREE - 32 RPG Animations (32)
CombatGirls_KatanaCharacterPack (28)
3D Scifi Kit Starter Kit (26)
Modular Multiplayer FPS Engine (Mi (25)
Endless Runner - Sample Game (23)
Castle Valley Collection 1 (22)
Infinite Runner Engine 3D (18)
Corgi Engine - 2D + 2.5D Platformer (17)
Chainsaw Attacks (14)
ARPG Pack (13)
Adventure Puzzle Kit (12)
Blaze AI Engine (12)
FPS Engine (11)
Elite Soldiers Pack (5)

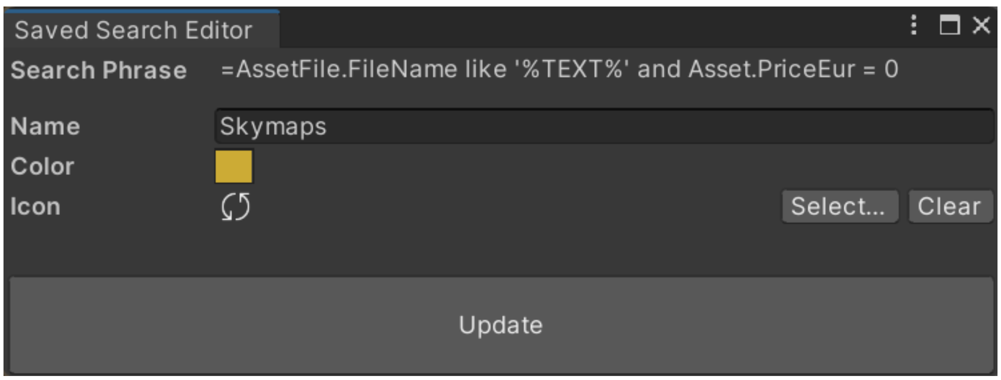
The **Settings** tab contains many properties that influence how the search results are collected, ordered and visualized. Specify the tile size, which text to display on the tiles and if assets should be pinged automatically upon selection.



SAVED SEARCHES

Once you have a search and filters setup you can save these input as a **Saved Search**. This allows you to later recall these easily. The search phrase and all filter settings will be stored. The results are not persisted but instead live from the database each time so any newly indexed items will also appear.



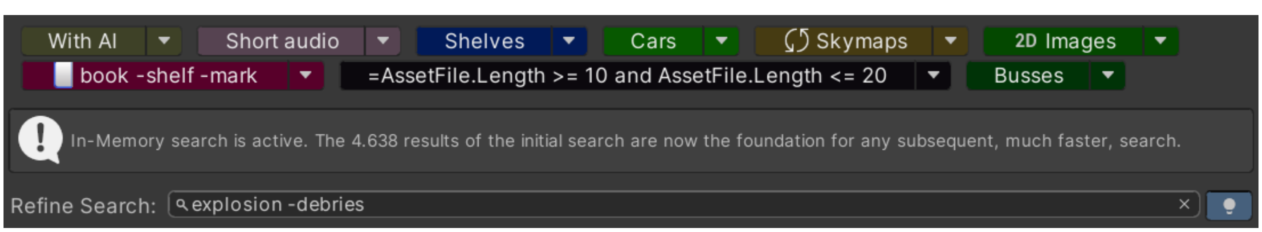


If you want to update a saved search with new search settings, select the dropdown next to the saved search and select "Override with Current Search".

HIGH-SPEED/IN-MEMORY MODE

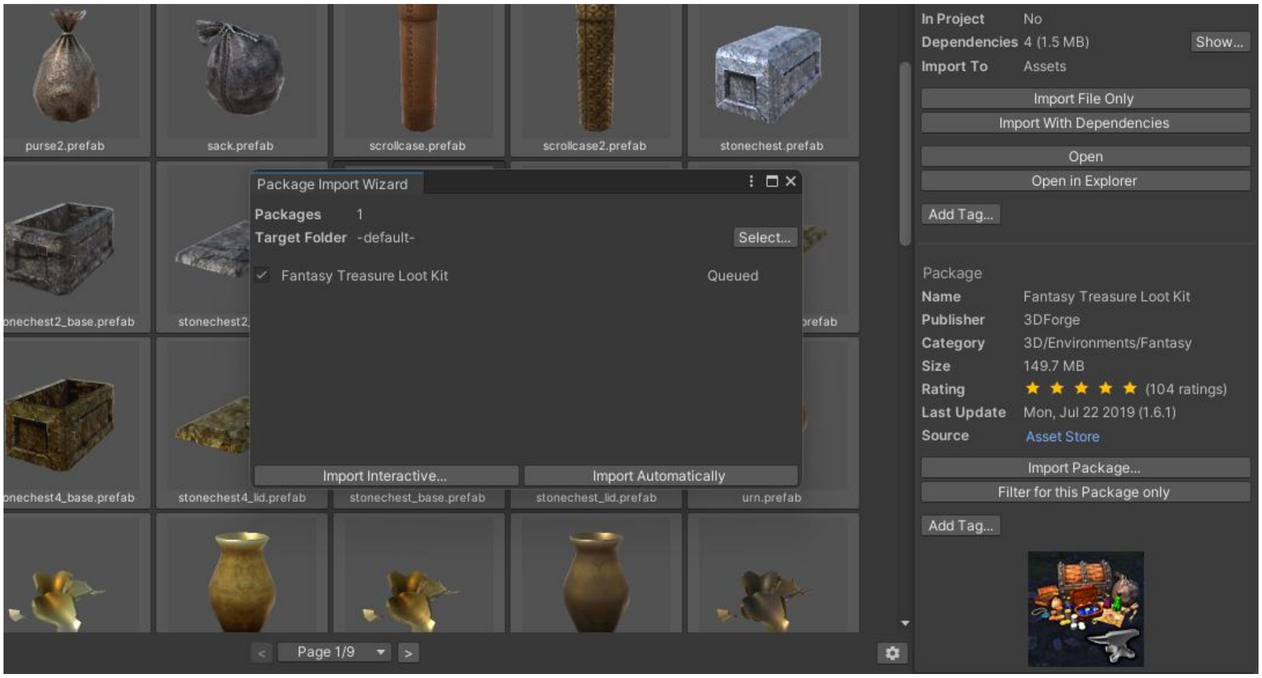
Browsing through search results, especially with many pages or complex queries, can be time consuming since many database operations are involved. When activating high-speed mode, all result pages will be loaded immediately and browsing and subsequent filtering by keywords will become lightning fast.

Once active, searching will only be done inside the initial search result, refining it further. The usual fuzzy search, +, - and ~ searches will be available but no extended features. Switching to a different saved search will retain in-memory mode.



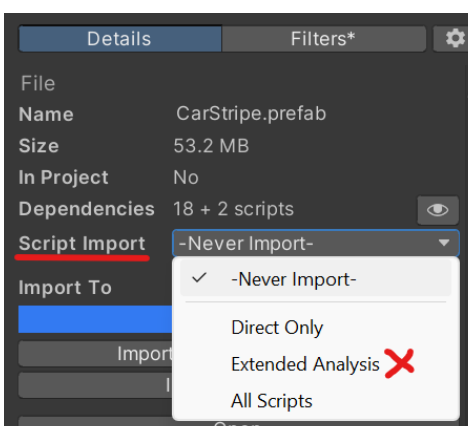
IMPORTING ASSETS

Assets can be imported individually or with all detected dependencies (default). The original directory structure of the files will be retained.



Clicking the *Show* button behind the dependency information will bring up the dependency information details, listing all files, their size and if they are already in the project or not.

In case there are **script dependencies** these can also be imported. A lot of effort has been invested to parse script dependencies, syntax, *asmdef* files and more when in *Extended Analysis* mode. This will work in a lot of cases but due to unforeseen dependencies in scripts, there might be compilation errors in rare cases.



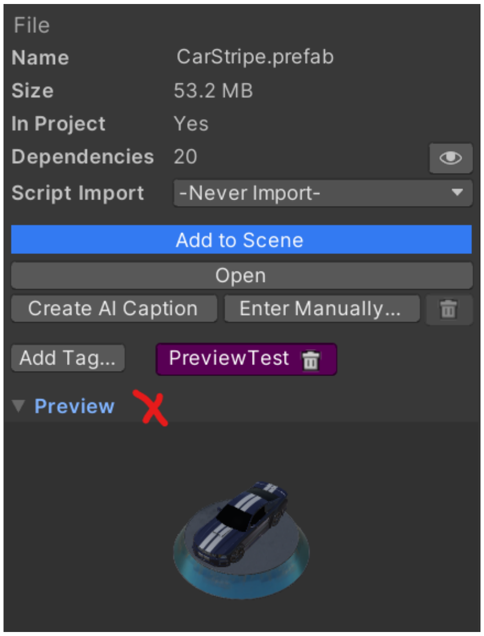
All dependencies and their relations can be seen in the graphical dependency viewer.

Import can also be triggered by **double-clicking** on an item (if activated in Search settings) or by **dragging** it into the Project, Hierarchy or Scene Window.

If an item already exists in the project, it can be **reimported** to restore the original in case it was modified. Reimport is also useful to reorganize assets. Assume you have imported a full package of UI components but only need a specific button. Since the button depends on textures and other files it is cumbersome to identify all these components.

The easier way is to reimport the button with the **Reorganize** flag (Settings/Import) active. This will move the button and all dependencies into the new target folder, extracting it from the original directory. Afterwards you can safely delete the full package and only the button will remain.

There is an inline preview available after import which allows to rotate 3d models.



Audio Files

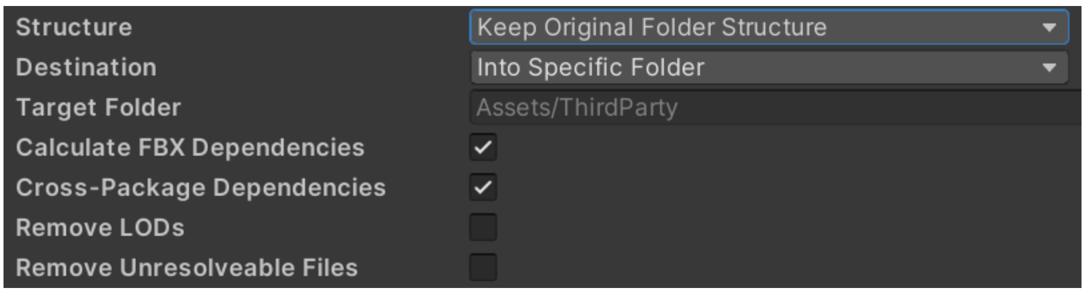
The tool has many additional capabilities specifically for audio files. When an audio file is selected, the *Editor Audio…* button will be shown. There you can select only parts of the audio to be imported and trim away silence.

Audio imported this way will always be in *wav* format.

Import Location, Folder Layout and Post-Import Actions

By default all assets will be imported under *Assets/ThirdParty*. This results in a clean root folder for your project. Most assets support this but a few might cause issues since they expect to find their files in the root. In that case either move them in the Project View or select a different **import destination**. The tool offers three options:

- Selected folder in Project View
- Asset Root
- Specific Folder (e.g. *ThirdParty*)



You can also specify the intended **target structure**:

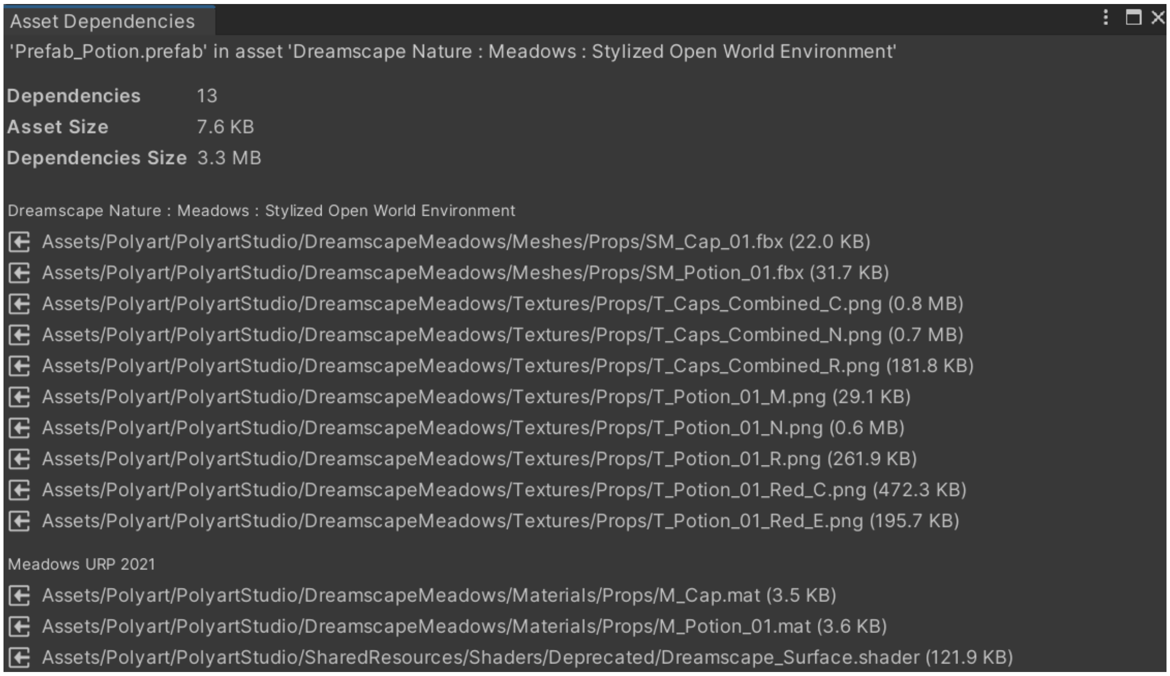
- By default even when importing single files they will materialize in the original folder structure of the asset. This is typically a good choice when importing multiple files or if files have dependencies like prefabs that will materialize model, material and other folders potentially.
- You can also select to have them all materialize in the same folder.
- Alternatively, files without dependencies can also be dragged into a target folder directly which will not create any additional structure.

Once items are imported, post-actions can be executed:

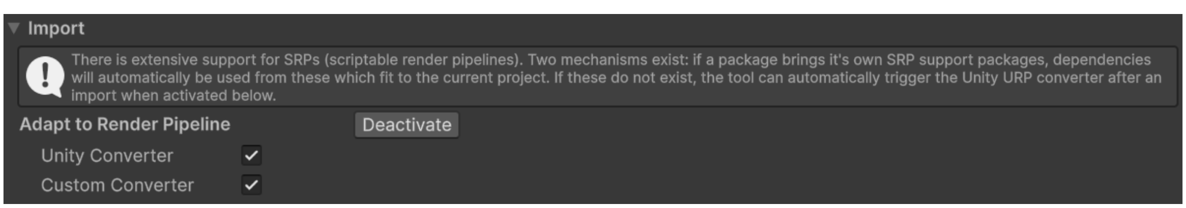
- Remove LODs will scan imported prefabs for LOD group components and remove it and all LODs except the first one.

Scriptable Render Pipeline (SRP) Support

When working in an **SRP** project, materials can appear with the pink error shader due to incompatibility. The tool provides extensive support for SRP projects to fix this. Two modes are available which can be combined for maximum coverage:

1. Many packages bring dedicated **SRP support packages** with them which contain additional files that need to be installed when working in a URP or HDRP project. These support packages will **automatically** be detected and when importing an asset, the tool will check which additional files are required from the support package. You will see the full dependency resolution in the dependency UI.


2. Packages without special SRP support packages can in many cases still be used in URP projects since Unity provides the **Render Pipeline Converter** which will adjust the imported materials to fit the URP materials. If activated under *Settings* (available in URP projects with URP 14+), the tool will **automatically trigger this conversion**, making it very convenient to import assets. If a dedicated SRP support package was found, this step will automatically be skipped. Since there is no programmatic way to influence which materials are converted, this setting will convert all project materials (typically not a problem) and will incur a minor speed penalty. It will also work when recreating previews.



EXPERT SEARCH

Tokens

A token is a name/value pair separated by a colon. A token can be put anywhere in the search field and will apply to basic and expert searches. They act as a shortcut to express sophisticated filter conditions. Use ' or " to escape token values containing spaces.

Available tokens:

- pt: Package tag
- ft: File tag

Behind a token one or multiple values can be stated, separated by coma. Tokens can be repeated and tokens can be combined through multiple ways (shown for 'ft' but use analog for 'pt').

- ft, WithAnyFt: At least one value should match
- WithAllFt: All values must match
- WithNoFt: No values must match

<u>Examples</u>

"red pt:car" will search for all files that contain the word "red" and have the package tag "car".

"highres nature ft:'asset management'" will search for all files that contain the words "highres" and "nature" and have the file tag "asset management".

"red withanypt:boat,vehicle withnopt:aircraft" will search for all files that contain the word "red" and have the package tags "boat" or "vehicle" and no package tags "aircraft".

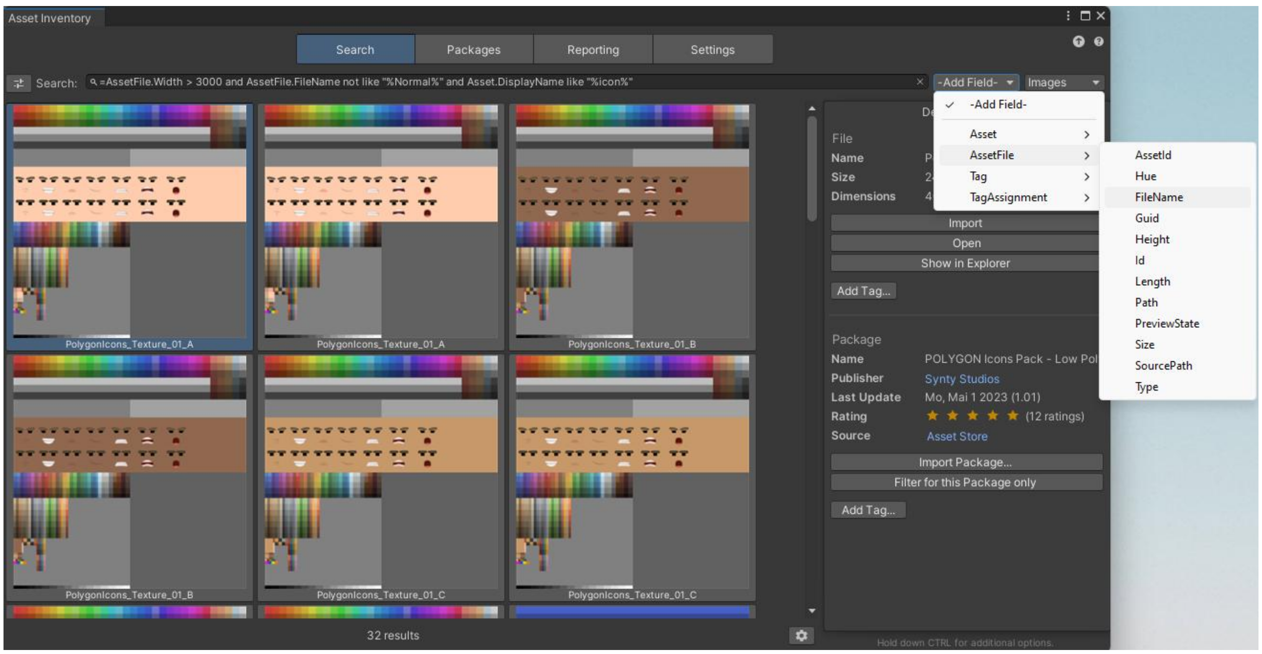
SQL

It is possible to use nearly the full feature set of SQLite 3 to search. This mode is activated when starting the search with "=". Afterwards the database fields and conditions can be stated.

Examples

```
=Asset.PackageSize > 0 and AssetFile.Type="wav"
```

```
=AssetFile.Width > 3000 and AssetFile.FileName not like "%Normal%" and Asset.DisplayName like "%icon%"
```

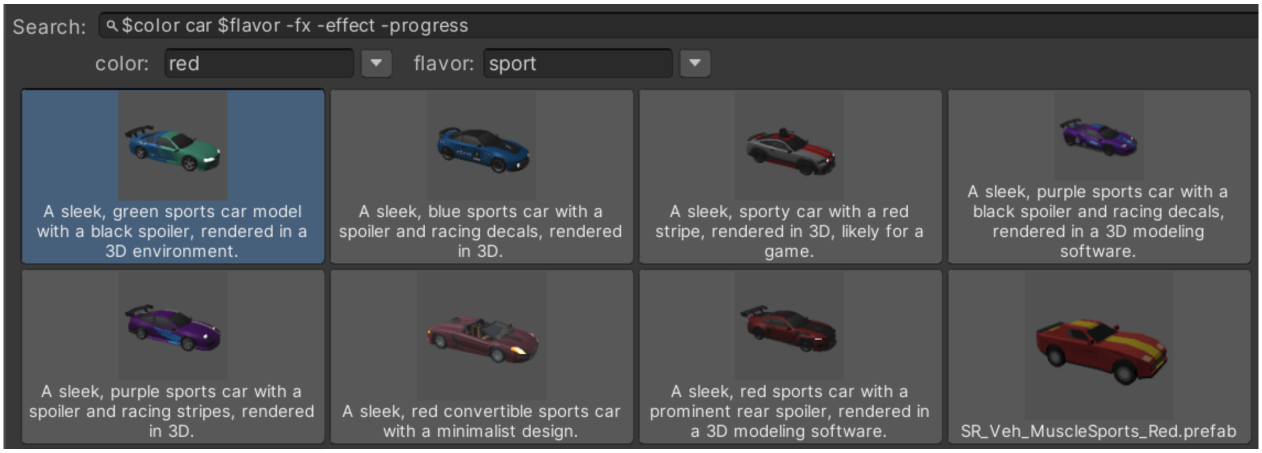


Useable fields can conveniently be picked from the dropdown behind the search field.

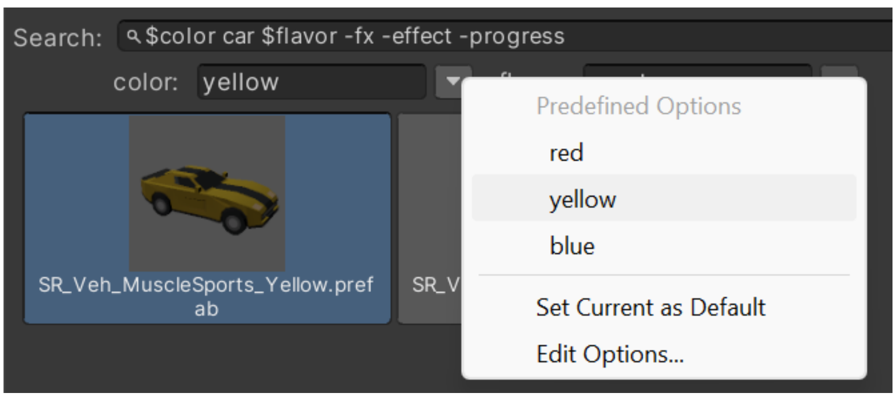
Parameters

SQL searches can require specifying text values multiple times or the query can become more complex. Parameters offer a way to simplify such queries and also make them more usable, especially when combined with Saved Searches.

To specify a parameter, simply prefix any word with $, e.g. $color. This will result in a new text field appearing under the search field where you can now enter the actual value for $color. This will then replace every occurrence of $color in the search with that value.

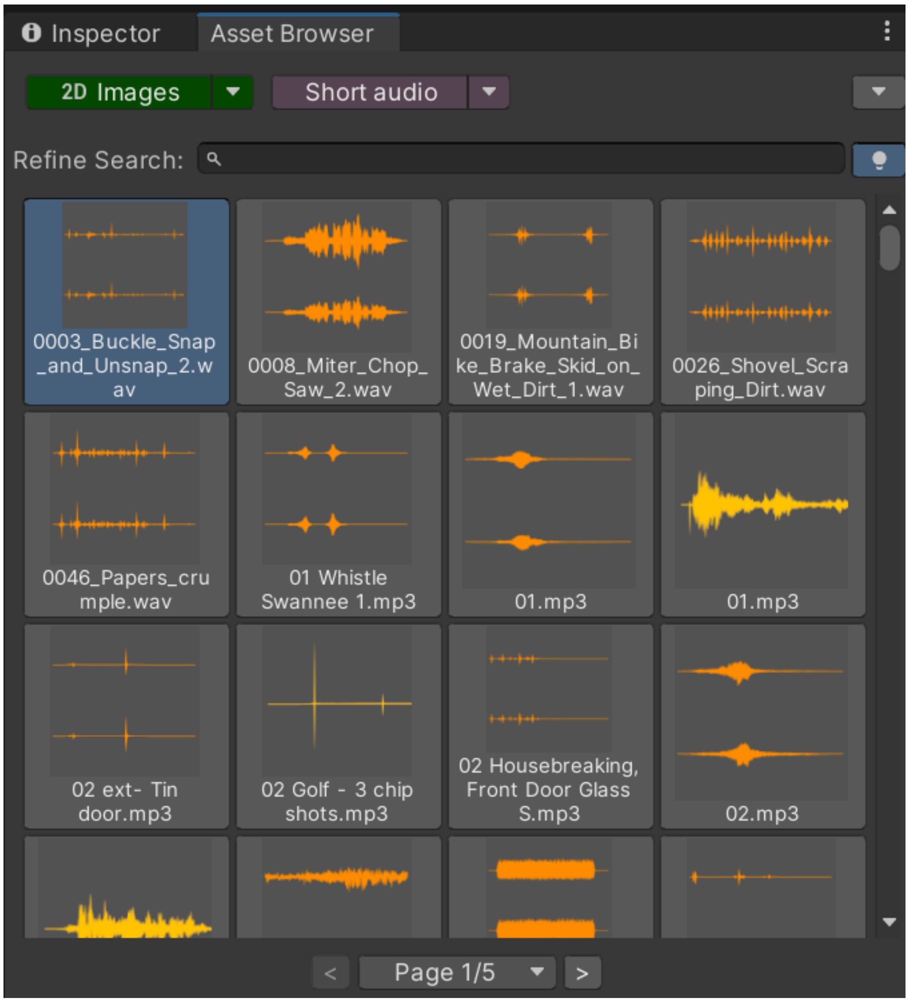


If you save such a search, you will now get the option to go even further and specify a **default value** and **predefined values** to select from via the dropdown menu which are then persisted in the Saved Search.



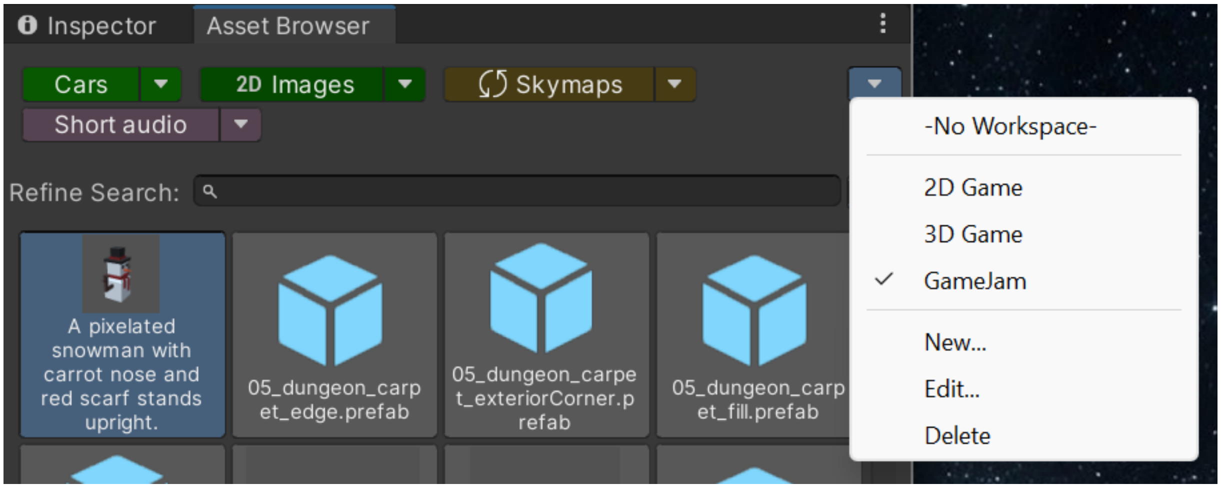
ASSET BROWSER & WORKSPACES

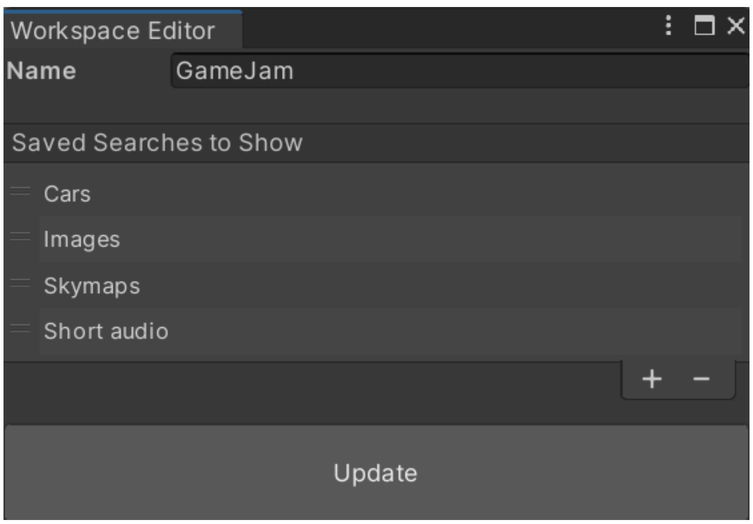
When you are only focusing on bringing assets into your project, the normal window with the packages list, settings and more can be too big or cumbersome to work with. There is a dedicated *Asset Browser* window that is streamlined and highly optimized for quickly brining in assets.



It is utilized best in combination with saved searches as it will automatically load the first saved search and always activate in-memory mode.

Since not all saved searches might apply to the current project you are working on, it is possible to group these into **workspaces**.





PACKAGES

OVERVIEW

The packages overview will show a list of all packages, indexed or detected in the current project. These can come from multiple sources:

Automatic

- Asset Store purchases
- Registry packages
- Unity Asset Manager
- The local Unity asset cache
- The local Unity package cache

Additional folders stated under *Settings*

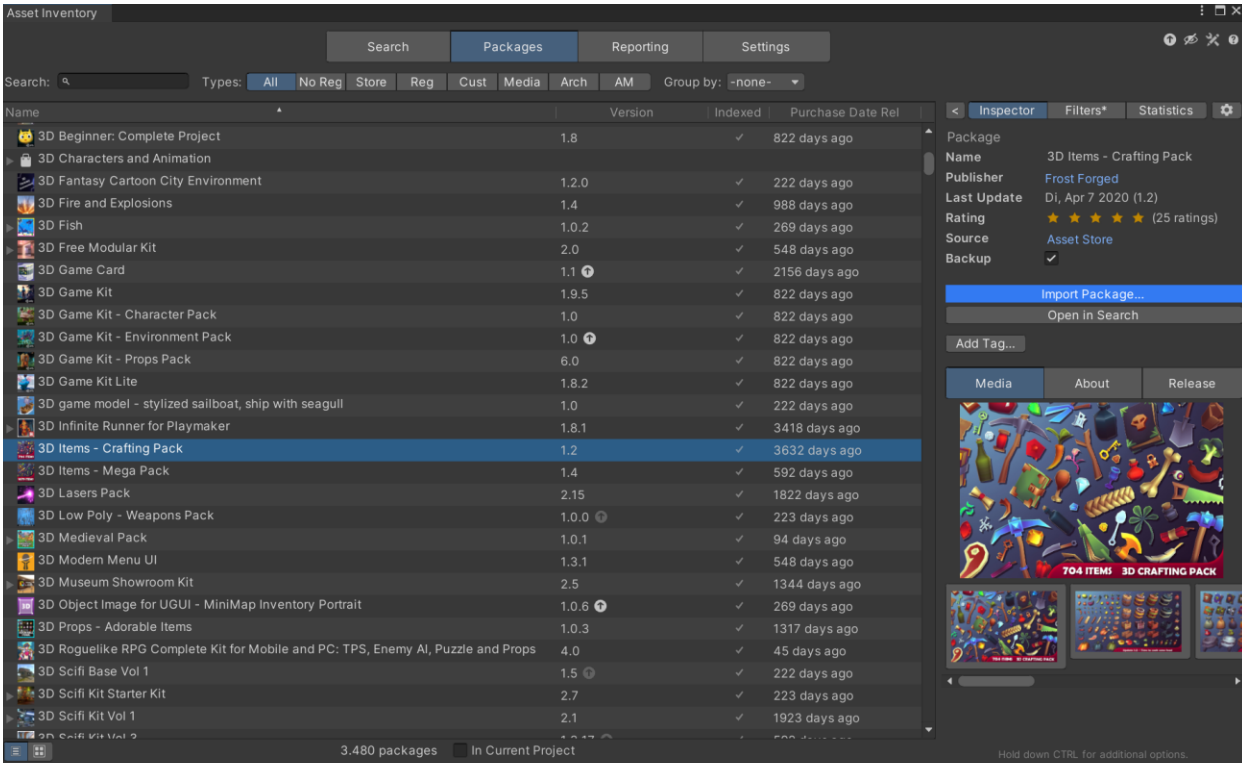
- Unity packages from other sources (e.g. *Synty*)
- Local asset packages under development (via *package.json*)
- Local media libraries (sound files, textures, models…)
- Unity projects
- Archives (zip, rar & 7z contents)

The list will indicate which of these were already indexed. To index more packages, download them into the cache or add more additional folders. Display is possible as a plain list or grouped by categories, publishers, state or tags.

In the case of registry packages, only the latest version of the package will be indexed. If a package contains **sub-packages** (also recursively), they will be shown directly under the package in a tree structure. **Double-clicking** any package will show the contents in the search.

After selecting a package, multiple options will appear. It is possible to fully import it from here. If you enabled backups and previous versions exist, also an older version of a package can be selected to go back to if newer versions contain issues.

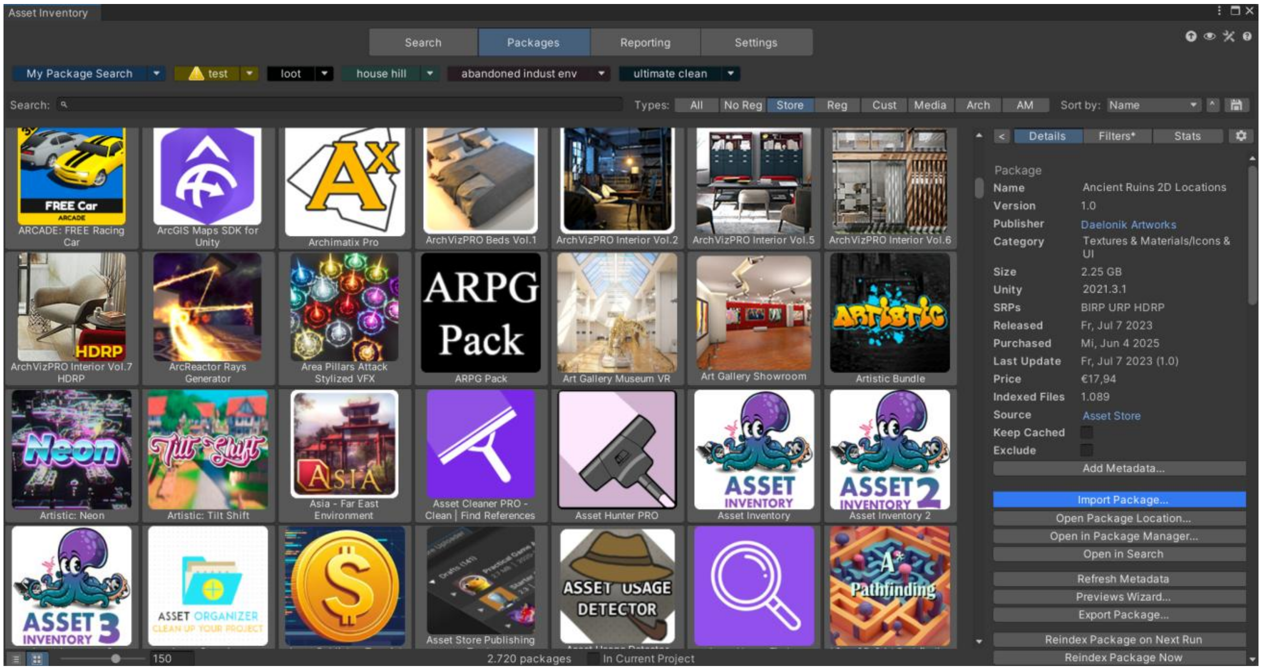
Packages can also be partially or completely removed again from the project and also from the database which can be useful after cleaning up outdated assets.



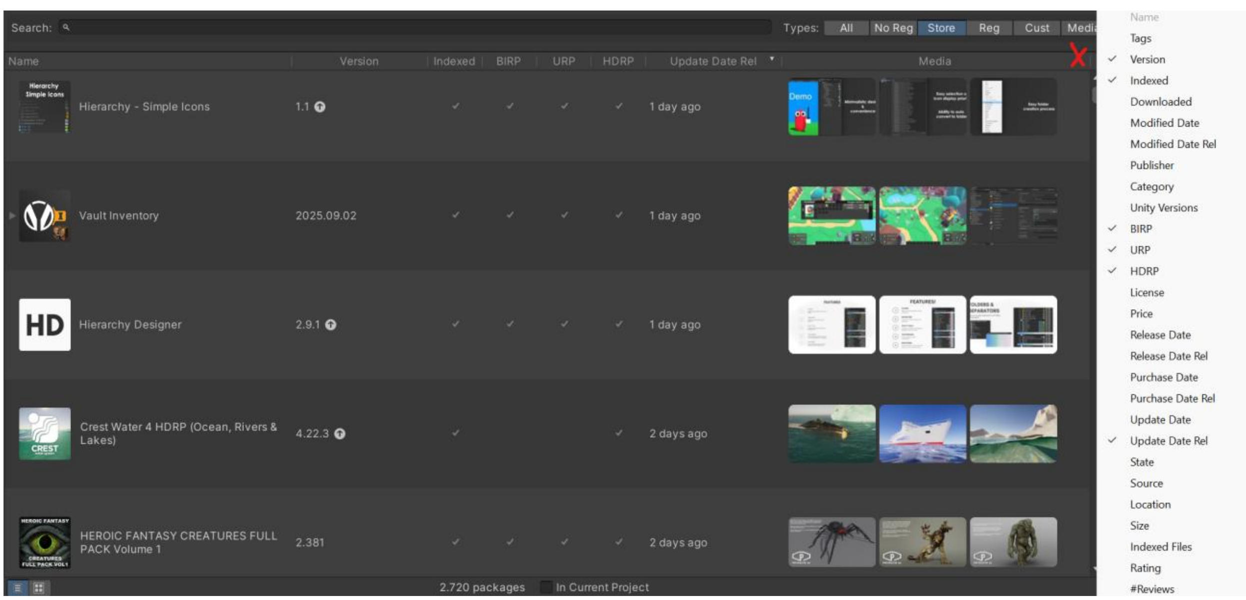
Depending on the package type additional options become visible:

- Registry Packages
  - Support for managing the installed version and setting an **update strategy** (recommended, latest stable compatible, latest compatible, manually).
- Development Packages
  - Use directly via **file link**
- Custom Packages
  - Support for **connecting to metadata** from the Asset Store. This will load the name, description, rating etc. and is especially useful when having downloaded packages from alternative sources like the Humble Bundle to still have all metadata available. See details in Custom Metadata section below.
  - Support for disconnecting from the Asset Store again

Using the buttons under the package list, the display can be switched between **list and grid style**.

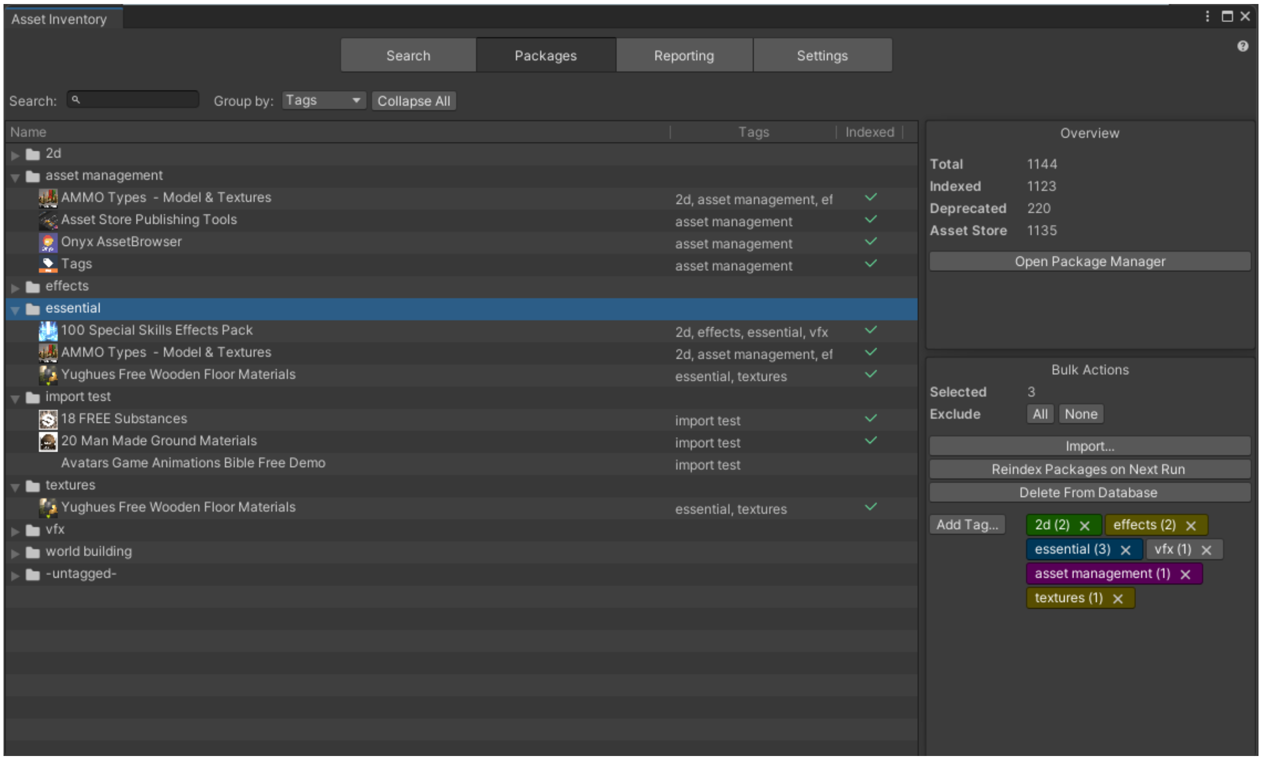


Right-click on any column header to **add or remove columns**. One special column is *Media* which will enlarge the list height to show package images right inside the list.

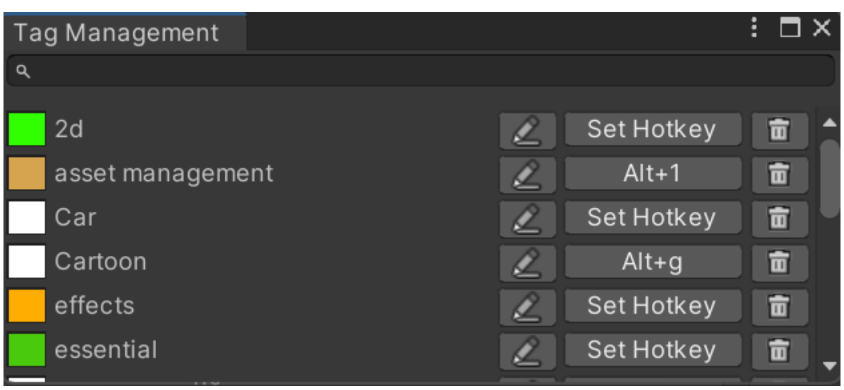


TAGGING

Tags will be imported from the Asset Store in case any are set there. In addition, local tags can be added and removed here as well (they are not synchronized back to the Asset Store yet). Bulk editing of tags is possible when selecting multiple items in the tree.



When adding tags, the **Tag Management** window can be opened through the small cog wheel. There, tags can be created, colored, renamed and deleted. You can also set **hotkeys**. The size of the tag selection window can be changed under *Advanced* settings.



You can drag a tag onto any other tag to create a **hierarchy**.

There are some **reserved** tags which when assigned trigger additional logic:

- **NoIndex**: will skip indexing the package but perform all other actions, e.g. backup

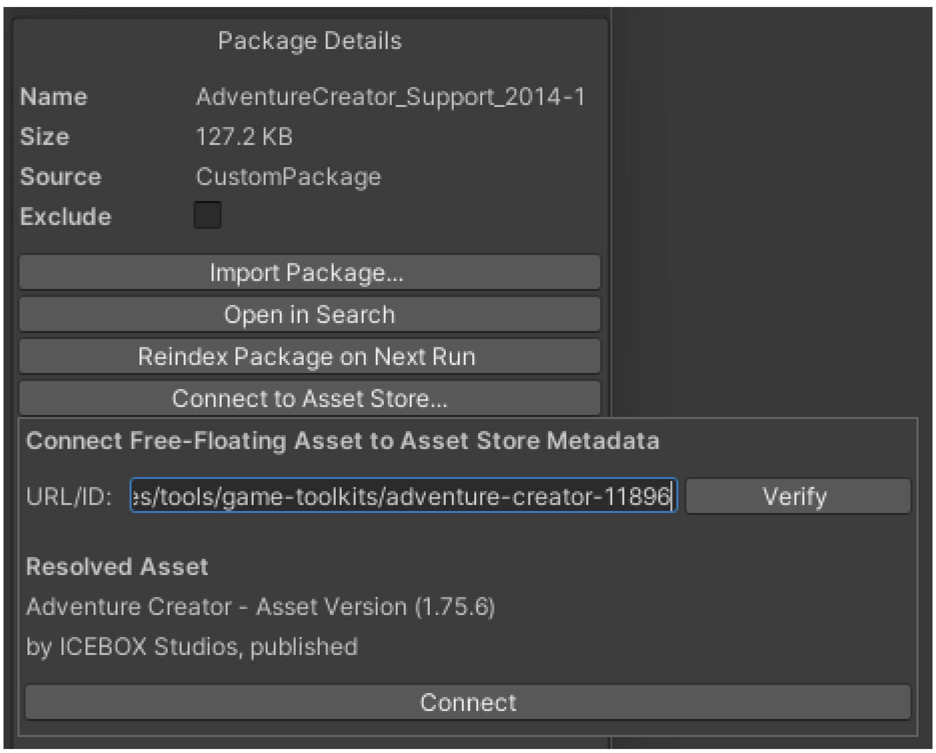
CUSTOM METADATA

Package metadata like name, publisher, category etc. is extracted directly from the package if the information is contained in there. If it is missing there are four options.

Link to Asset Store

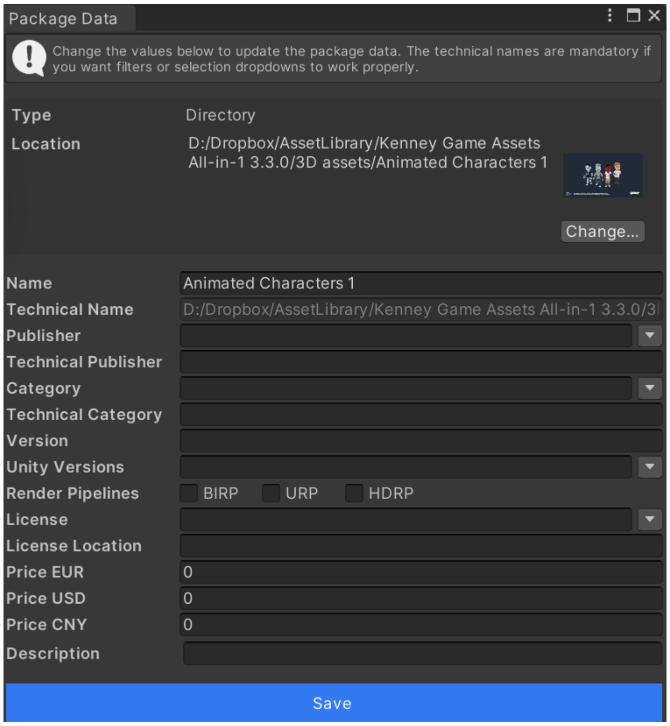
If the package exists on the Asset Store, the package can be linked to the entry there and a lot of metadata will be loaded automatically and also regularly to fetch updates to it.

Copy/paste the URL from the Asset Store website of the asset you want to link, *Verify* it is detected correctly and click *Connect*.



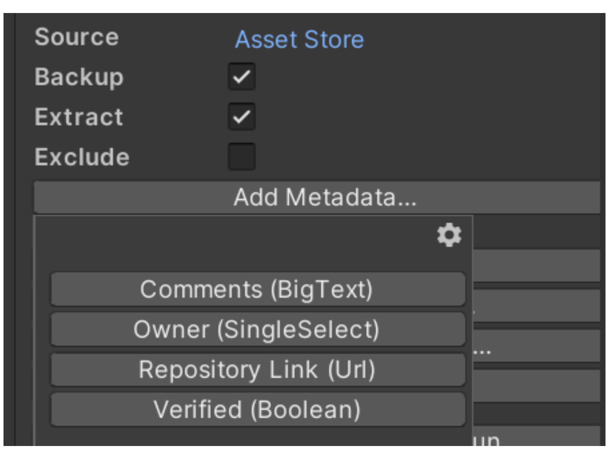
Manual Data Entry

Package data can also be edited manually using the "*Edit Data…*" command visible for packages that are not linked to the Asset Store. This way also custom preview images can be set for archives and custom packages.

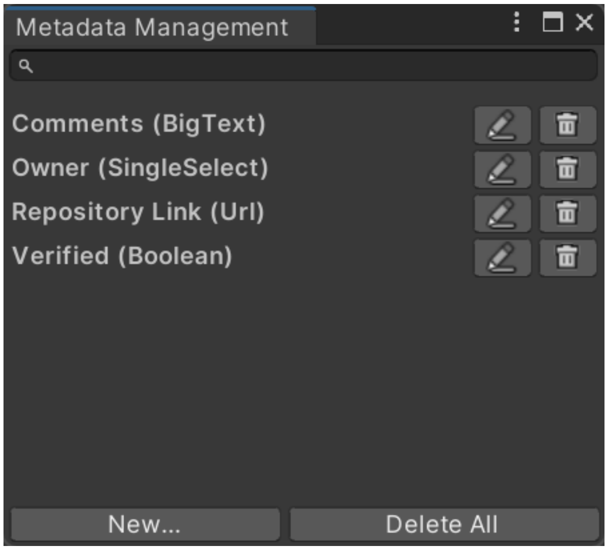


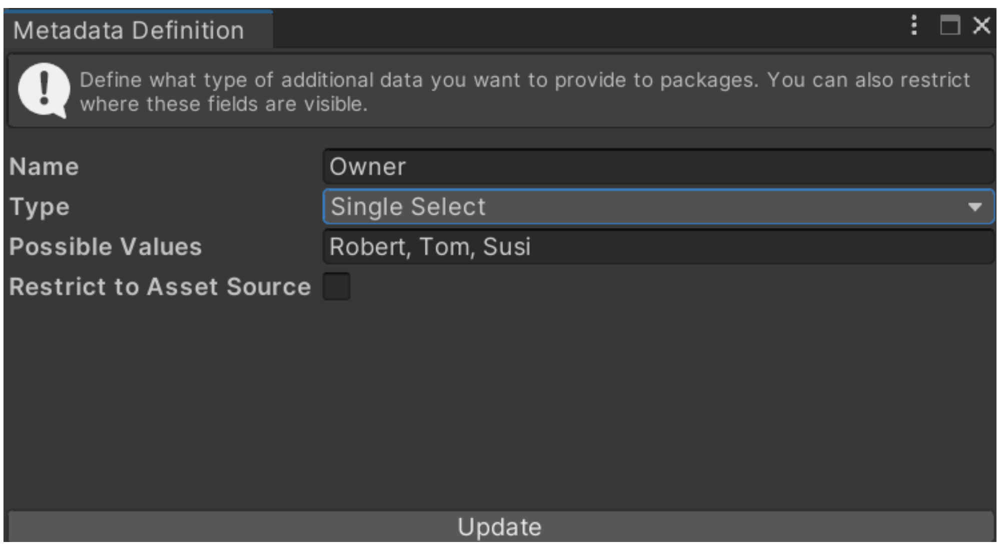
Define Custom Fields

If the metadata fields that are available in the database do not fit or you want to capture additional information, you can define additional fields. These can be of various types to render checkboxes, selection lists, texts, numbers, links or dates. Metadata fields can be restricted to only appear for packages of specific sources, e.g. archives.

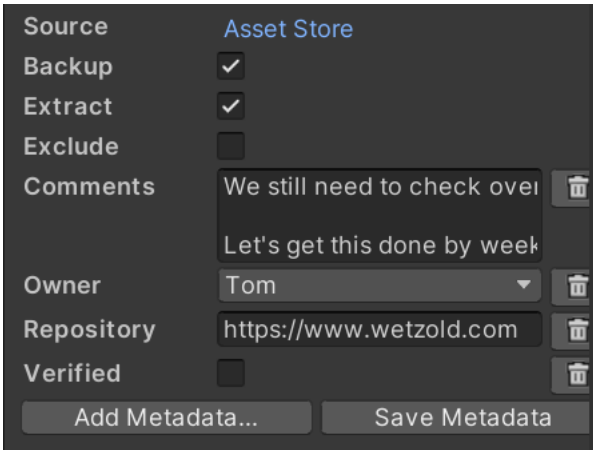


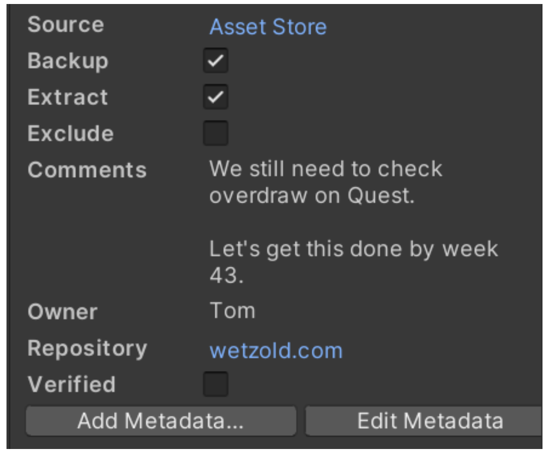
Existing fields can quickly be added from the dropdown. To define new fields, go to the management UI using the cog wheel. There you can create, edit and delete all custom fields.





Once metadata is added to a package the UI will be in either view or edit mode (except for checkboxes). During edit mode, values can be changed and fields removed. This will only remove them from this specific package.





Custom Preview Images

Usually Asset Store packages will bring a preview image for the package included. If you want to override it or if you have custom packages you can associate a dedicated preview image by placing an image next to the package file. This needs to be named like the package file with the suffix *".icon.png"*.

Override Files

The last option is to create an Override *Json* file containing metadata. This needs to be named like the package file with the suffix *".overrides.json"*. This overrides file can contain any data a package can hold and also a list of tags in addition. If defined in there, this data will override any existing data read from the Asset Store or the header file.

The advantage of this method is that such files can be put in a central location, so everybody who will index the location has the same metadata. This ensures for example the common use of tags and categories if working in a team.

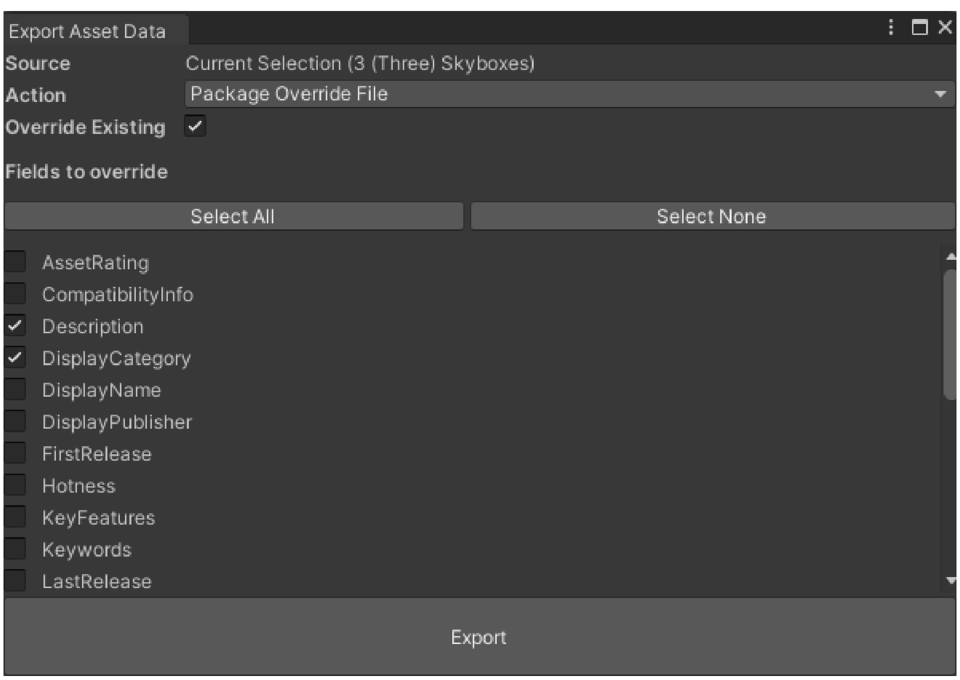

```
{
    "displayName": "Three Skyboxes",
    "tags": ["sky", "limit"]
}
```

Supported fields: foreignId, *displayName, displayCategory, safeCategory, displayPublisher, safePublisher, publisherId, slug, revision, description, keyFeatures, compatibilityInfo, supportedUnityVersions, keywords, version, latestVersion, license, licenseLocation, purchaseDate, firstRelease, lastRelease, assetRating, ratingCount, hotness, priceEur, priceUsd, priceCny, requirements, releaseNotes, registry, repository, officialState, tags*

Consider that filtering and grouping works on the *Safe* versions of category and publisher so these should also be set when changing these values. *Safe* versions are named in a way that they could also be used in the file system, without special characters like *&* or */*.

SafeCategory	DisplayCategory
Filter	Filter
Complete ProjectsSystems	Complete Projects/Templates
3D ModelsProps	3D Models/Props
AudioSound FX	Audio/Sound FX
Textures Materials	Textures & Materials
Textures MaterialsGround	Textures & Materials/Ground
3D ModelsVehiclesLand	3D Models/Vehicles/Land
Textures Materials	Textures & Materials
Editor ExtensionsGame Toolkits	Editor Extensions/Game Toolkits
Complete ProjectsPacks	Complete Projects/Packs

To make it easier to create package override files, the Package Export supports to create these conveniently for an individual or also a list of packages.

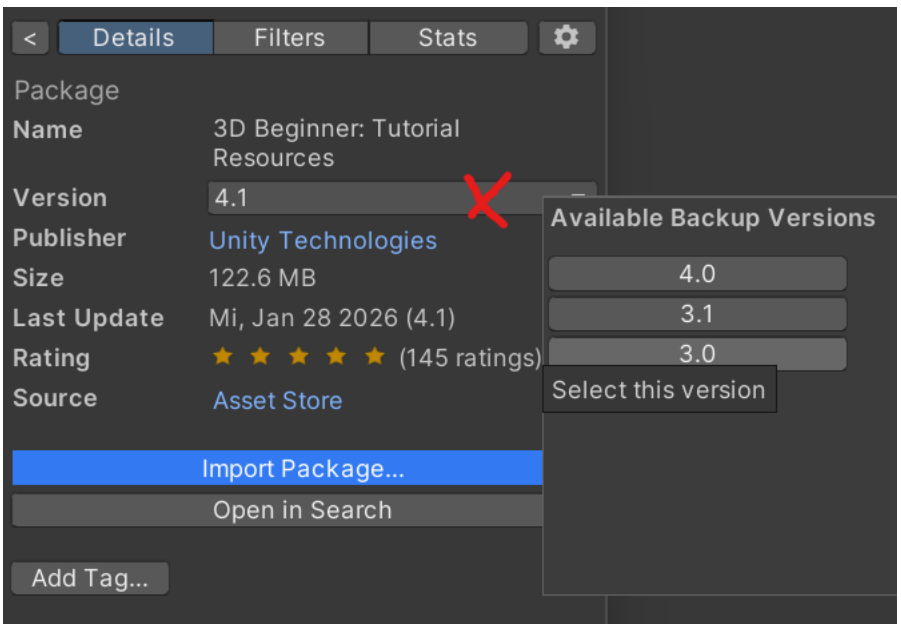


IMPORT & BULK IMPORT

Importing packages can happen one-by-one and in bulk while multiple packages are selected. Both interactive and automatic mode is available. In addition, a **custom root folder** under which assets should be imported can be selected. This is especially helpful if the "*/Assets*" root should be kept clean and organized.

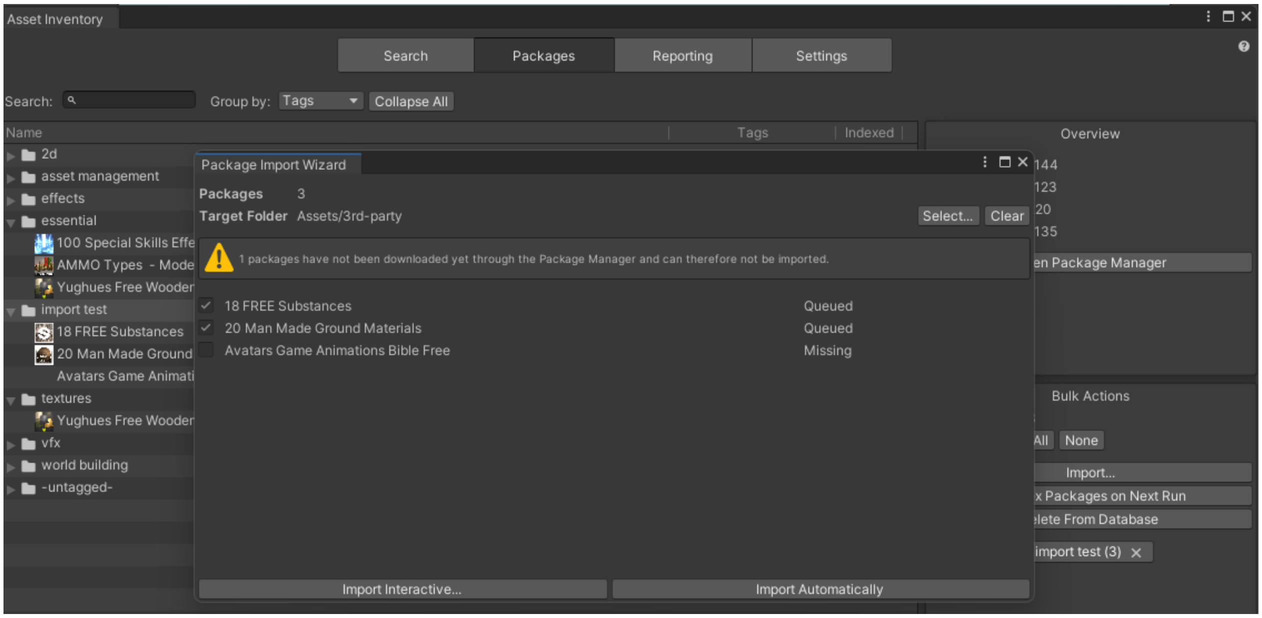
Some packages do not support custom locations and always need to be installed into the */Assets* folder, e.g. when using hard-coded paths. There is an exclusion list inside the tool to handle such cases automatically. If you encounter a package that does not work in a custom folder, please submit the package name & link on the support Discord.

If backups are available, you can go back to previous versions of an asset which can be helpful if the package author changed something incompatibly or there are issues. If multiple versions exist, the version label will change into a dropdown field.



Packages from registries will be added to the project manifest. If registry packages need a custom scoped registry this will also be added automatically.

Importing an archive will extract the archive into the target folder.

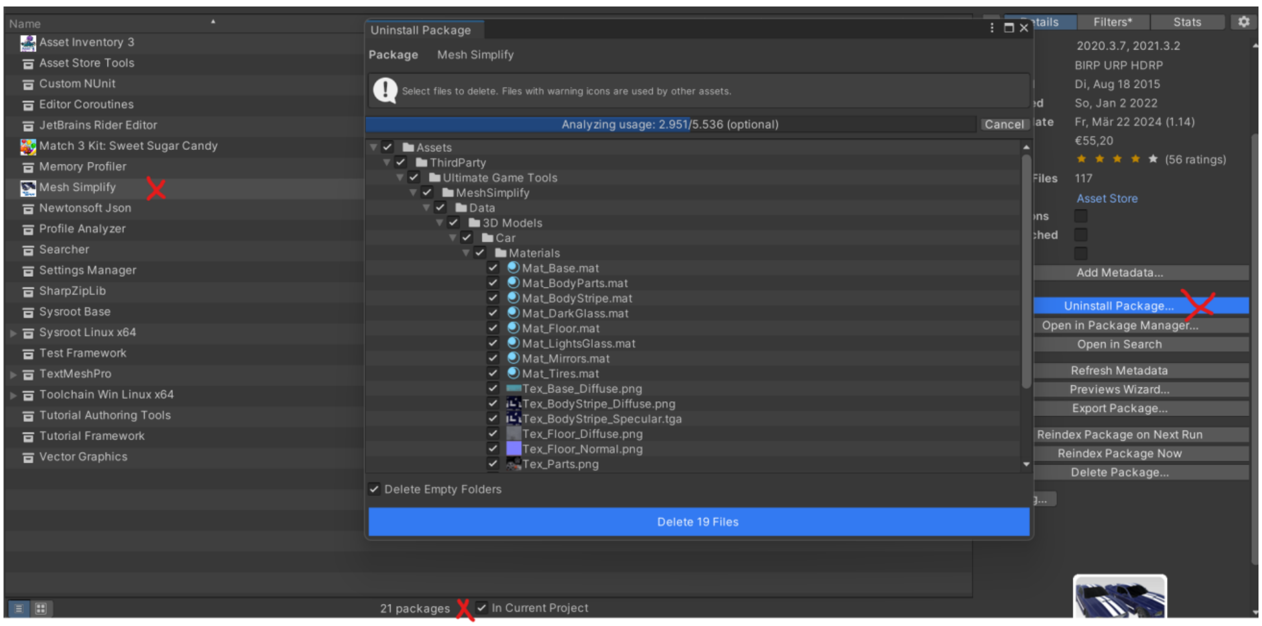


If items are not downloaded yet this can happen right in the dialog.

UNINSTALLING PACKAGES

Once a package is imported into the project, it can also be uninstalled again. For the full functionality to be available it is recommended to activate *"In Current Project"* under the packages list.

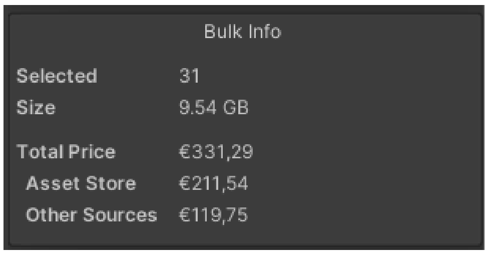
Registry packages will be uninstalled completely. In case of other assets, An *Uninstall Package…* button will appear. This will bring up a detailed view of all files installed from that package, their current location and it will perform an **analysis** which files are in use by other files in your project and show a warning if so. This way you can **fully or partially** uninstall packages



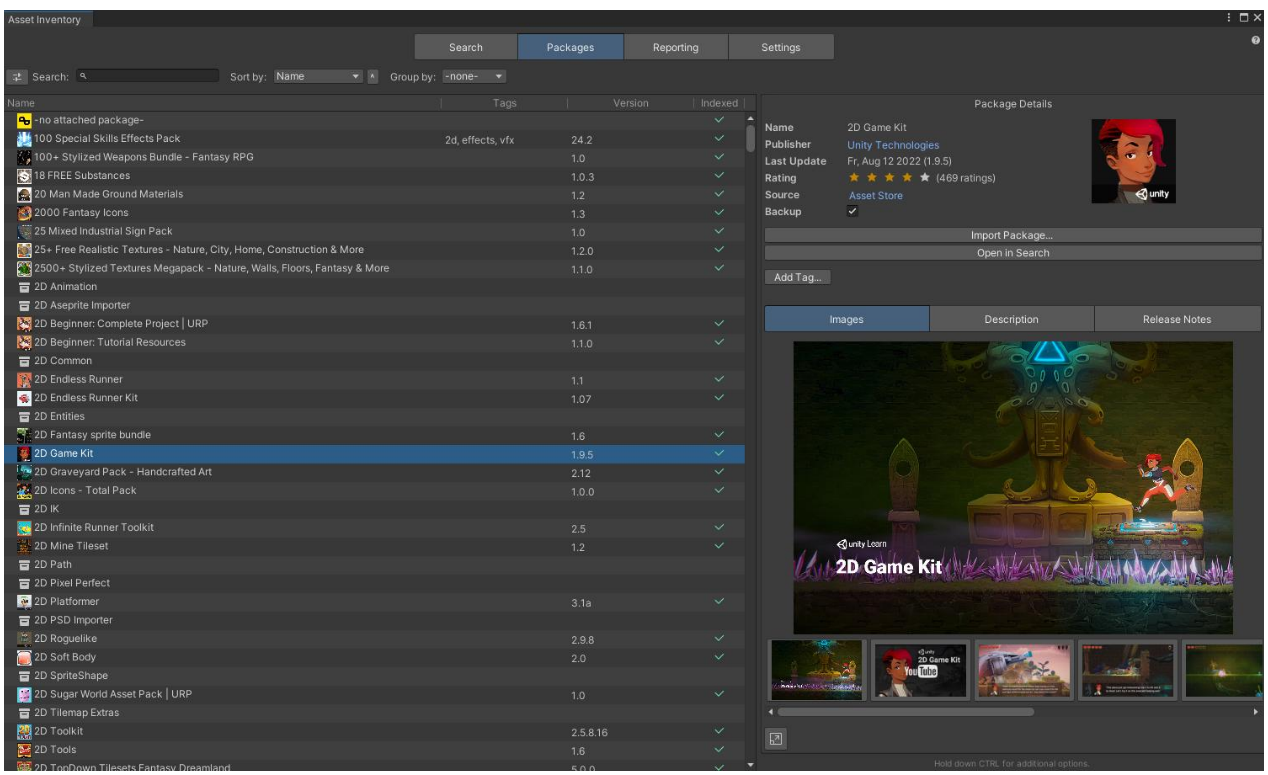
ADVANCED

In case packages should not appear in the search results and not be indexed, the **Exclude** toggle can be used in the package details. Excluded items will disappear from the list and can be reactivated using the "*Excluded*" maintenance view.

Bulk selection will show the total size of the compressed assets before they are downloaded so that you can estimate if temporarily downloading all for indexing would still fit on the hard drive. If available also costs are shown (current, non-discounted).



Clicking the icon in the lower right corner will toggle the details view into an **expanded view** twice as big. Depending on availability it will now also display media, description, release notes and dependencies (for registry packages).

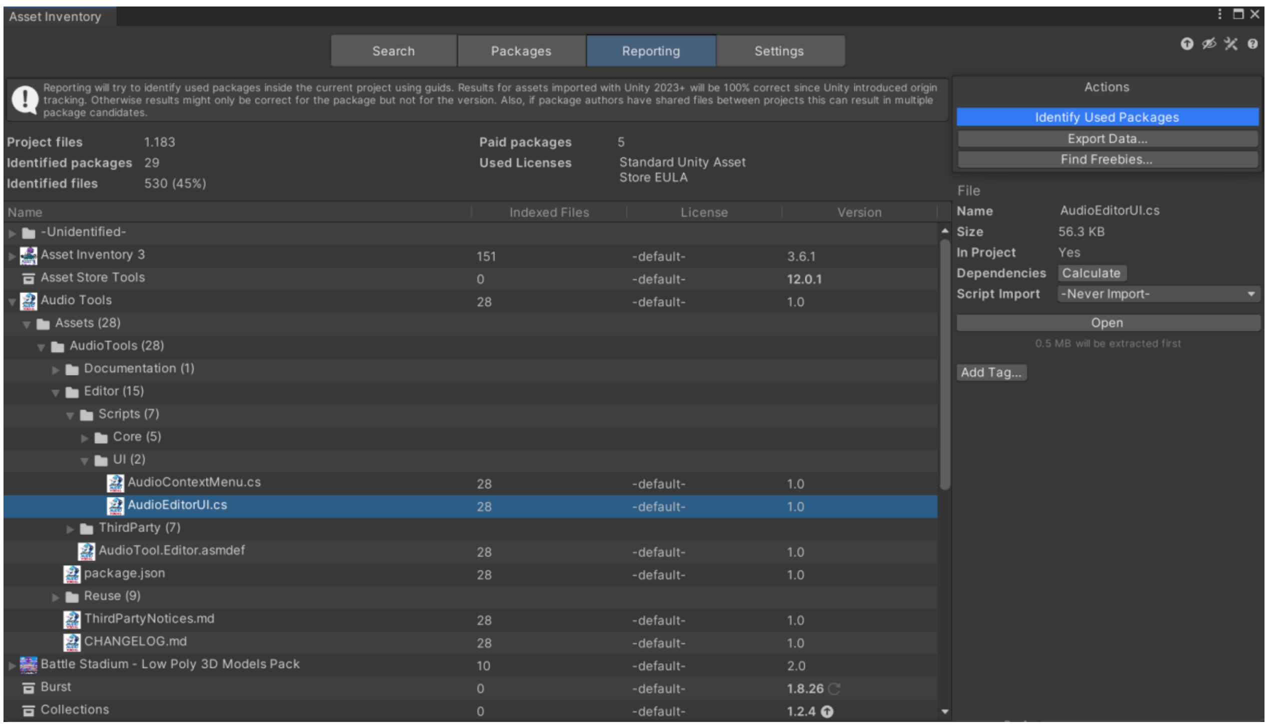


Holding **CTRL** will reveal a lot of additional package data and also many actions.

REPORTING

OVERVIEW

This module will scan the complete project and try to identify packages that were being used. It also supports exporting your data and listing used licenses.



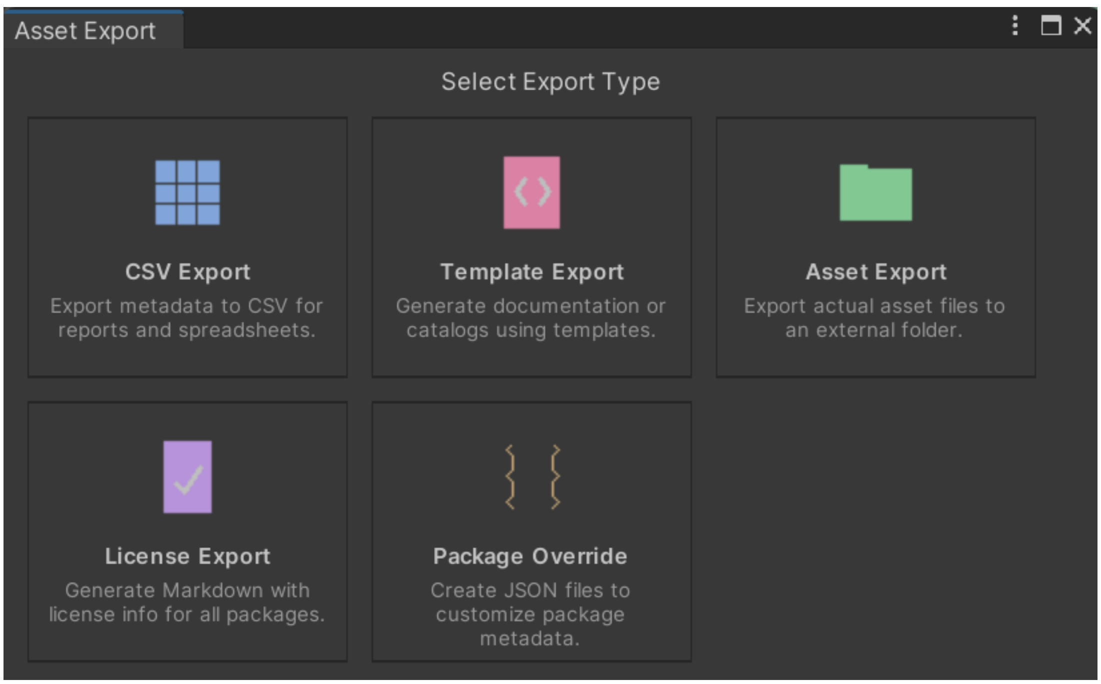
Packages that could be identified with 100% **confidence**, where also the version is guaranteed to be correct, will be shown with a **bold** version column. All registry packages automatically fall into this category. From Unity 2023+ also local assets are typically identified correctly if imported through Unity 2023+, since it will store the asset origin in the meta data files allowing to pinpoint the exact version correctly.

When selecting an asset in the Unity *Project View*, the reporting tab will try to identify the associated package in the lower right info box.

ASSET EXPORT

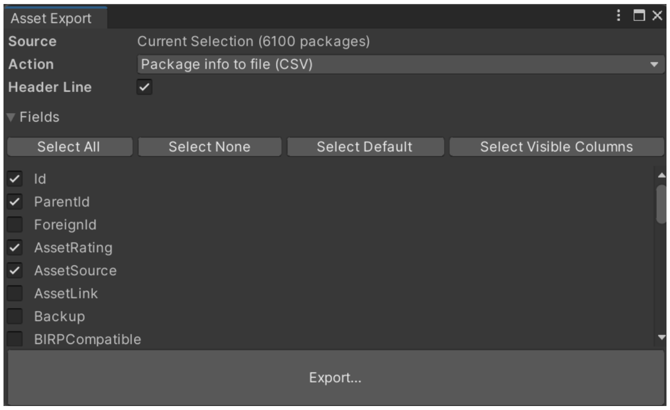
In case you want to perform your own analysis or create reports it is possible to export the database contents into various formats. Exporting packages can be done one-by-one or in bulk. Supported options are:

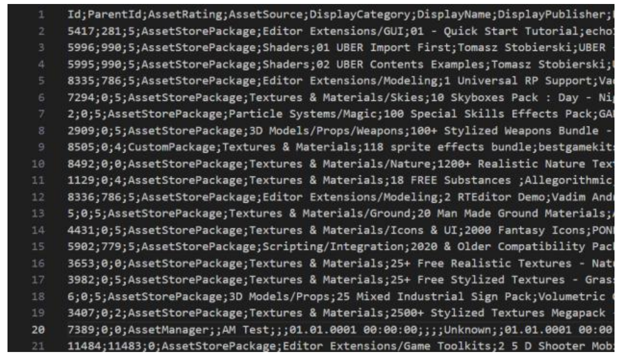
- **CSV**: for easy processing in Excel and other compatible software
- **Template-Based**: complex export formats like HTML webpages or any custom format utilizing an easy but powerful scripting language
- **Package Contents**: will extract packages and copy their files to another directory
- **Licenses**: will list all packages with a non-default license in MD format
- **Overrides**: JSON files which can be used to override package-specific metadata in team scenarios (e.g. to enforce common tags & categories)



CSV Export

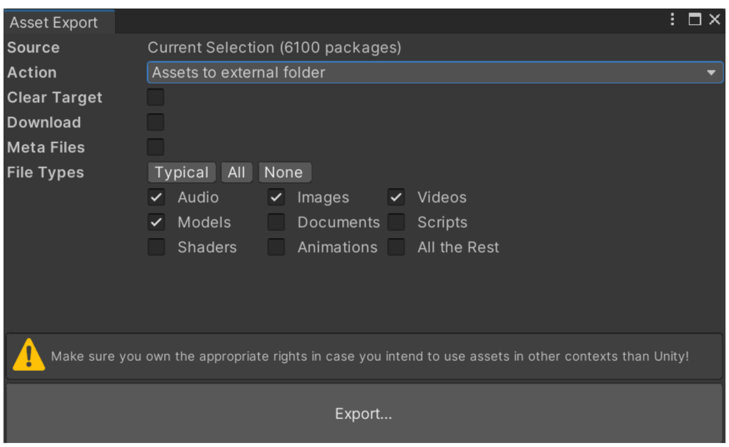
Will export package metadata to a single file which can be read by spreadsheet software or other tools.





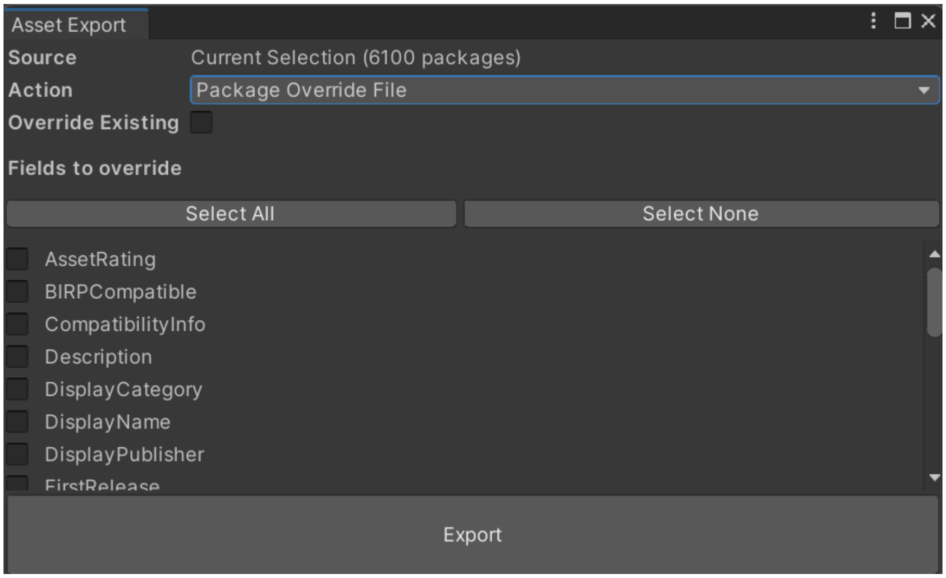
Package Content Export

Exporting files from a package will allow you to reuse these easily in other contexts.



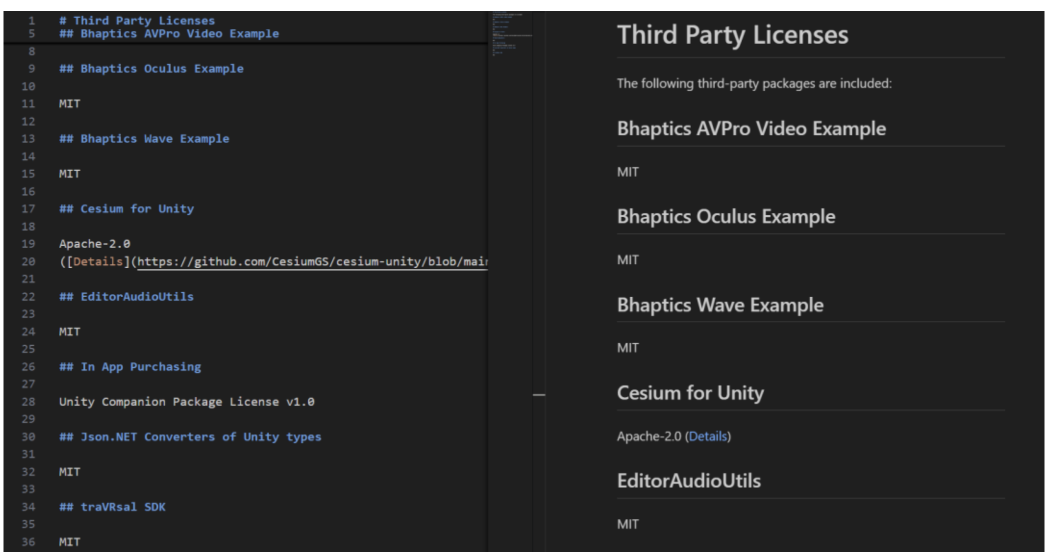
Overrides

Override *Json* files are a way of providing additional information for packages during indexing. This is helpful if e.g. a whole team is indexing and you want to ensure the same category or certain tags are set the same everywhere. To have a better starting point these files can be created with the existing data so that yo-u only need to adjust them instead of creating them from scratch.



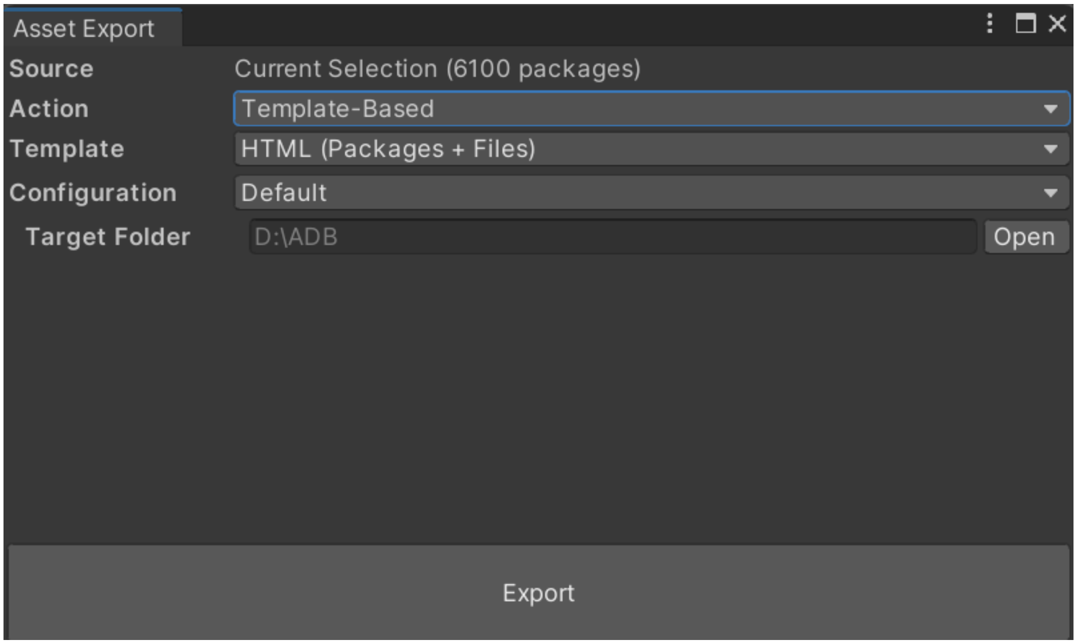
Licenses

It can be very useful for compliance to list all third party licenses involved in a project. The license export will create an MD containing all packages with a non-default license.



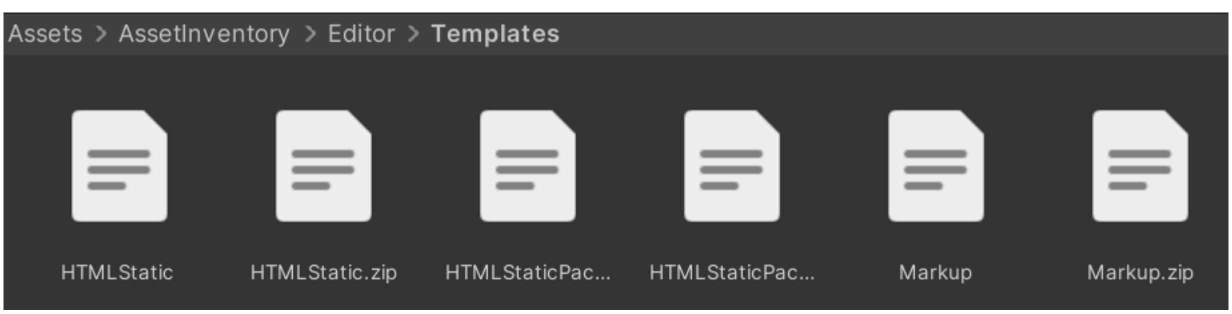
Template-Based

The most powerful export mechanism is the template based approach. It processes a whole set of input files, utilizes *Scriban*, a template scripting language, to embed data and export the result to a custom directory. Some powerful templates are already included and custom ones can be added easily.



Configurations allow to use different paths depending on the target environment, e.g. for local testing or when deployed remotely on a server. In one case you might want to load data & images from specific folders, another time from a CDN URL.

Depending on the type of template, only a subset of features might be visible. The included HTML template points to the *Previews* folder and therefore will always export next to that directory. What features are possible is defined in the **template descriptor**, an optional *Json* file with the same name as the template.

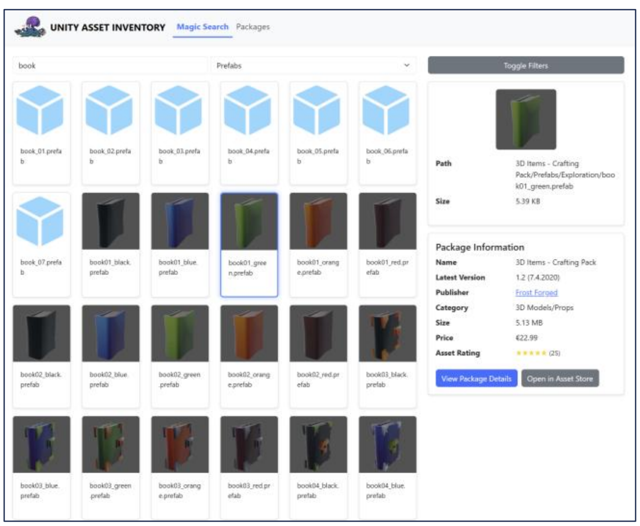


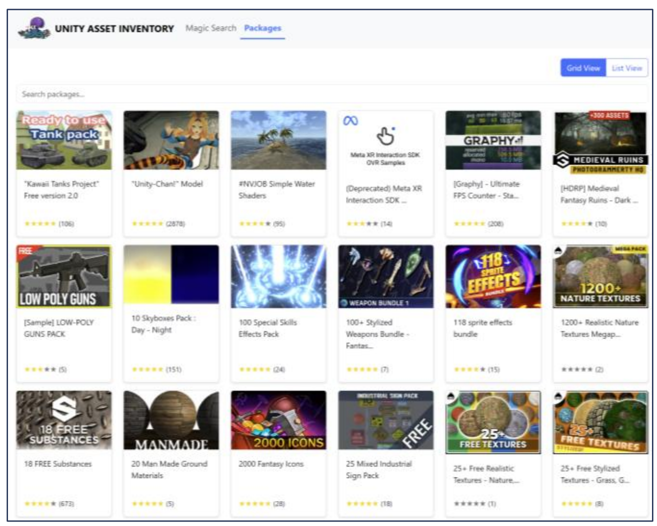
Below is an example template descriptor for the package-only HTML export based on the full HTML export template, extending it but deleting files and moving others around.

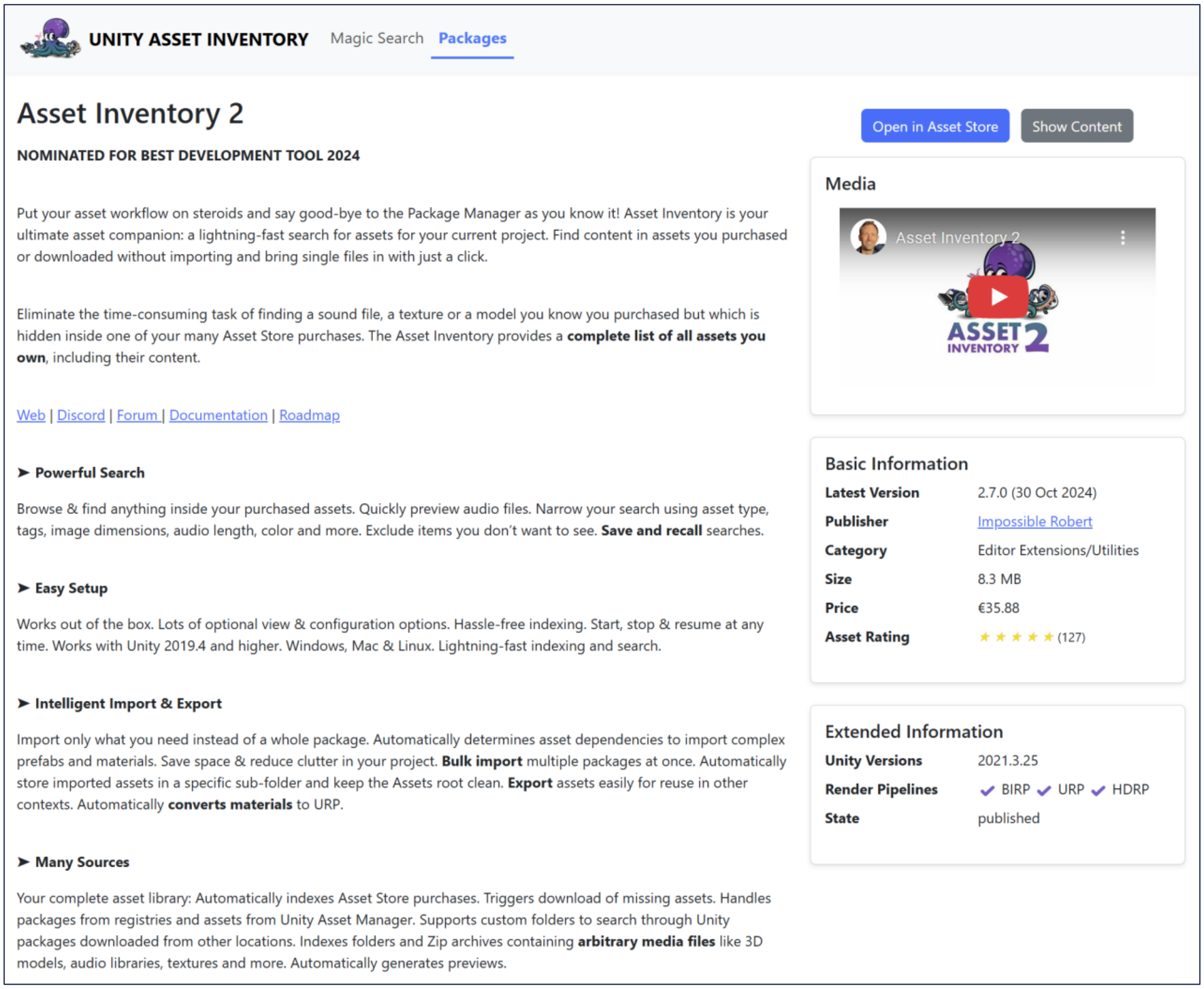
```
{
  "name": "HTML (Packages)",
  "description": null,
  "version": 1,
  "date": "2025-02-20T18:13:33.1939877+01:00",
  "readOnly": true,
  "inheritFrom": "HTMLStatic",
  "needsDataPath": true,
  "moveFiles": [
    "favicon.ico>site/favicon.ico"
  ],
  "deleteFiles": [
    "index.html",
    "site/js/search.js"
  ]
}
```

Available properties: *name, description, version, date, readOnly, isSample, fixedTargetFolder, entryPath, needsDataPath, needsImagePath, packageFields, fileFields, inheritFrom, moveFiles, deleteFiles*

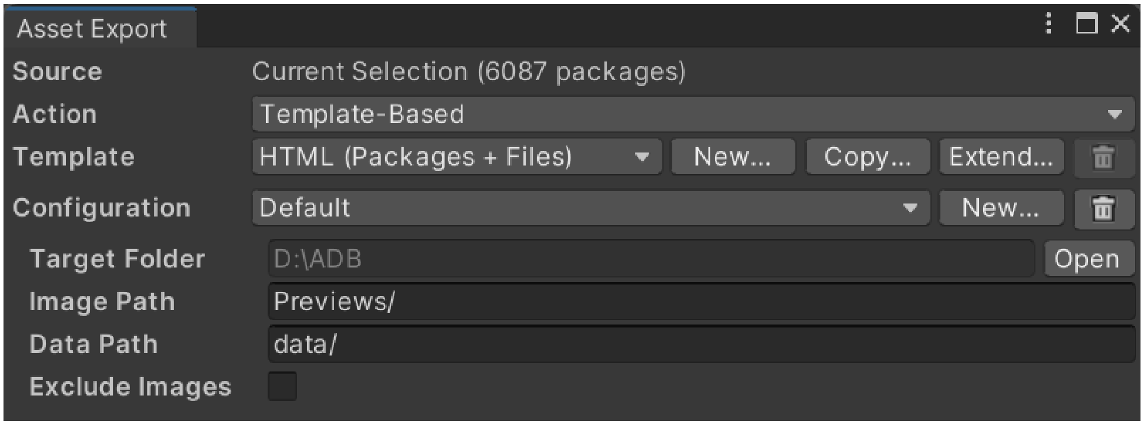
The full **HTML export** provides a serverless file search right inside the HTML export with powerful filter options, paging and rich visualization. The package view lists all packages and allows to see package details.



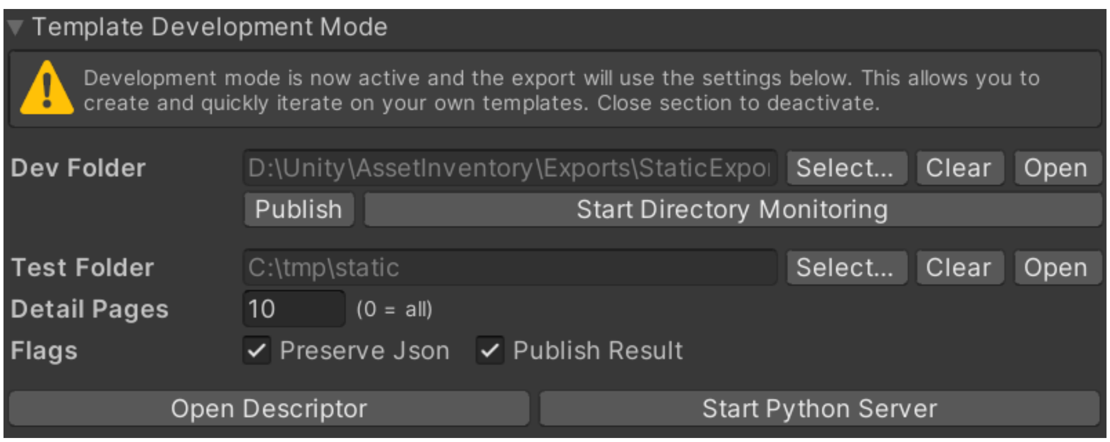



UNITY ASSET INVENTORY
Magic Search
Packages
Asset Inventory 2
NOMINATED FOR BEST DEVELOPMENT TOOL 2024
Put your asset workflow on steroids and say good-bye to the Package Manager as you know it! Asset Inventory is your ultimate asset companion: a lightning-fast search for assets for your current project. Find content in assets you purchased or downloaded without importing and bring single files in with just a click.
Eliminate the time-consuming task of finding a sound file, a texture or a model you know you purchased but which is hidden inside one of your many Asset Store purchases. The Asset Inventory provides a complete list of all assets you own, including their content.
Web | Discord | Forum | Documentation | Roadmap
➤ Powerful Search
Browse & find anything inside your purchased assets. Quickly preview audio files. Narrow your search using asset type, tags, image dimensions, audio length, color and more. Exclude items you don't want to see. Save and recall searches.
➤ Easy Setup
Works out of the box. Lots of optional view & configuration options. Hassle-free indexing. Start, stop & resume at any time. Works with Unity 2019.4 and higher. Windows, Mac & Linux. Lightning-fast indexing and search.
➤ Intelligent Import & Export
Import only what you need instead of a whole package. Automatically determines asset dependencies to import complex prefabs and materials. Save space & reduce clutter in your project. Bulk import multiple packages at once. Automatically store imported assets in a specific sub-folder and keep the Assets root clean. Export assets easily for reuse in other contexts. Automatically converts materials to URP.
➤ Many Sources
Your complete asset library: Automatically indexes Asset Store purchases. Triggers download of missing assets. Handles packages from registries and assets from Unity Asset Manager. Supports custom folders to search through Unity packages downloaded from other locations. Indexes folders and Zip archives containing arbitrary media files like 3D models, audio libraries, textures and more. Automatically generates previews.
Open in Asset Store
Show Content
Media
Asset Inventory 2
ASSET INVENTORY 2
Basic Information
Latest Version 2.7.0 (30 Oct 2024)
Publisher Impossible Robert
Category Editor Extensions/Utilities
Size 8.3 MB
Price €35.88
Asset Rating (127)
Extended Information
Unity Versions 2021.3.25
Render Pipelines BIRP URP HDRP
State published

Using the advanced features (hold CTRL), you can **create or extend** templates. Extending a template means that all files of the original template will be materialized upon export plus the files you put into the new template and the changes expressed in the descriptor.



You can also activate the **template development mode**. This is powerful to quickly iterate on templates, reduce the amount of data processed, automatically trigger exports upon changes in the dev directory and more. Setting a dev folder will export the template from this folder instead from the zip. Setting a test folder will always export to this folder first before copying the files over to the target (or skip this step if *Publish Result* is deactivated).



Once done developing, press *Publish* to package the folder into a *zip* and copy it into the *Templates* directory.

It is strongly recommended to use a **dedicated external folder** though for storing custom templates since the tool is usually updated by removing the full installation, which would then also remove the custom templates. This can be specified under *Settings/Locations → Custom HTML Templates*. Once set, all new and modified templates will automatically be stored there.

Template Language

Files of type *.html*, *.js*, *.csv*, *.txt* and *.md* will be parsed by the template engine and any placeholders will be replaced. The engine used is [Scriban]. It is a simple but powerful language supporting many programming constructs, variables, include files and more. Below is an example how to iterate over all packages.

```
<div class="row" id="cardView">
  {{ for package in packages }}
    {{ if package.parent_id == 0 }}
      <div class="col-md-2 mb-3 d-flex align-items-stretch">
        <div class="card w-100 h-100 position-relative">
          <img loading="lazy" src="{{ package.slug }}"
               class="card-img-top" alt="{{ package.display_name }}" onerror="this.src = 'images/nocover.png';">
          <div class="card-body d-flex flex-column h-100">
            <div>
              {{ package.display_name | string.truncate 40 }}
            </div>
            <div class="mt-auto">
              {{ include 'assetrating.html' }}
            </div>
            <a href="package_{{ package.asset_id }}.html" class="stretched-link"></a>
          </div>
        </div>
      </div>
    {{ end }}
  {{ end }}
</div>
```

A number of predefined variables exist which can be accessed:

- **dataPath**: path to the json files
- **imagePath**: path to the preview images
- **pageSize**: number of results per page
- **hasFilesData**: true if the current package has files indexed
- **internalIdsOnly**: true if only the internal id should be used for detail pages
- **packages**: all packages selected for export
- **package**: the currently selected package (e.g. on detail pages)
- **packageFiles**: all files of the current package

Some files inside a template trigger special functionality:

- **packages.json**: will contain the data about all packages
- **files.json**: will contain minified data about all packages
- **package_details.html**: will be replaced with files of the form *package_412.html* where 412 is the internal Id of a package for each package. If the asset is from the

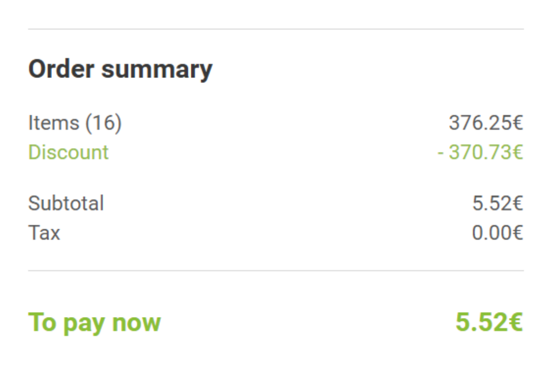
Asset Store, the name fill be *package_f72632.html* using the foreign Id of the package if not set otherwise by the environment setting *internalIdsOnly*.

SCANNING FOR FREEBIES

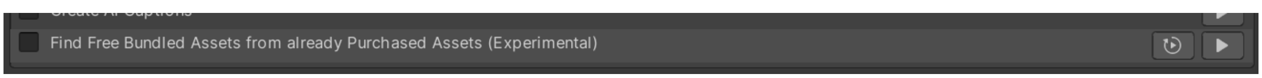
When purchasing Asset Store packages, authors sometimes grant **reduced or even free access to other packages** of them. Also, some authors sell **bundles**. When purchasing a bundle, a linked list of other packages becomes available for free. These packages are typically listed in the description.

The Asset Store does not show these free packs though easily and one has to actually go to every linked package in order to claim them.

This can lead to the common situation, that many packages remain unclaimed although they are already owned. When not claimed, the tool will not index them to make the contents available for use.



Using the Freebie scanner action will check all your purchased packages, if they contain any links to other assets in the description and will show these as potential candidates that you can claim.



Since the Asset Store does not provide an API to detect this automatically, you will need to check the results manually on the Asset Store website but can easily open the identified candidates from the log file or, if using the *force* mode, all will be opened automatically as tabs. Keep in mind, that only a fully loaded tab will show the correct result as the Asset Store changes the price only right at the end of the loading process.

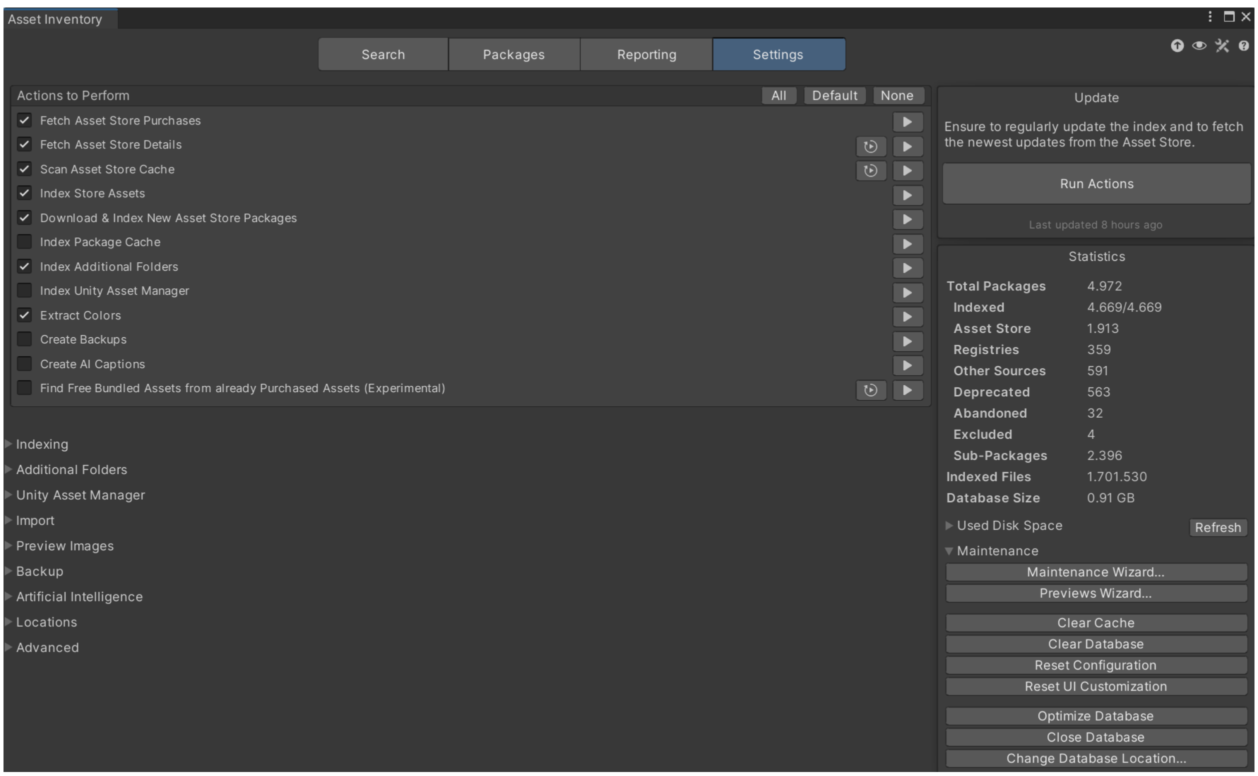
SETTINGS

OVERVIEW

Use this section to configure what and how should be indexed. Indexing can be started and stopped at any time and will pick off again where it last stopped.

ACTIONS

All features are separated into **actions**. Using the *Run Actions* command will run all selected ones at once. Each action can also be triggered individually. They will be executed from top to bottom.



Indexing the Asset Store cache is activated by default and the main source for your data. There are two options available:

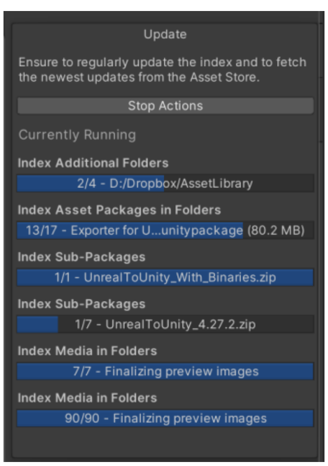
Index everything that is already downloaded into the local cache.

- That means you only need to click the **Download** button in the **Package Manager** window or right in the Asset Inventory for each asset you own (without importing them) to make them available for the index (bulk selection is also possible).
- If you have specified a **custom asset cache directory** in Unity 2022.1 or higher, it will typically be auto-detected. Press CTRL to see the resolved path and verify. You can manually override the location by switching the Asset Cache Location to *Custom*. If the environment variable ASSETSTORE_CACHE_PATH is set it will override the default asset cache location.

Automatically download purchased assets for indexing and delete them again afterwards.

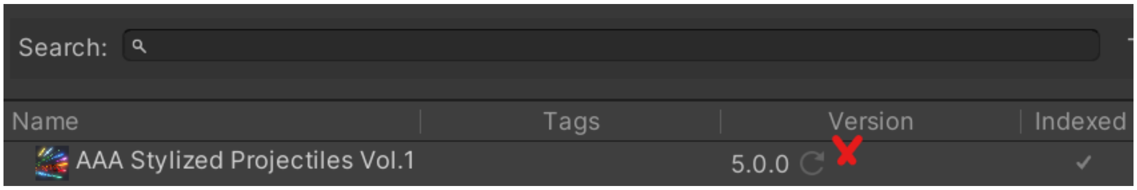
- With this option you will **index everything you have** without the need to download everything, saving a lot of disk space, at the expense of a potentially very long indexing due to all the necessary downloads.

Nearly all actions are incremental, meaning you can **interrupt and restart these at any time** and they will mostly continue where they left off.



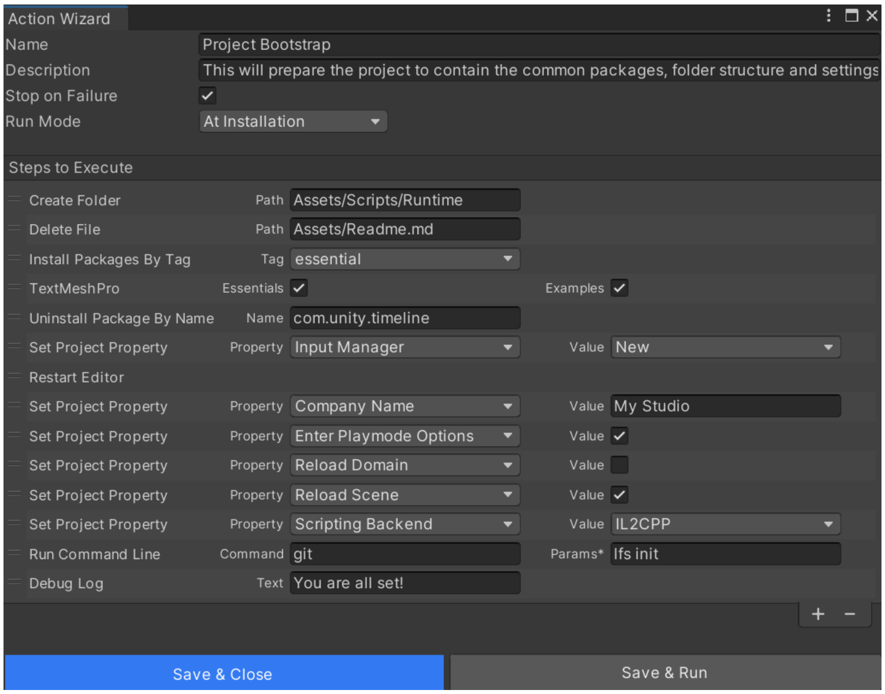
Once an asset is indexed it does not need to remain on your hard drive if you just want to search for it. Only when importing, it needs to be available.

If an asset is still scheduled to be indexed or a new version was downloaded, an indicator will appear in the version column. In that case run the actions and it will automatically index all remaining packages.



Custom Actions

In addition to the predefined actions it is possible to define your own ones. These can have any number of steps. Use the buttons at the bottom of the actions list to add or remove actions.



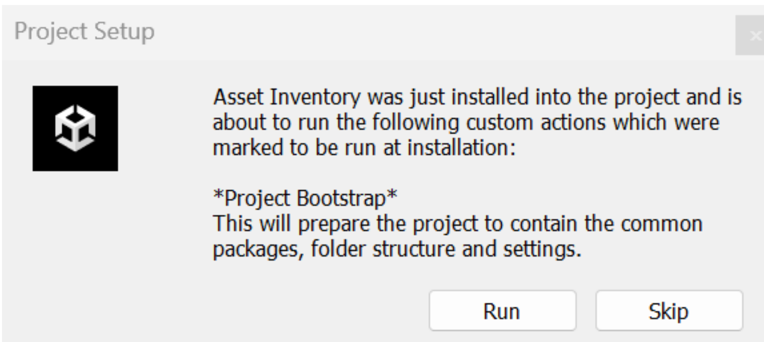
When selecting an action, you can run it individually using the Play button at the end. Or you run all steps with the *Save & Run* command at the bottom.

An important setting is what to do in case of errors. This can be defined for the whole action.

--- Files And Folders ---
Create Folder
Move Folder
Delete Folder
Copy File
Move File
Delete File
Compress Folder
Extract Folder
--- Importing ---
Install Package By Name
Install Package By Path
Install Packages By Tag
Uninstall Package By Name
Uninstall Feature By Name
Update Package
TextMeshPro
--- Actions ---
HTML Export
FTP Upload
FTP Delete
--- Settings ---
Set Project Property
Add Define Symbol
Remove Define Symbol
Add Compiler Arg
Remove Compiler Arg
--- Misc ---
Run Action
Run Command Line
Set Text Variable
Debug Log
Message Dialog
Restart Editor

There are two run modes available:

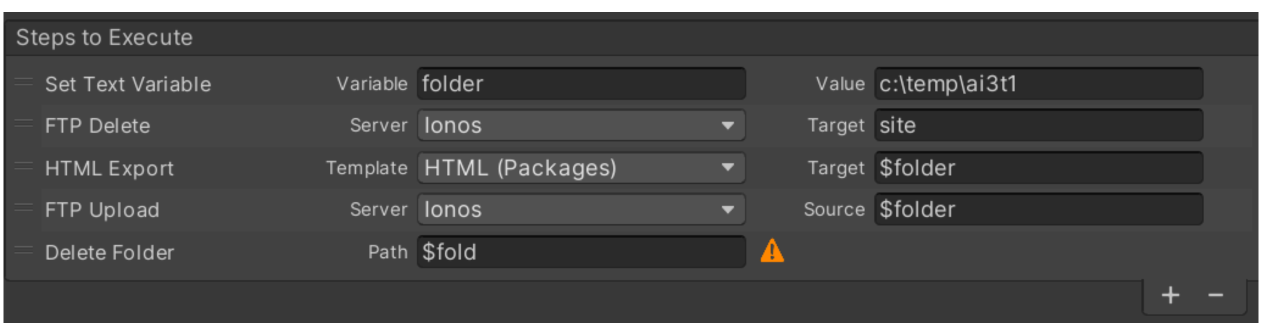
- **Manual**: Execute the action manually from the UI or as part of the currently selected actions
- **At Installation**: In addition to manual, the action will bring up a dialog after the initial installation of the tool in a new project to be automatically run.



Variables

When actions become more complex, e.g. to automate certain folder tasks, or perform combined steps like exporting, compressing, uploading, clean-up, some items can become repetitive and through that error-prone, e.g. the name of a folder.

This can be simplified with variables where you only specify a value once and then reuse it. Use the *Set Text Variable* step for your own variables and use them via $name afterwards. Variables which could not be resolved will be marked with an exclamation mark.



In addition to your own variables, many predefined variables can be accessed. These have the naming scheme *"group.value"*. You can look up the possible values in the linked documentation below.

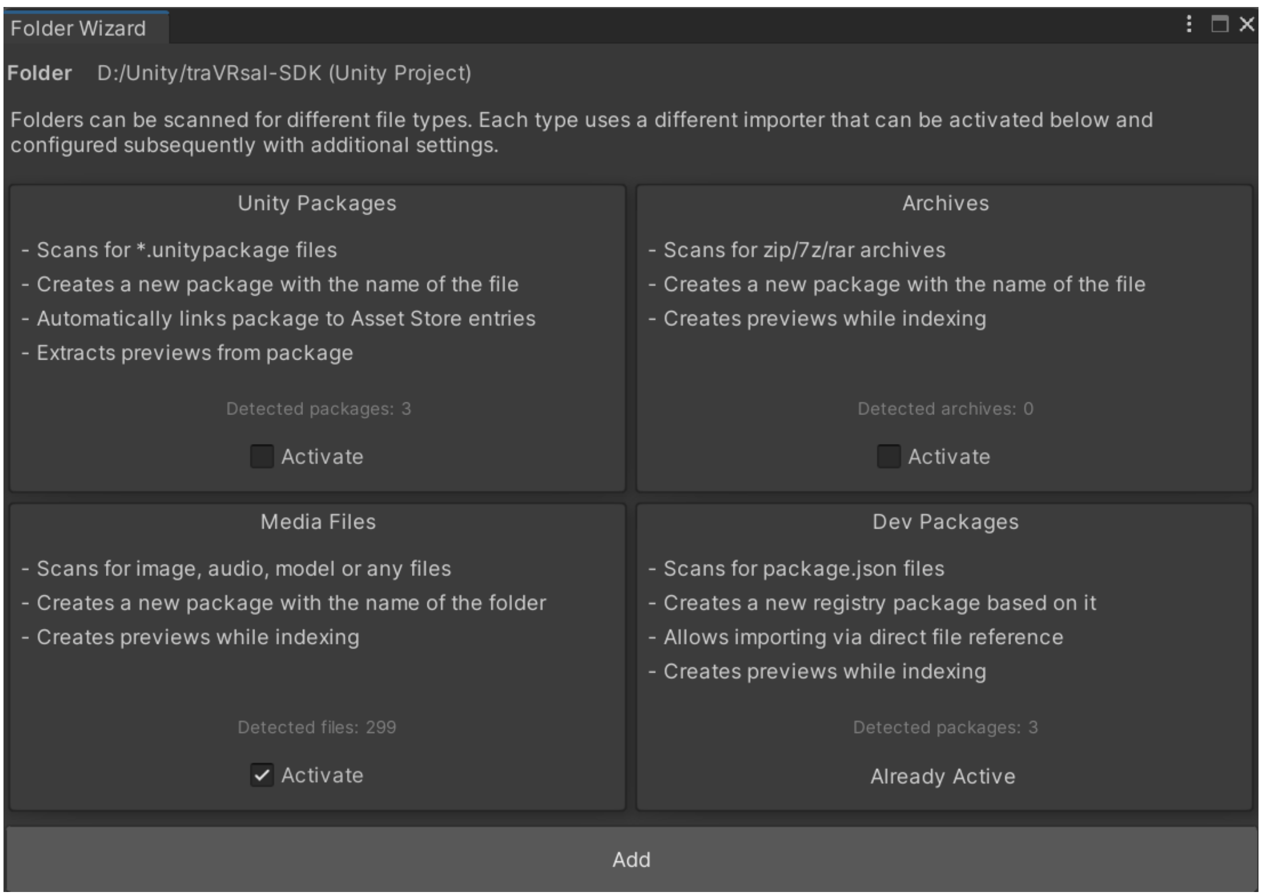
Available predefined variables:

- Application (e.g. Application.dataPath)
- SystemInfo (e.g. SystemInfo.deviceName)
- Environment (e.g. Environment.MachineName)
- Config (e.g. Config.exportFolder)
- DateTime (e.g. DateTime.Now)
- PlayerSettings (e.g. PlayerSettings.companyName)
- EditorApplication (e.g. EditorApplication.applicationPath)
- QualitySettings (e.g. QualitySettings.maximumLODLevel)
- Screen (e.g. Screen.height)
- BuildTarget

LOCAL FOLDERS

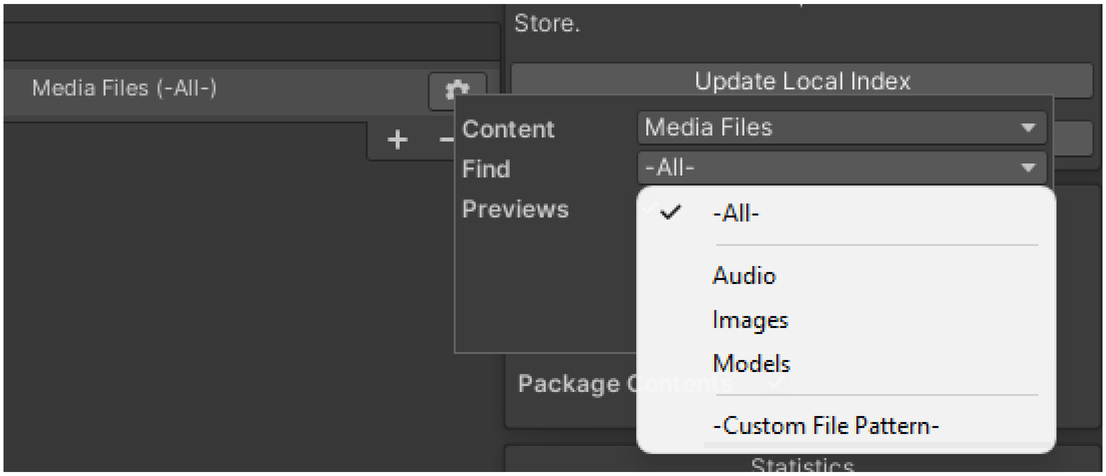
If you want to index Unity packages downloaded from **other sources**, simply add the folders to the **additional folders** list. When selecting an entry from the list of additional folders, more options can be specified on the right-hand side. This way it is possible to select what types of files should be searched for:

- **Unity Packages**: Finds any *.unitypackage* files in the folder and indexes the content.
- **Development Packages**: Finds any *package.json* files and treats those as local packages.
- **Media Files**: Allows to index all kinds of other, **free-floating** assets, like images, models, audio libraries etc. Also Unity projects can be indexed this way.
- **Archives**: Will extract archives (zip, rar, 7z) on the fly and index all contents.

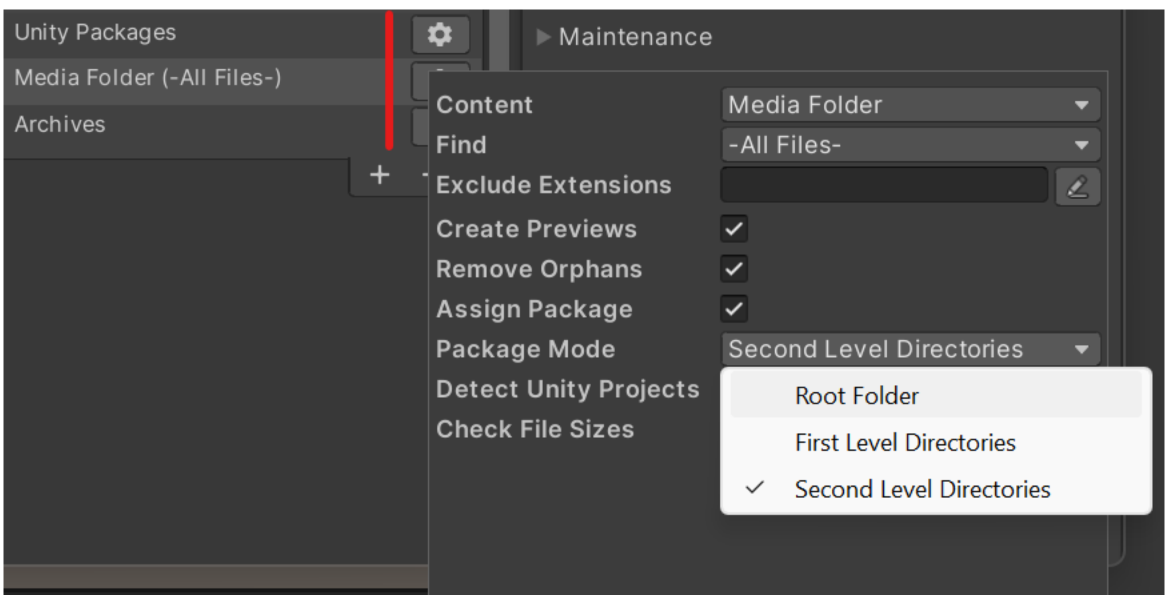


Using additional media folders will let Asset Inventory act as the go-to place for asset management and quickly finding any available asset on your drives from a single, consistent UI. Media files can optionally have no connected asset but otherwise behave the same. If they have a *.meta* file next to them, their GUIDs will also be stored.

The order in which additional folders are processed can be changed by dragging them up or down in the list. Deactivating or removing an additional folder entry will not remove it from the index.



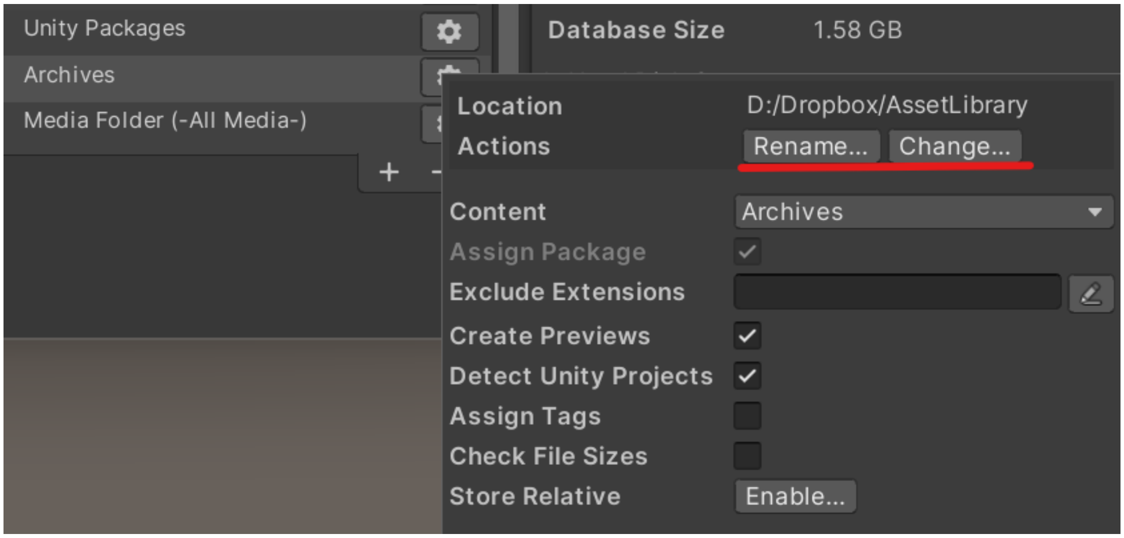
To find the contents of the folder easily later, it is recommended to let the tool create a package. If not done so all files will be stored under the global *-No Package Assigned-* package. There are different **package modes** available which you can select depending on your folder contents:



In root folder mode, the name of the additional folder will become the package name. If first or second level directories is selected, a new package for each identified subdirectory will be created.

If you remove an additional folder setting, the indexed data will still remain.

If you rename an additional folder or move it somewhere else, you can use the settings button to point the entry to the new valid location. Otherwise your index would be broken. You can also rename the folder right there as well instead of going to the file system.



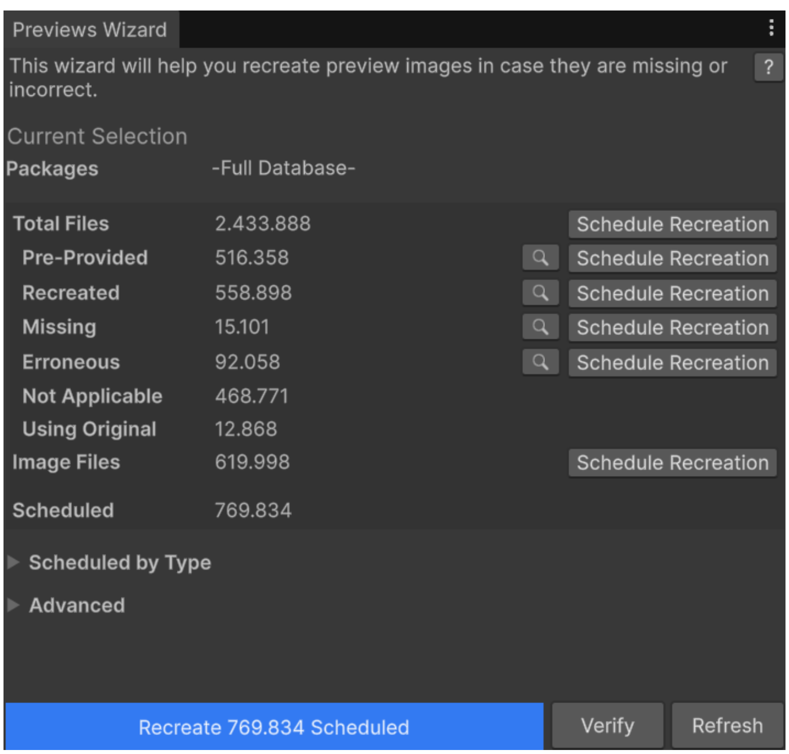
The options are only available in advanced mode so make sure to click the eye icon before.

PREVIEWS

The search result will show previews of the assets. These previews can come from three sources:

- Included in the package and created at the time the asset was built (state: pre-provided)
    - This is done by the Asset Store tooling and the reason why e.g. previews for sound files can look different depending on which version of Unity was used.
    - Sometimes these previews can be empty when exported incorrectly.
- Created by Asset Inventory during import (state: recreated)
    - This happens automatically for atomic assets without dependencies, e.g. sound files, textures or models.
- Created by Asset Inventory on-demand (state: recreated)
    - This mode can also create previews for prefabs and materials but will take **much** longer since all dependencies need to be materialized first.

It is possible to recreate a single preview from inside the search view, all previews for a package or perform recreation for missing previews in the complete database. There is a **Preview Wizard** to help with all steps. Previews can also easily be switched between provided and recreated there in case of issues, e.g. incompatible render pipelines.



Preview file generation will work by temporarily copying each file into the current Unity project, letting Unity create a preview and removing it again. While creating previews it might happen that a new folder called *_TerrainAutoUpgrade* will appear under Assets as a side-effect. This process will take a bit of time but will allow Asset Inventory to show preview images and also additional meta data for many file types.

Image files will be handled separately for common file types, bypassing Unity, which is much faster and will, if active, automatically scale preview images to a higher resolution already during indexing.

Image **verification** is a recommended additional step after preview creation to detect if a preview was correctly created or e.g. the materials are pink due to the wrong SRP in use.

The tool will by default not recreate previews for audio files. These usually take a very long time with rather limited benefit. This can be changed with the *Exclude Extensions* setting. The field allows to enter any arbitrary file extension like *wav* but also accepts tool-internal type groups which bundle multiple extensions at once. Available type groups are:

- *Audio, Images, Videos, Prefabs, Materials, Shaders, Models, Effects, Animations, Fonts, Scripts, Libraries, Documents, Scenes*

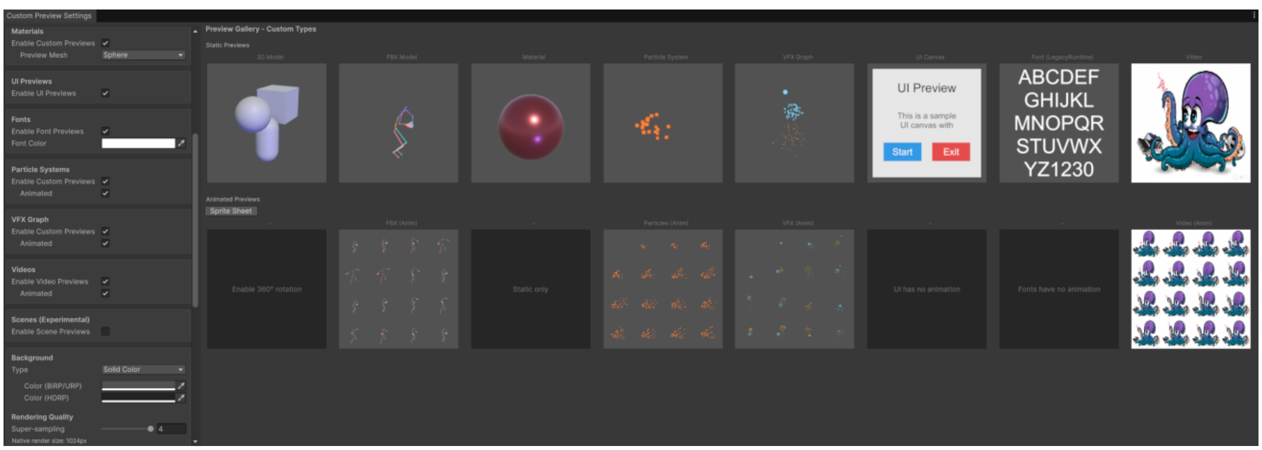
Custom Preview Pipeline

Unity's own previews are limited in multiple ways:

- Limited to 128x128 pixels, with often bad lighting, especially in SRPs
- No support for UI, fonts, videos, particle and VFX systems
- No animation preview

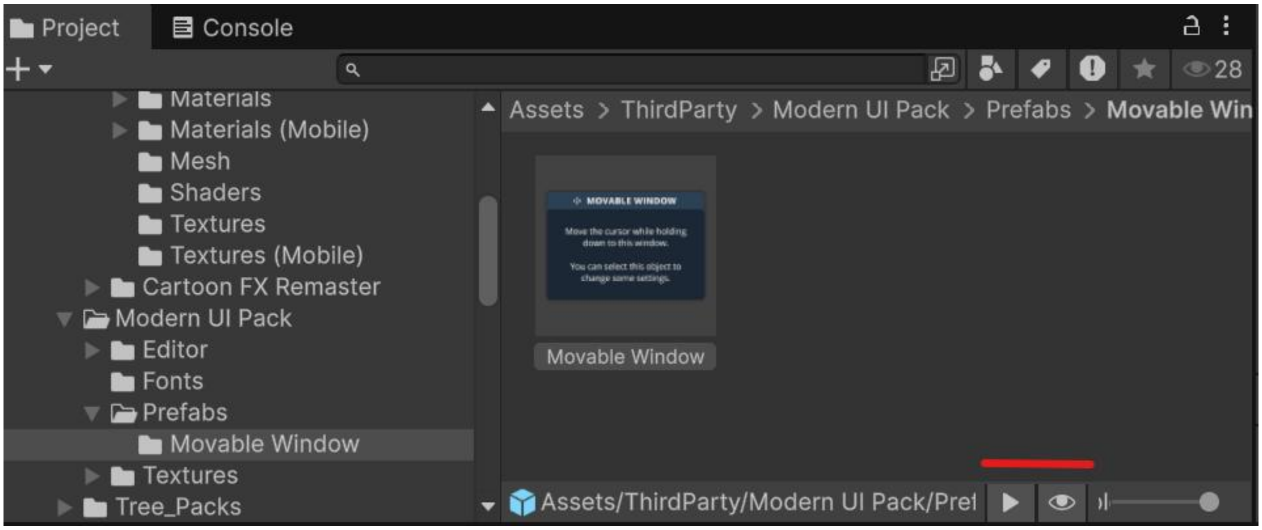
The tool includes a custom infrastructure to create better previews which solve all of the above. All settings for it can be found under *Settings/Preview Images/Configure Custom previews*.

The window will show a **preview** of all supported custom preview types. This also includes Unity-supported types like *3d prefabs, fbx and materials* as it significantly enhances their representation. There are many parameters to tweak the previews the way you want.



Animated previews will play directly in the search results when an item is selected. FBX animations will show the first non T-Pose animation as a preview.

Once an item is imported, the tool will also automatically **replace the preview that Unity shows in the Project window**, including animated previews. You can toggle this behavior using the two additional buttons at the lower right of the Project window.



BACKUP

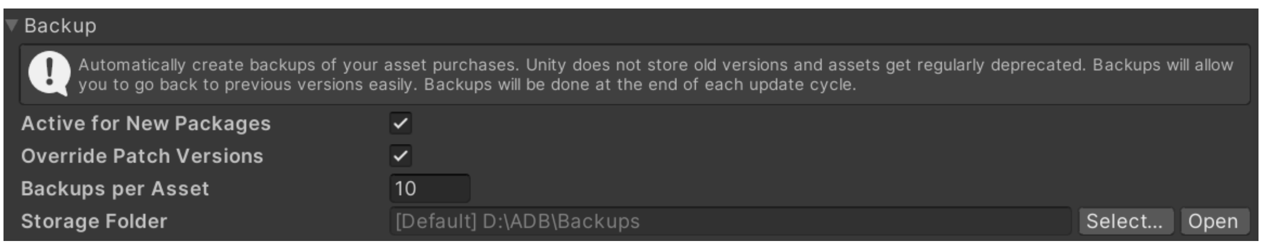
The tool supports two backup mechanisms, one for packages from the Asset Store and one for the database of the tool itself.

Unity Package Backup

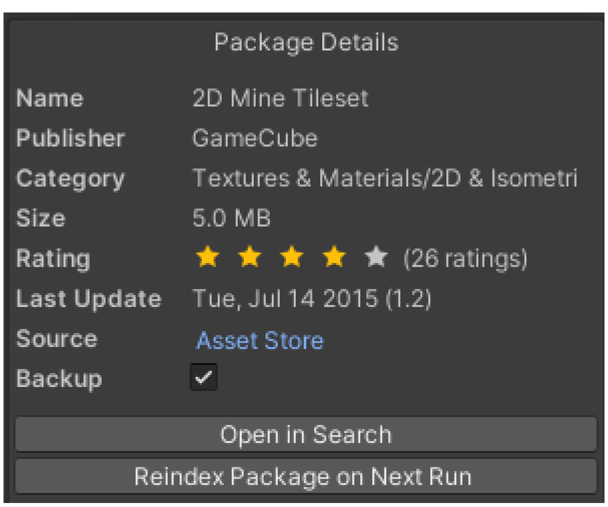
This will copy packages to a dedicated place for a permanent backup. This way you are not affected by package deprecation by Unity where it might be impossible to download older (but for you working or compatible) versions of assets anymore.

The backup tool can keep multiple versions of packages and will automatically handle the life cycle, version comparison and more.

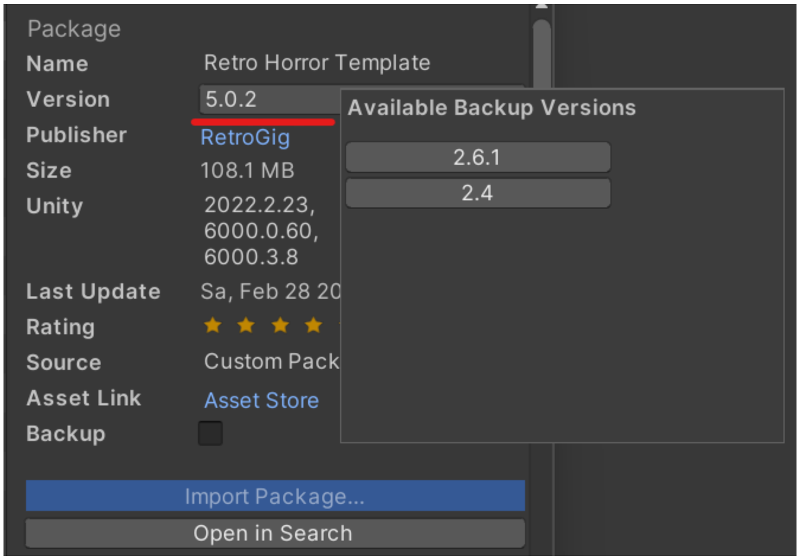
Per default, all files are stored flat in a single folder. If you need some sort of structuring inside the folder to better support replication scenarios, you can move arbitrary files into **sub-folders**. The tool will honor this dynamically and put new backup revisions into the respective folders then.



Once backup is configured and the corresponding **action activated**, it can be set individually per package in the package details (also in bulk).



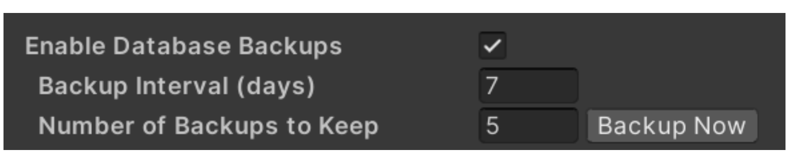
If backups are available for a package, the version will become a dropdown and allow to import packages of older versions.



Tool Backup

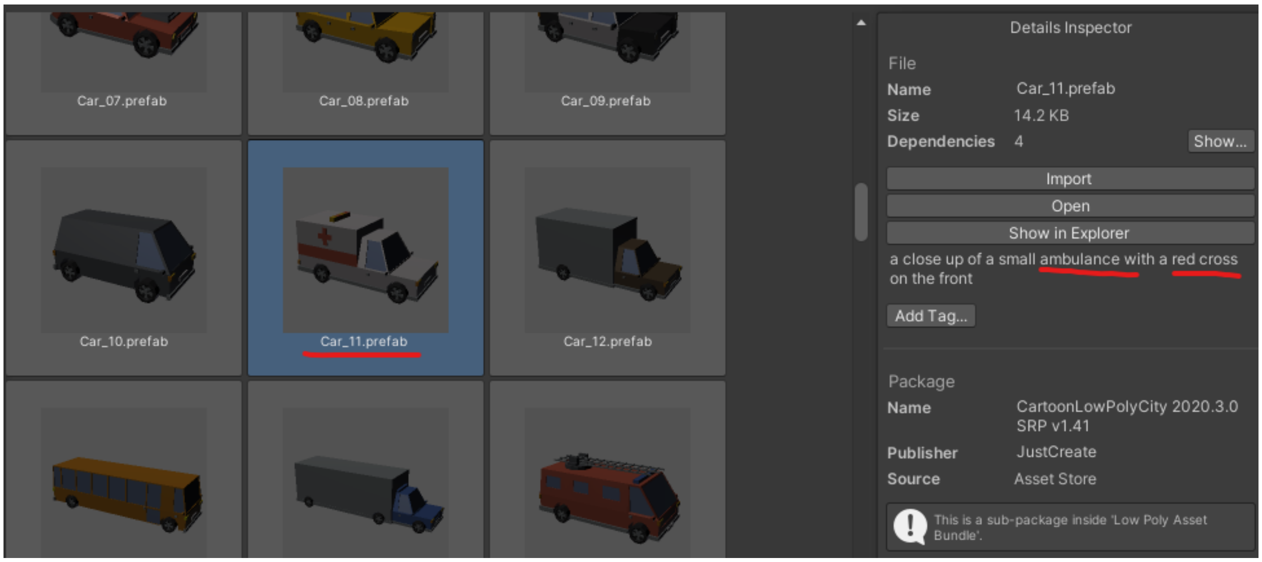
The tool will store all data, index information, AI captions and more in a central database. It is prudent to regularly do a backup in case of accidental deletion of packages or running destructive operations. This way you can go back to a working state and only need to redo the delta since then and not potentially the whole database.

This type of backup is active by default and will also be stored in the central backup folder. There you find the *.db* files with their timestamp. Right now this is only available when using *SQLite* as the database backend. A backup of the current configuration will also be made.



ARTIFICIAL INTELLIGENCE

Using AI it is possible to automatically create captions that describe individual asset files. This way you are also able to find files which are labeled inappropriate or insufficient. See the example below:



While the asset is simply called *"Car_11"*, the AI has captioned it *"a close up of a small ambulance with a red cross on the front"*.

Searching for *"ambulance"* now and using the AI generated captions during the search will also yield *"Car_11"* as a result, which would have been impossible to find before.

Setup

The basis for AI captioning are the existing preview images. Make sure you have performed preview recreation where applicable.

AI inferencing will happen on your own device. This means it does not use any remote server and also does not incur costs for a third party service. It requires a potent PC though as the process is resource intensive. An exception is a remote model configured through LM Studio. In that case it is strongly advised to keep the costs under watch.

There are three technologies supported for captioning:

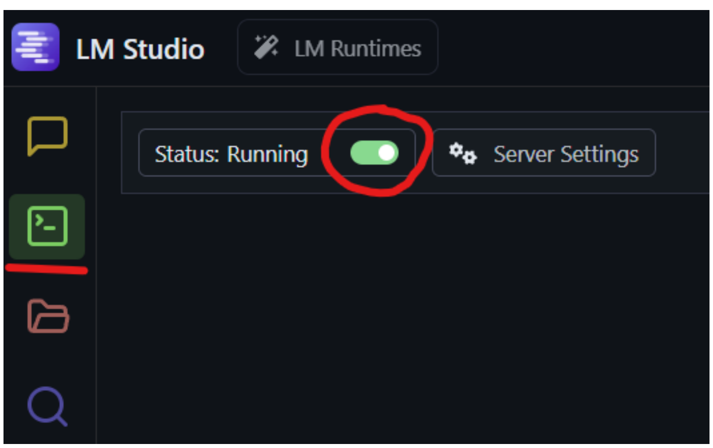
- A command line tool called blip-caption using the freely available image captioning model Salesforce Blip, coming in a *base* and a *large* version. This is the legacy backend now and still contained for compatibility and users who prefer it.
- Ollama, which is a modern and sophisticated solution to run many different models locally. It is much easier to set up, supports a continuously updated list of new models, runs very efficient and also very fast. This is the **recommended** way. The only drawback is that it needs multiple additional dependencies, which might cause conflicts in your project. In this case either perform the AI captioning in a dedicated separate project or use another backend.
- LM Studio, which is a good alternative to Ollama in case of dependency issue, if you want to use more models or if you have it installed anyway.

Ollama Setup

Follow the installation instructions on https://ollama.com/. Once installed, the tool will detect the installed service automatically.

LM Studio Setup

Follow the installation instructions on https://lmstudio.ai/. Once installed, start the local server.



The tool will detect the installed service and list the available models afterwards automatically.

Blip Setup

To use it locally, Python 3, pipx and the blip-caption tool are required to be installed. These are the basic steps assuming basic familiarity with the computer system:

- Install PipX
  - Windows
    - Install Scoop
      - Open a new PowerShell window (no admin required)
      - Run: *Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser*
      - Run: *Invoke-RestMethod -Uri https://get.scoop.sh | Invoke-Expression*
    - Switch to a normal Command line window
      - Run: *scoop install pipx*
      - Run: *pipx ensurepath*
  - MacOS
    - Run: *brew install pipx*
    - Run: *pipx ensurepath*
- Install tooling
  - Run: *pipx install blip-caption*
    - *In case of errors on MacOS, most likely need torch in an older Python version (use an environment therefore and adapt global path)*
      - Run: */usr/bin/python3 -m venv .venv*
      - Run: . *.venv/bin/activate*
      - Run: *pip install blip-caption*
      - Edit ~/.zshrc and add: *export PATH="$PATH:/Users/NAME/.venv/bin"*
      - Run: *source ~/.zshrc*
  - Run: *pipx ensurepath*
- Restart the system
- Click "Create Caption" in the Settings/AI section to test if all if working as expected. The first time the tool starts will take a significant amount of time since it will download the model file once.

```
C:\Users\[redacted]>scoop install pipx
Installing 'pipx' (1.7.1) [64bit] from 'main' bucket
pipx.pyz (321.6 KB) [==========================================================================================] 100%
Checking hash of pipx.pyz ... ok.
Running pre_install script...done.
Linking ~\scoop\apps\pipx\current => ~\scoop\apps\pipx\1.7.1
Creating shim for 'pipx'.
'pipx' (1.7.1) was installed successfully!
'pipx' suggests installing 'python'.

C:\Users\[redacted]>pipx ensurepath
Success! Added C:\Users\[redacted]\.local\bin to the PATH environment variable.

Consider adding shell completions for pipx. Run 'pipx completions' for instructions.

You will need to open a new terminal or re-login for the PATH changes to take effect. Alternatively, you can source
your shell's config file with e.g. 'source ~/.bashrc'.

Otherwise pipx is ready to go! ✨ 🌟 ✨

C:\Users\[redacted]>pipx install blip-caption
⣾ installing blip-caption
```

There is support for GPU acceleration but it requires a [custom patch of the blip tool] to be installed. You can install this custom version via:

pipx install git+https://github.com/mutherr/blip-caption-gpu/ --force

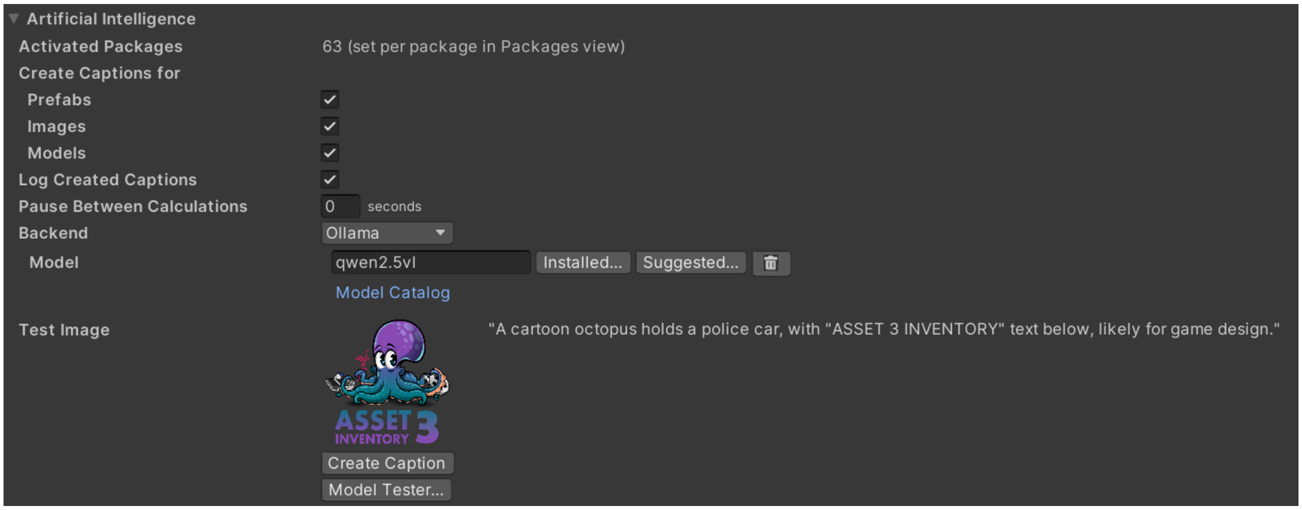
You also need to make sure to have PyTorch with GPU support as well as a CUDA runtime installed. Instructions can be found on the [official website].

Once that is done you can activate *GPU* (advanced option, hold CTRL), which dramatically speeds up the generation of AI captions (especially with larger bulk sizes).

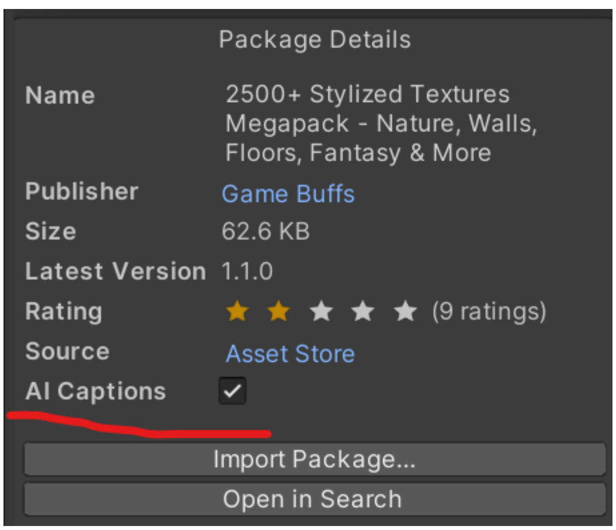
Good bulk sizes are 50 to 100. Higher is more efficient, but will produce less progress information.

Usage

On the *Settings* tab the successful installation can be tested.



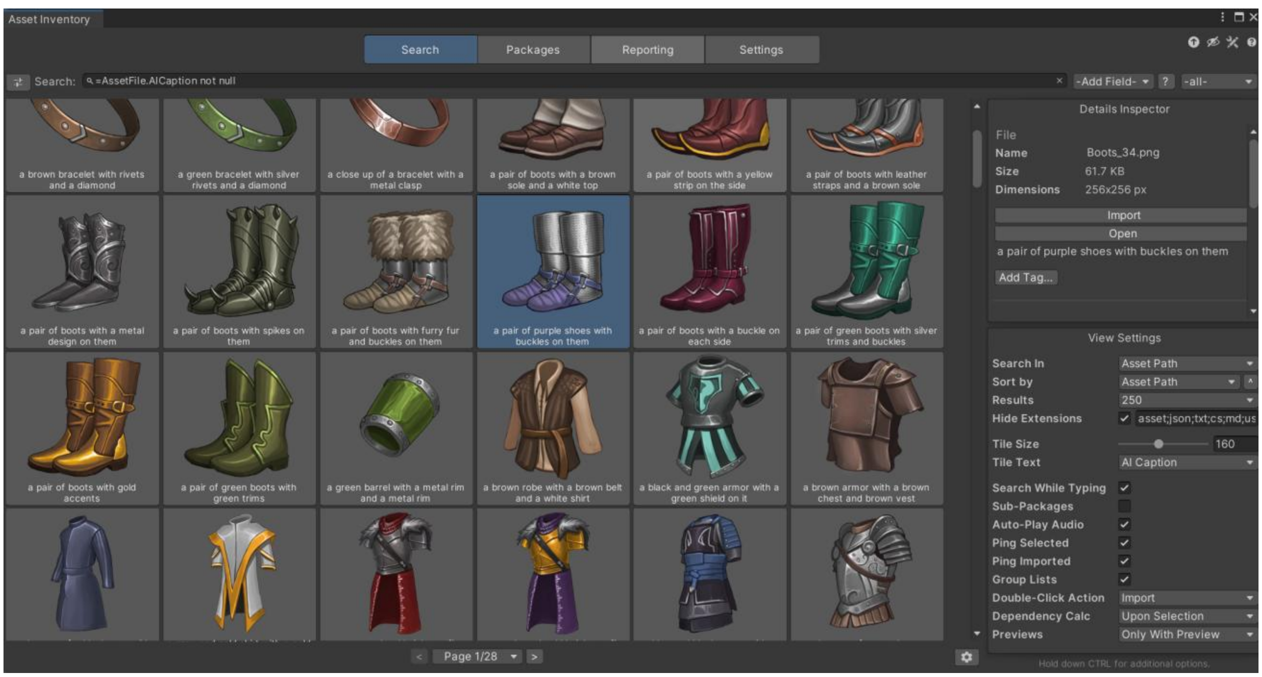
Once activated, the tool will create AI captions during the next Update cycle for packages where *AI Caption* was turned on in the *Package* view.



Important: AI options in the UI will only be visible whenever the **AI Caption action is active**.

AI captions will be shown in the *Search* results when selecting a file. You can also create individual captions directly from the *Search* results, edit them manually or delete them again.

There is also an option to search in the AI captions in the search settings and the AI caption can also be shown as the tile text if wanted.



Using an expert search like the one below will return all files for which a caption was already created:

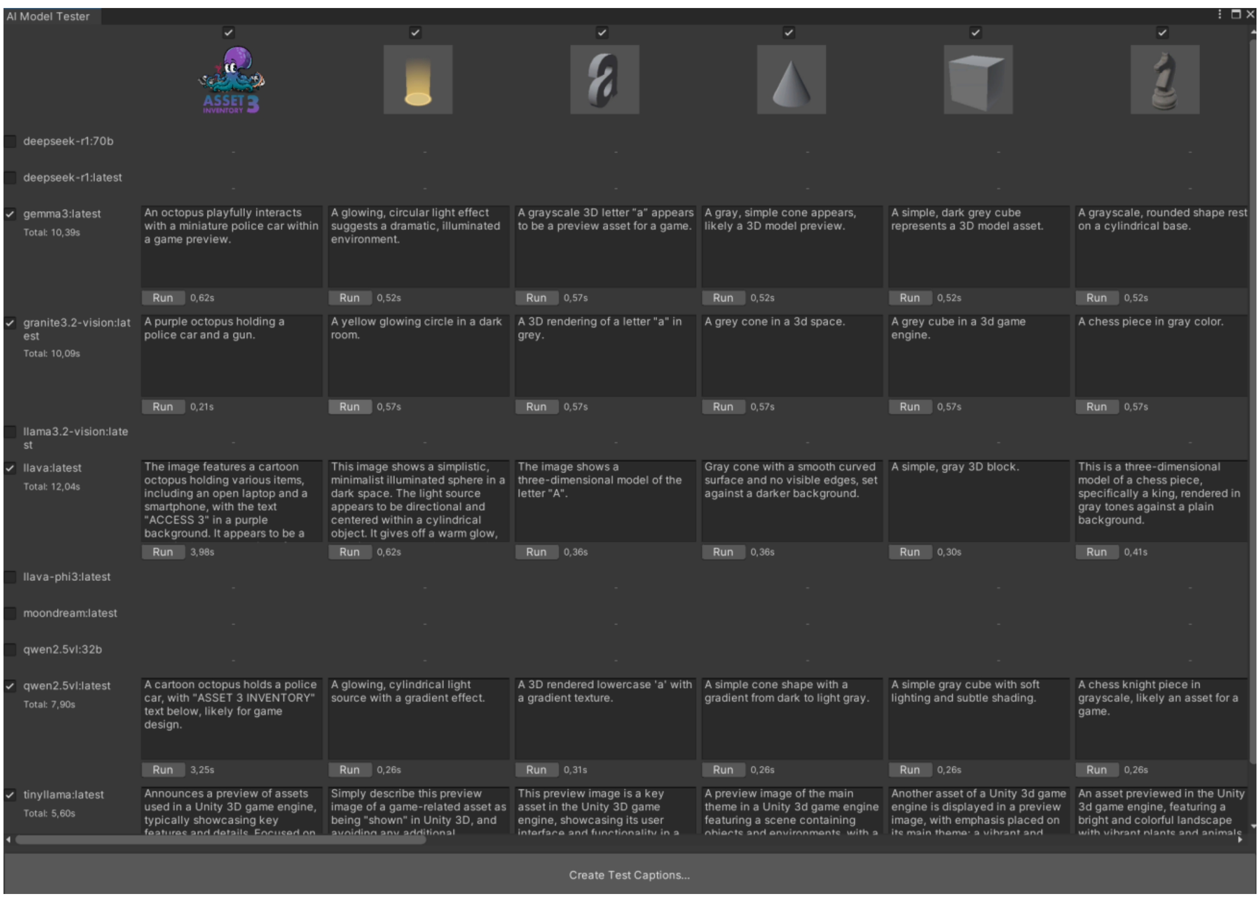
=AssetFile.AICaption not null

Model and Prompt Selection

When using the *Ollama* or *LM Studio* backend, you can select between different models to use for captioning. Models are trained for specific tasks and to follow different performance characteristics. The most important choice is to select a model that is tagged with *Vision* as only these allow to use images as input.

So far the best model to use seems to be *qwen2.5vl* which seems to have an exceptional understanding of game assets and follows a given prompt very closely, while being very performant.

To make it easier for you to compare models, a **Model Tester** module is available allowing you to caption a series of sample images and compare their output and timings (the first time is always higher since it needs to load the model into memory).

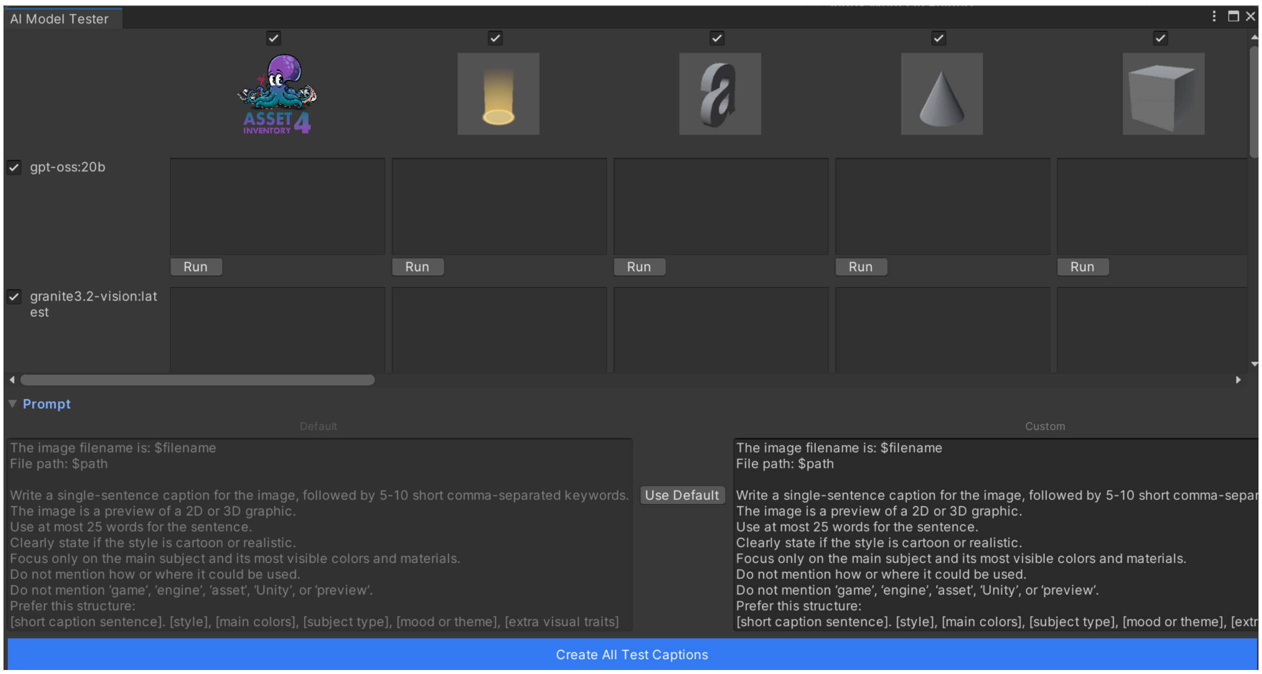


Adjusting the Prompt

The default prompt for captioning can be overridden using the *Customize* button in the *Prompt* section in the bottom of the *Model Tester*. You can instantly see the changes by recreating the test captions.

For additional context, the captions support variables which come from the currently processed file:

- *filename, path, sourcepath, type, size, width, height, length, assetid, guid*

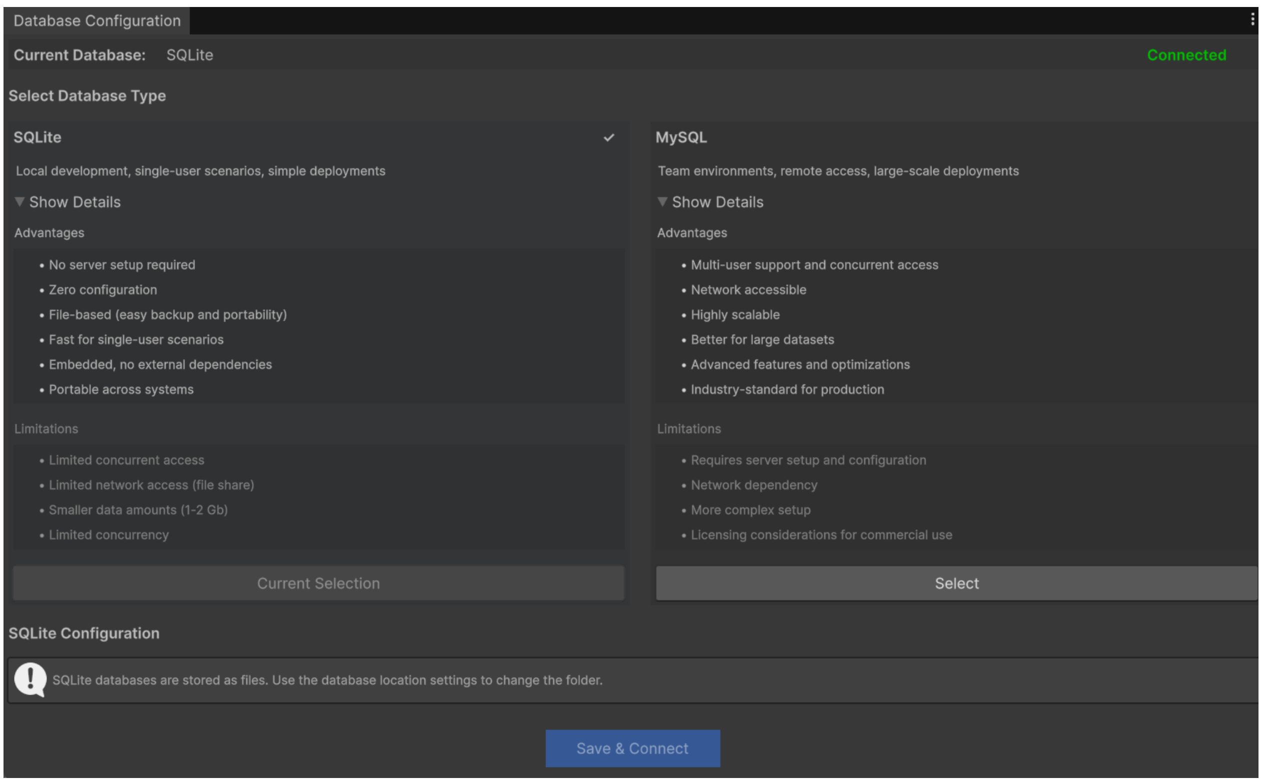


DATABASE CONFIGURATION

The tool supports two distinct database types which you can select from:

- **SQLite**: the recommended default, fast & reliable, no setup required, works out of the box, highly portable, can handle millions of records, limited network capabilities
- **MySQL/MariaDB**: for professional setups with multiple people accessing the database concurrently, highly scalable, good concurrent access support, network accessible

You can switch between these under *Settings/Locations/Database/Configure*. Existing data will not be copied to the new database.

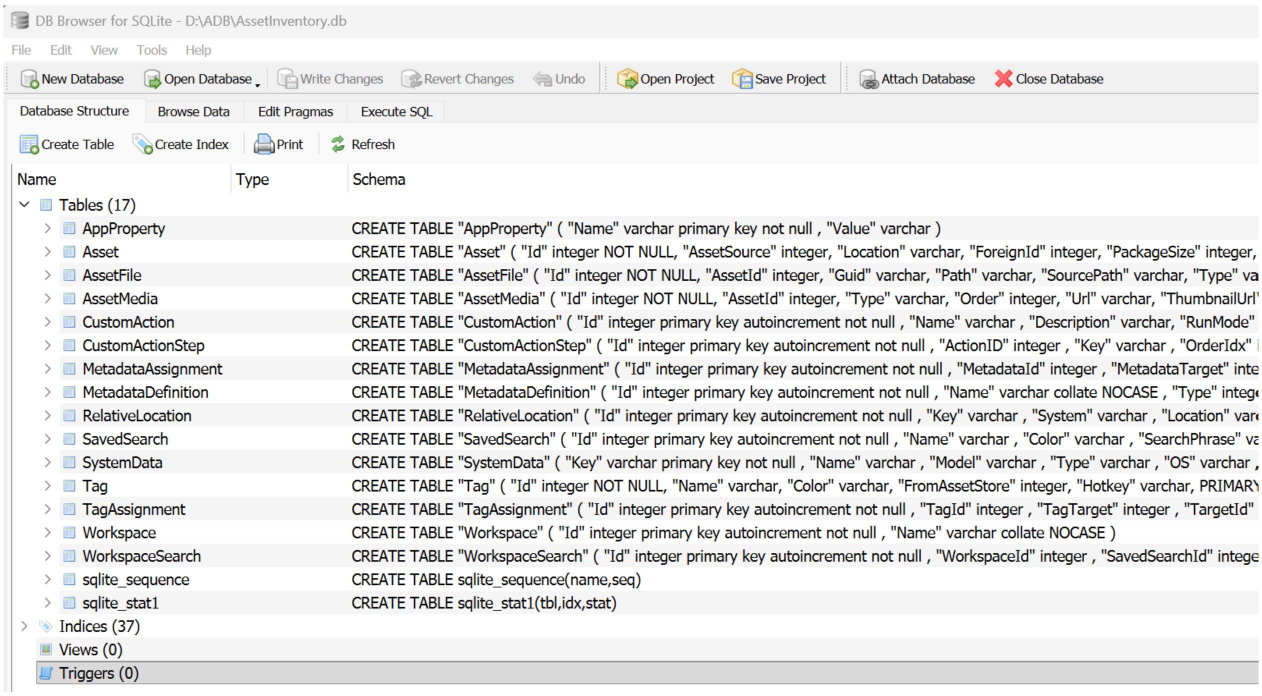


In case of SQLite the database will automatically be created upon first usage. The default location is in the *Documents/AssetInventory* directory. The size should be rather small but increases with the number of indexed assets. As a rule of thumb assume 20Mb database + 200Mb preview images per 30k files (if not upscaled). With 2 million files indexed it will be around 1.2Gb in size.

Database Access

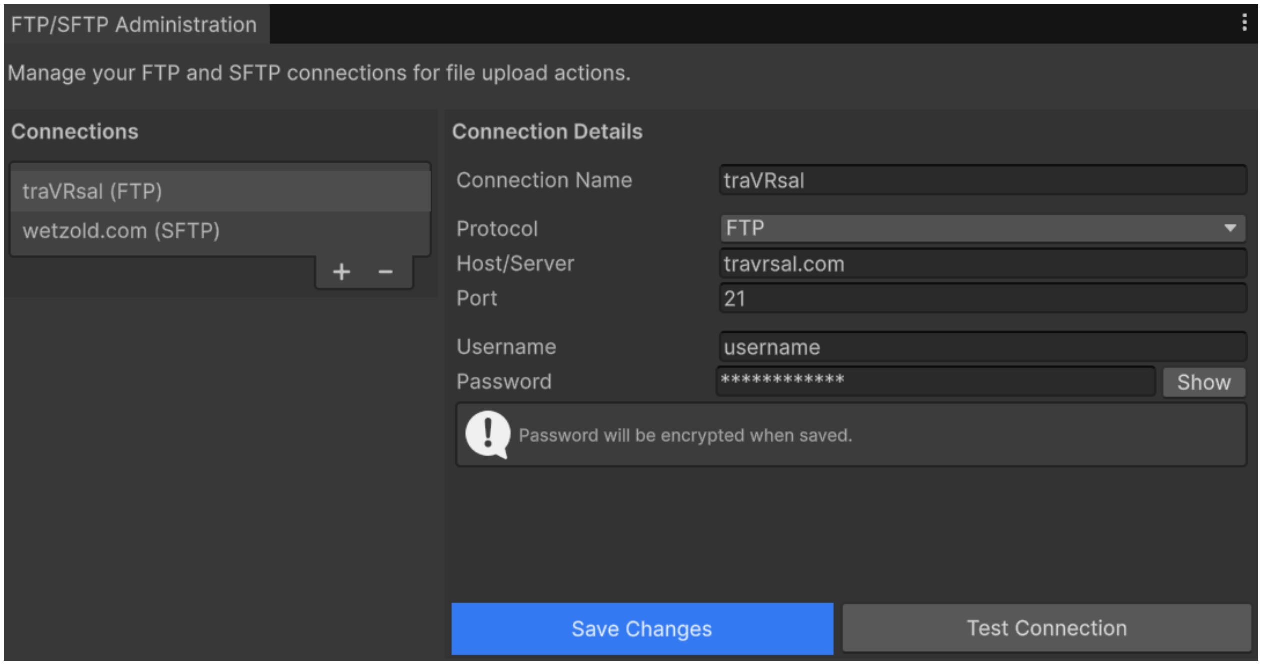
It is possible to **connect to the database** directly and perform your own analysis and queries with the collected data.

- For **SQLite** format, a good viewer is [DB Browser for SQLite].
- For **MySQL/MariaDB** [HeidiSQL] does a great job.



FTP CONFIGURATION

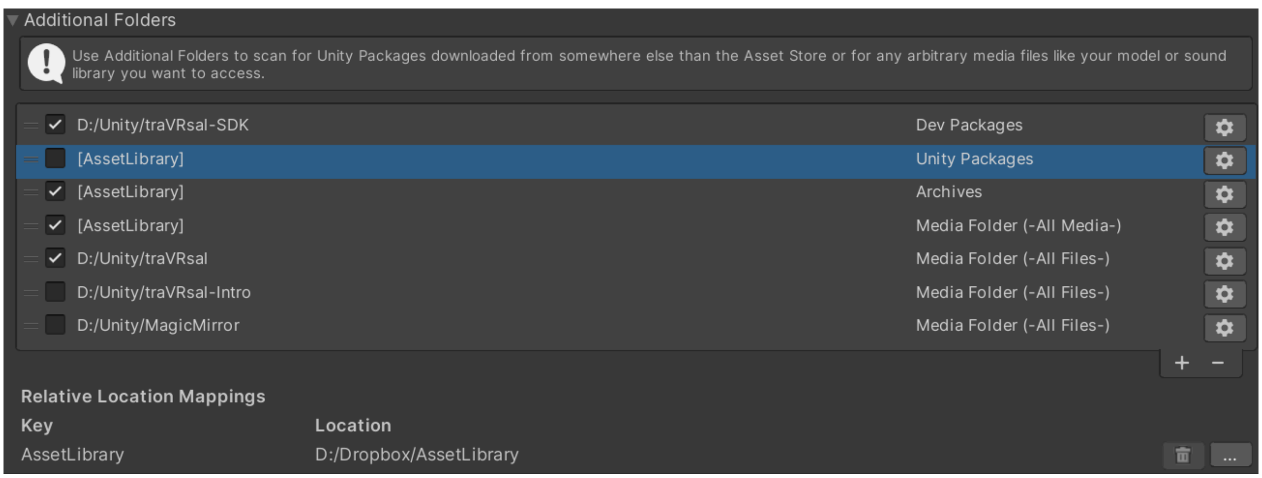
There are custom action steps that support connecting to an FTP server. You can configure these under *Settings/Locations*. The dedicated FTP UI allows to manage multiple connections. For protocols, FTP, FTPS and SFTP are supported. Your passwords will be encrypted in the configuration file and are only readable on your machine.



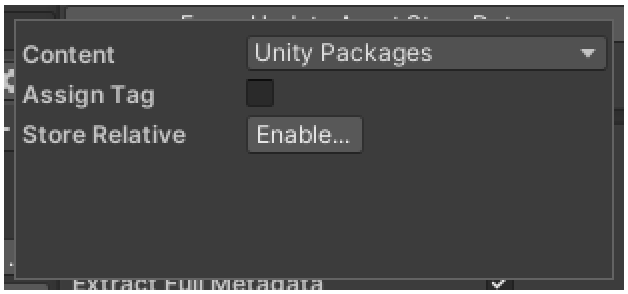
CROSS-DEVICE USAGE

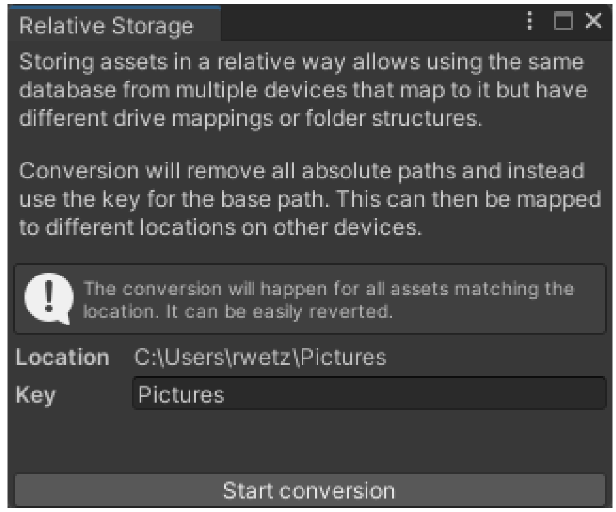
It is possible to use the same database from multiple devices. This way the indexed assets can be browsed and seen from every device. The easiest way to achieve this is to make the database and all asset folders available through a mounted network drive. If the drives and folders are identical from each device, it will instantly work this way. Also if Asset and Package cache folders differ the database will be compatible since these paths will be stored with *[ac]* and *[pc]* keys (see below). Furthermore, all paths are stored with forward slashes, making the database compatible between different operating systems.

Sometimes drives or folder mappings cannot be kept identical for various reasons. The tool supports this scenario through a mechanism called **Relative Persistence**. Each additional folder entry is replaced with a **key** and all occurrences in the database to this folder are also replaced with this key. On each device the key can then be **mapped** to the required drive and folder. This is a one-time action and once done the tool can be used as always.

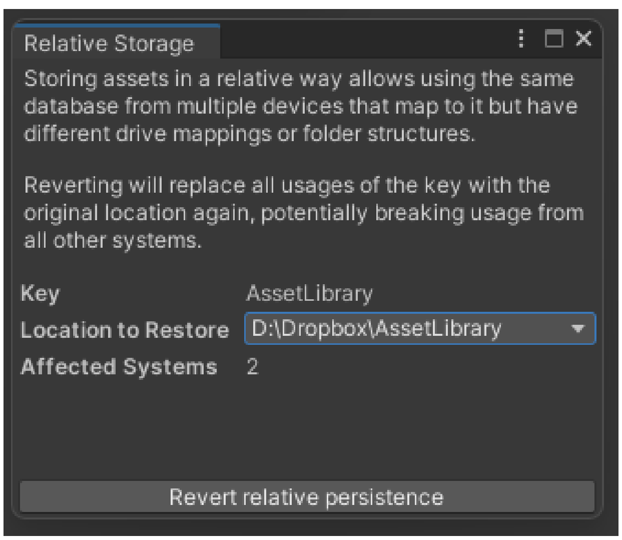


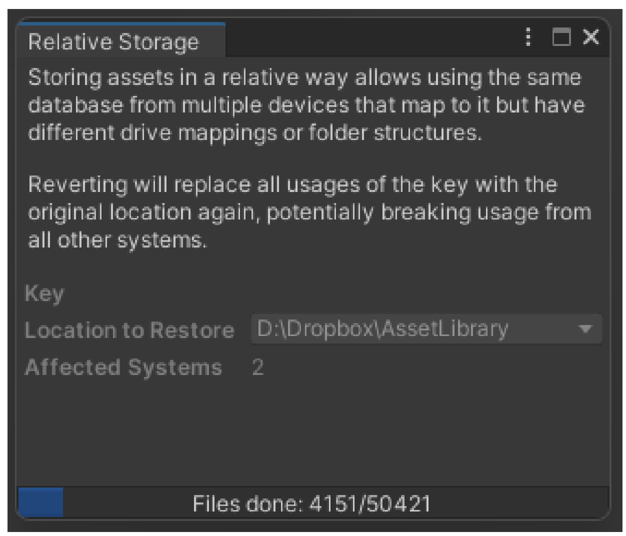
In order to convert a folder to relative format, open the folder settings, hold down CTRL and select **Enable…**





If a folder is already converted it can be **switched back** again through the same mechanism. This will remove all mappings and reorganize the database to contain the full paths again.



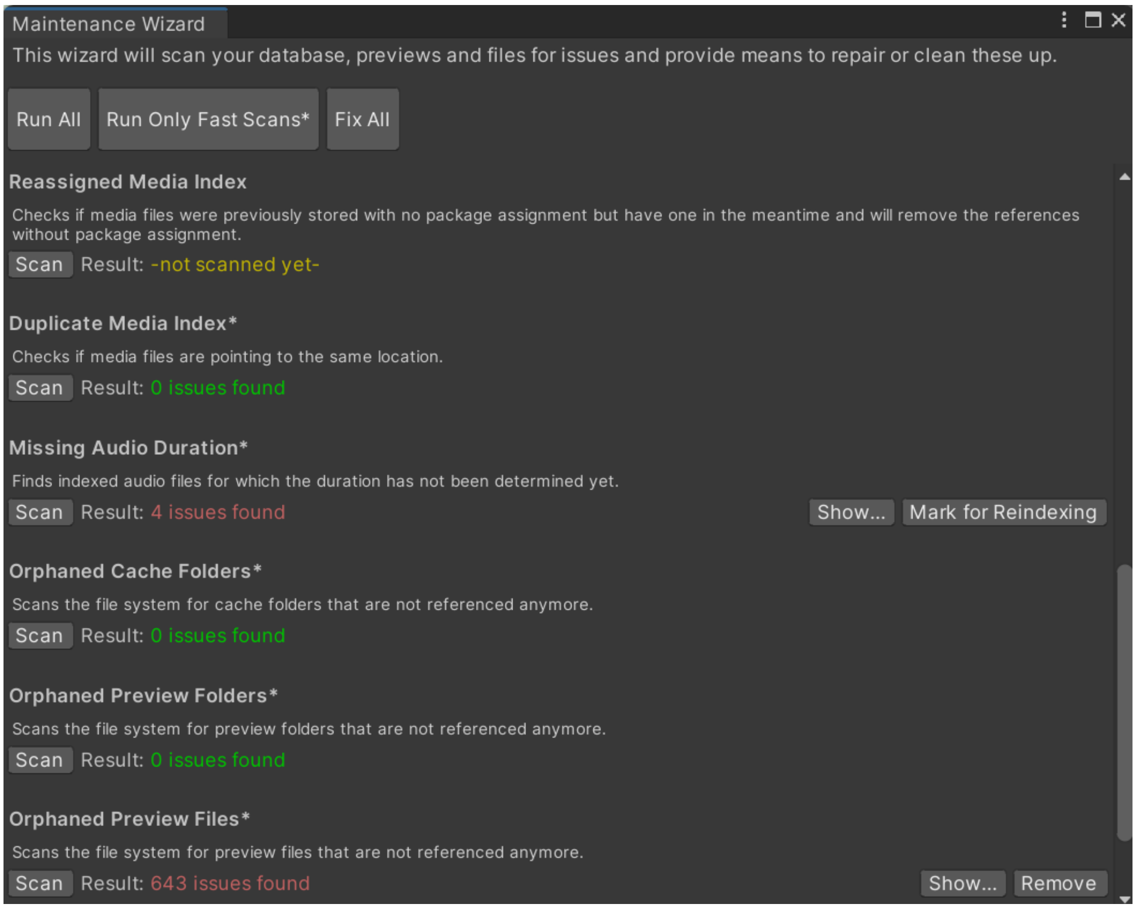


MAINTENANCE

Maintenance functions are located on the *Settings* tab:

Harmless & recommended to run regularly

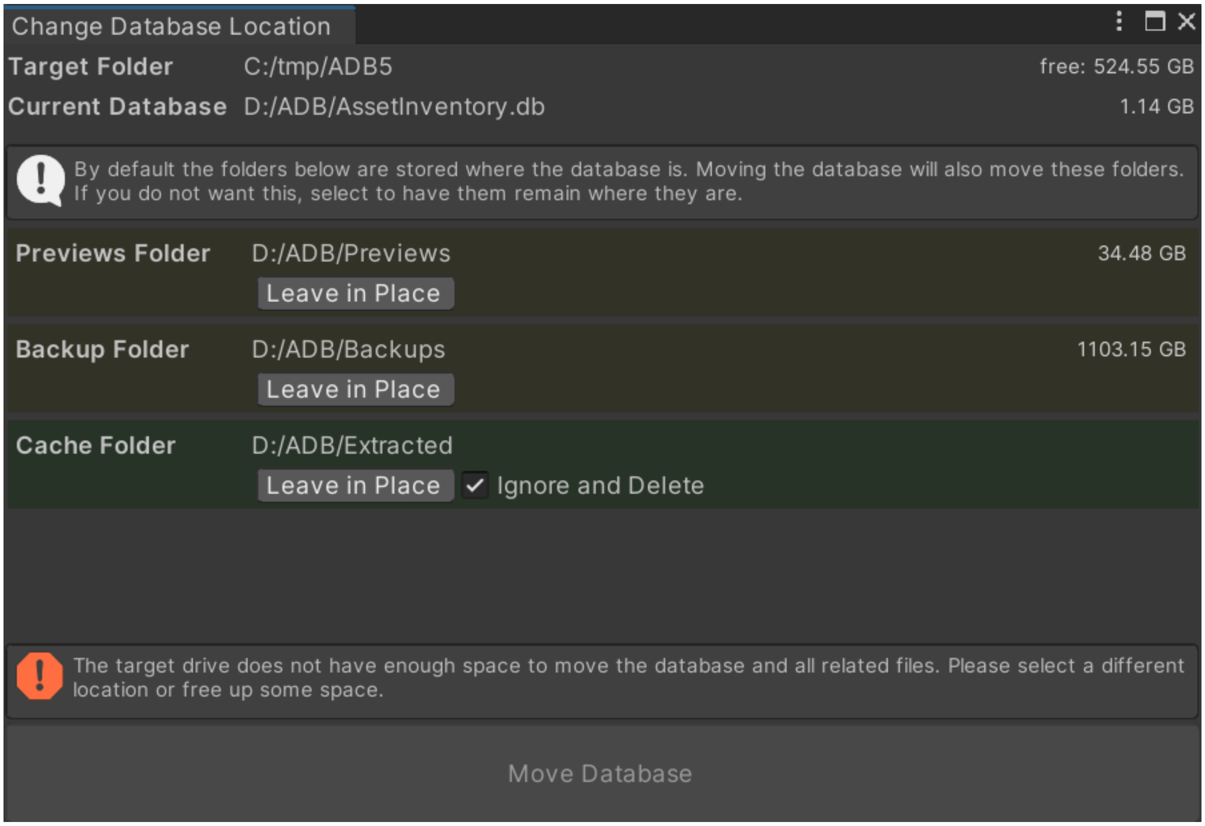
- ⇨ Maintenance Wizard: will scan for common database and file system inconsistencies and offer automatic solutions



- ⇨ Previews Wizard: will try to create preview images for asset files which do not have ones so far or were flagged to be recreated. This can take quite some time especially for complex prefabs.
- ⇨ Optimize Database: compacts and reorganizes the database for more performance and less required space, should be done after bigger indexing activities

Harmless

- ⇨ Close Database: allow backing up or copying the file
- ⇨ Clear Cache: deletes the temporary files to save space, can be done without any side-effects



Destructive

- ⇨ Clear Database: deletes the complete database to start over
- ⇨ Reset Configuration: set everything to like it was initially, removing any additional configuration that was done

ADVANCED

In order not to clutter the UI, some seldom used settings are to be set directly in the configuration file. It is in formatted JSON format and can be opened in any text editor when Unity is closed. The following properties do only appear there:

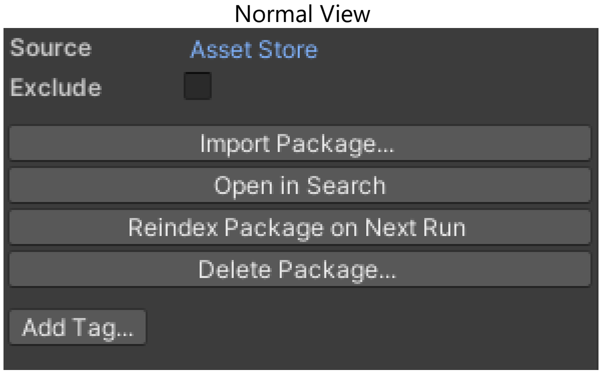
- **centerTiles** *(bool)*: will center grid views instead of left-aligning within window
- **dbJournalMode** *(string)*: defines the journal mode to be used with the SQLite database. Default is "WAL". Use "DELETE" for a more conservative method in case of database issues. Other options see link.
- **hueRange** *(float)*: degrees to widen color search
- **inMemorySearchDelay** *(float)*: seconds after typing a character to wait until triggering in-memory search
- **maxResultsLimit** *(int)*: hard limit of maximum results to be shown even if *-all-* is selected
- **mediaHeight** *(int)*: size in pixels for big media image
- **mediaThumbnailWidth** *(int)*: size in pixels for width of media thumbnails
- **mediaThumbnailHeight** *(int)*: size in pixels for height of media thumbnails
- **rowHeightMultiplier** *(float)*: adjustment factor to the height of table rows
- **searchDelay** *(float)*: seconds after typing a character to wait until triggering search
- **showExtensionList** *(bool)*: flag if to list all indexed extensions in the Types dropdown on the search. Can result in significant lag while typing if the database is large.
- **showHints** *(bool)*: flag if to show additional usability hints in the UI

ADVANCED FEATURES

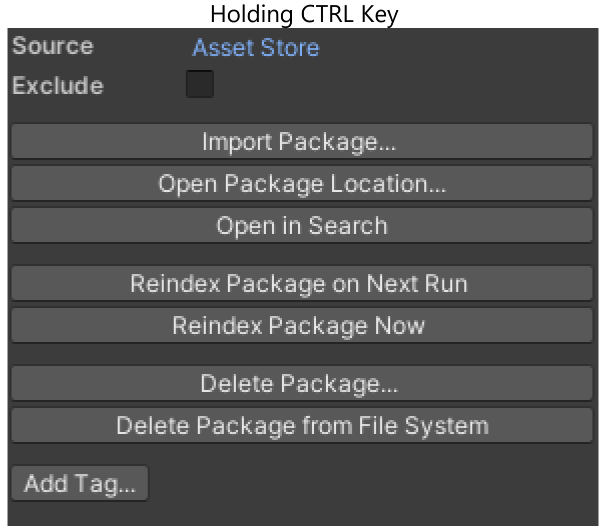
EXPERT FUNCTIONS

Many more seldom used and advanced features are initially hidden to not clutter the UI. They can be made visible by holding down the **CTRL key** or pressing the **eye icon** in the upper toolbar.

Normal View



Holding CTRL Key



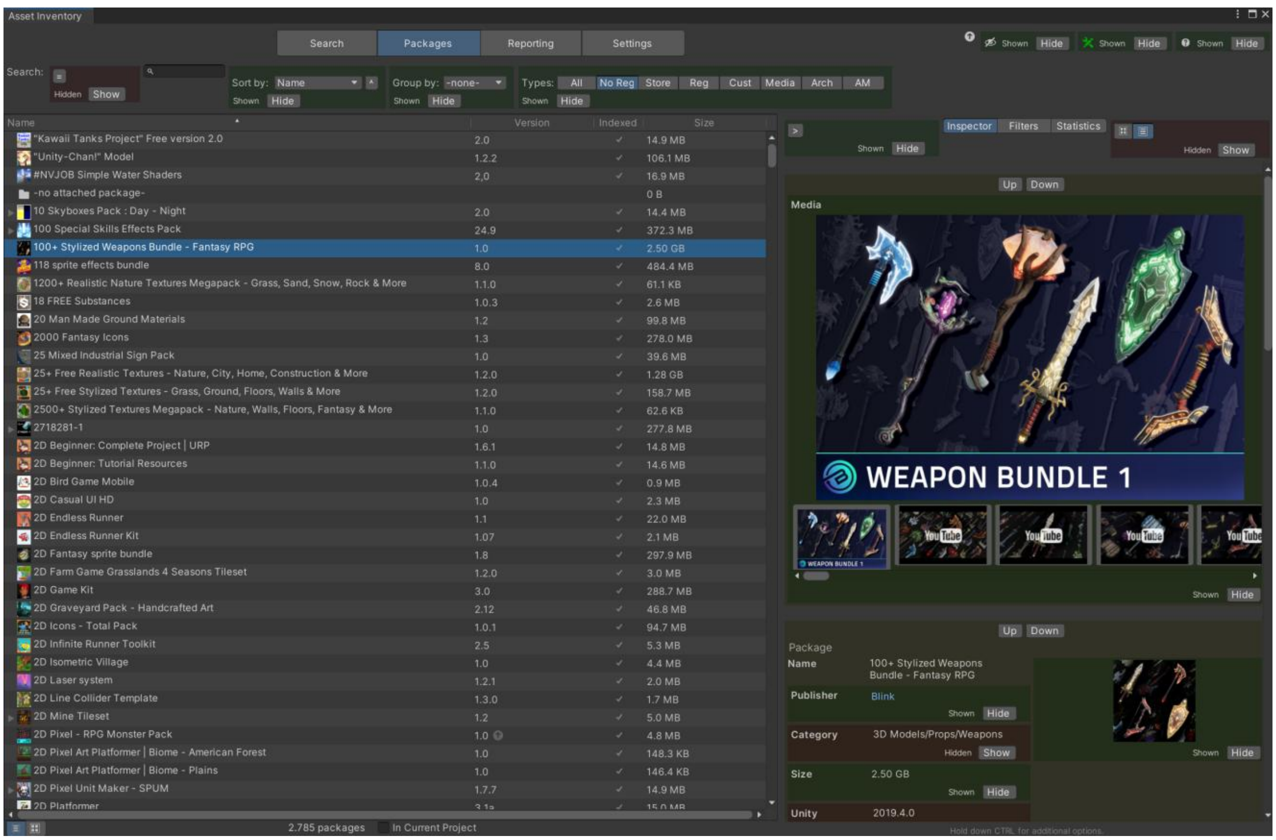
Expert function visibility will not only apply to buttons but also to **additional information** which is not often needed. This ensures the UI is clean for the 90% use-case while also catering to experts on-demand. There are two ways to influence the visibility of the advanced UI:

- Temporarily toggle it on via the eye icon in the upper toolbar or permanently via a setting in the *Advanced* section on the settings tab.
- Customize what should be considered “advanced” (see next section).

UI CUSTOMIZATION

Using the customization icon in the upper toolbar, the UI can be switched to *design-mode* (and back). When active, sections (labels, buttons, info boxes, panels…) that can be customized will be highlighted in **green, red and yellow** and additional actions will be visible. Green sections will always be shown. Red sections will be considered *advanced* and only shown when advanced UI is toggled. Yellow blocks can be moved up and down as a whole.

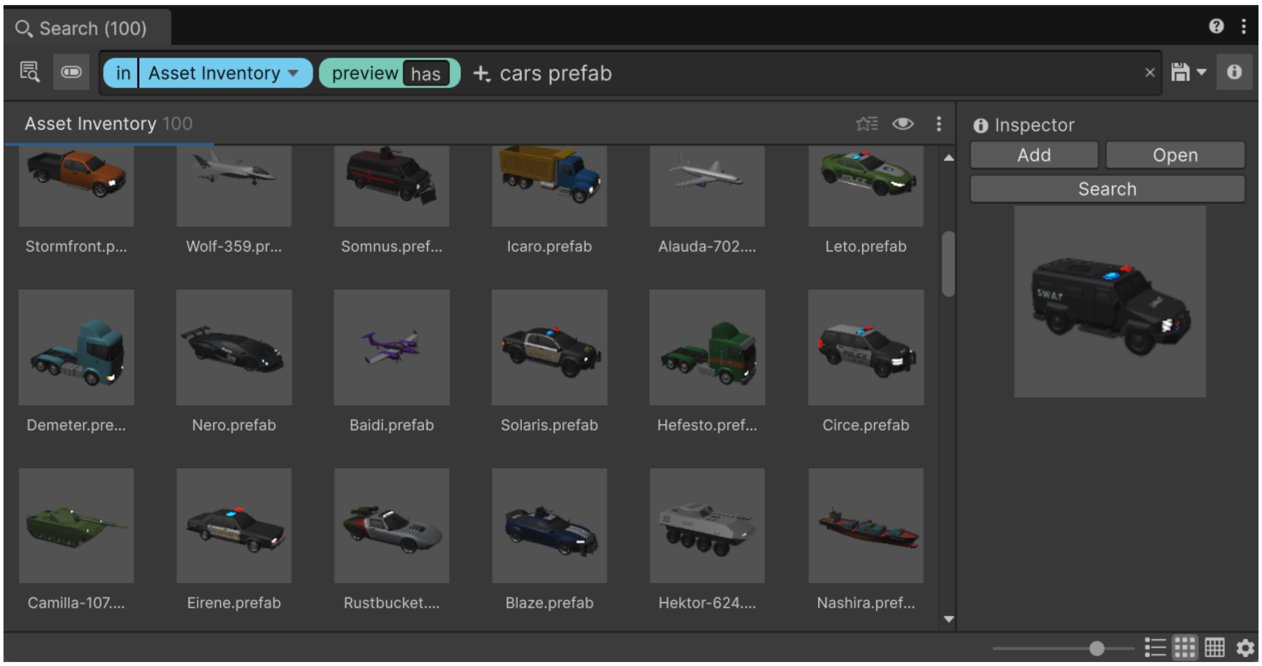
During design-mode, all advanced UI will automatically be shown. Depending on the current selection and context, not all UI elements will be visible (e.g. registry packages will show different actions than Asset Store packages).



UNITY SEARCH

The tool natively integrates into the Unity search. Press CTRL+K to call up the search window. The provider is called *Asset Inventory* ("assetinv:"). Available actions include importing & adding the selected item to the current scene, opening the main search window or opening the file in the system.

Grid views and list views are fully supported. In grid view you can add a multitude of additional columns.



AUTOMATION & API

The tool can be controlled via scripting. This happens through the static methods of the *AI.cs* class and through a dedicated search API. The existence of the tool can be detected via the automatically added script define *ASSET_INVENTORY*.

To use the powerful **search** UI features (through the *ResultPickerUI*), the following code can be used (it is also provided in the *Examples* folder). Optionally a search type and a search string can be supplied to narrow down the search.

Two modes are available: close upon selecting a result tile or explicitly using the *Select* button. Once the user selects something, a callback with the path of the asset inside the project will be invoked. The asset will automatically be materialized.

The *ResultPickerUI* offers multiple methods:

Show()

will return the path of the selected asset inside the current project.

ShowTextureSelection()

Will default to "Images" as search type and return a dictionary of files. The selected entry will be inside "original". The tool will do a heuristic analysis of additional adjacent files that belong to the same texture, like normal maps, metal maps etc. If found, they will also be materialized and returned.

Supported Types: albedo, normal, specular, metal, occlusion, displacement, height, emission, reflection, alpha, mask, roughness

The scripts in the *Examples* folder contain samples how to access the API both via C# and UI integration.

```
GUILayout.Label("UI Examples", EditorStyles.boldLabel);
if (GUILayout.Button("Search for a car..."))
{
    ResultPickerUI.Show(path =>
    {
        EditorUtility.DisplayDialog("Selection", path, "Close");
    }, "Prefabs", "car");
}
if (GUILayout.Button("Search with details..."))
{
    ResultPickerUI window = ResultPickerUI.Show(path =>
    {
        EditorUtility.DisplayDialog("Selection", path, "Close");
    });
    window.instantSelection = false;
    window.hideDetailsPane = false;
}
if (GUILayout.Button("Search for texture sets..."))
{
    ResultPickerUI window = ResultPickerUI.ShowTextureSelection(path =>
    {
        EditorUtility.DisplayDialog("Selection", string.Join("\n", path.Select(e => e.Key + ": " + e.Value)), "Close");
    });
    window.instantSelection = false;
    window.hideDetailsPane = false;
}
EditorGUILayout.Space();

GUILayout.Label("Programmatic Examples", EditorStyles.boldLabel);
if (GUILayout.Button("List all indexed packages"))
{
    List<AssetInfo> packages = AI.LoadAssets().Where(p => p.IsIndexed && p.SafeName != Asset.NONE).ToList();
    Debug.Log($"Indexed packages: {packages.Count}");
    foreach (AssetInfo package in packages)
    {
        Debug.Log(package.DisplayName);
    }
}
if (GUILayout.Button("Search for images less than 256 width"))
{
    AssetSearch.Options searchOptions = new AssetSearch.Options
    {
        SearchPhrase = "flower",
        MaxResults = 10,
        CurrentPage = 1,
        RawSearchType = "Images",
        CheckMaxWidth = true,
        SearchWidth = "256"
    };

    AssetSearch.Result result = AssetSearch.Execute(searchOptions);
    Debug.Log($"<color=cyan>Found {result.ResultCount:N0} image files with width < 256px (showing first {result.Files.Count}):</color>");
    foreach (AssetInfo file in result.Files)
    {
        Debug.Log($"<color=white>-</color> <color=yellow>{file.FileName}</color> <color=gray>({file.Type})</color> <color=green>{file.Width}x{file.Height}
    }
}
```

CONFIGURATION

OVERVIEW

Settings will be saved automatically in a file called *AssetInventoryConfig.json*. This happens by default in the *Documents* folder of the currently logged in user. This way the tool can work independently of the Unity project and the asset search is available in an identical way everywhere. Two alternatives are available:

- It is possible to force the tool to use a project-specific configuration. To do so, copy the *AssetInventoryConfig.json* file anywhere into the project and restart Unity. The detected location will be shown on the **Settings tab** under **Maintenance Functions**.
- Using the environment variable "*ASSETINVENTORY_CONFIG_PATH*" a custom location (just the folder name) can be specified.

Some advanced settings can only be set in the *.json* file and are not otherwise accessible through the UI. An example for this is the delay after which the search will start searching. Feel free to adjust these values as well. Both for changing the config location as well as editing the *json* file Unity needs to be restarted.

INSTALLING AS A PACKAGE

Especially when managing many Unity projects keeping Asset Inventory up-to-date can mean spending quite some time on manual package updates. An alternative is to install the tool in one Unity project only and then referencing this single installation via package reference in the *manifest.json*. This way all Unity installations always use the same tool version.

The easiest way to set this up is to open the manifests from the *Packages* folder where the tool should be referenced and adding it directly there. The path can either be absolute or relative. Example:

"com.wetzold.asset-inventory": "file:../../MySourceProject/Assets/AssetInventory",

THIRD PARTY

USAGE

Asset Inventory uses multiple third-party libraries that really make it shine:

- Package2Folder by Code Stage uses an ingenious idea to intercept the Unity package manager and adjust the target folder where to install assets to.
- SQLite is a great way to store data in a single file while providing easy and fast database operations.
- SQLite-Net is an amazing extension to SQLite offering convenient high-level functions when executing SQL queries.
- Editor Audio Utils does a great job to allow playing audio files also while not in play mode.
- SharpCompress for extracting *zip* and *rar* archives.
- LibGit2Sharp for working with Git repositories.
- Scriban for templating capabilities.
- OllamaSharp for easy access to the local Ollama API.
- Microsoft AI as an abstraction for interacting with local and remote language models.
- ImageSharp provides an alternative to System.Drawing which works on Mac and Linux.

INTEGRATION

Assets that already integrate with Asset Inventory:

Gaia from Procedural Worlds

Makes it really easy to pick new prefabs to spawn and textures to use in your procedural content. Access all your assets quickly with just a few clicks. The corresponding selectors will appear automatically once you have both tools imported in your project.

TECHNICAL DETAILS

UNITY COMPATIBILITY

Most features of the tool are usually compatible with Unity from version 2021.3 up to the latest beta and tested on Windows and Mac. Linux is also reported to work just fine but is not in active testing. Some features require specific minimal Unity versions. See all exceptions below:

Feature	Restrictions
Automatically detect custom asset cache location	2022.1+
Integration into Unity Search (CTRL + K)	2022.2+
Asset Manager integration	2022.3+
Unity Asset Origin	2023.1+
VFX previews	6000.0+
Main toolbar button	6000.3+

FOLDERS

Inside the *AssetInventory* directory, three additional folders will be created:

- **Previews**: containing all images for the asset catalog
- **Extracted**: containing temporarily extracted files. You can safely delete the *Extracted* folder at any time if needed. The size of the folder can be limited with the Limiter feature under *Settings/Locations*.
- **Backup**: acting as the default location for package backups

Some operations require Unity to perform work, e.g. creating previews. For such tasks, the folder *_AssetInventoryPreviewsTemp* will appear under *Assets* for a short time and will be automatically removed again once done.

LIMITATIONS

- Dependencies can only be calculated in packages that use *text/yaml* serialization. A few legacy packages still use *binary* serialization. These packages also work in most cases except if Unity cannot resolve them, e.g. because of missing scripts on prefabs.
- Running the game view maximized on play and having the Asset Inventory window docked will cause it to reinitialize, resetting all search settings.

FAQ

SQLITE IS ALREADY IN MY PROJECT (CONFLICT)

The only requirement is that SQLite is available. In this case it is safe to delete the one supplied by Asset Inventory. Close Unity and delete either the full *ThirdParty/SQLite* folder or if your own package doesn't bring *System.Data.SQLite* only delete *ThirdParty/SQLite/Plugins/x64 & x86*.

ODIN SHOWS VALIDATION ERRORS

If the Unity tutorials package is not installed, the tutorials will be detected as "unknown type". This is a false positive. To solve, configure an exception for the tool.

SOME PREFABS SHOW NO DEPENDENCIES

This is due to the serialization format. Only assets serialized as text can properly be parsed right now.

SOME AUDIO FILES USE DIFFERENT COLORS IN THE PREVIEW THAN OTHERS

Preview colors depend on which Unity version an asset author used to upload his asset. Unity seems to change colors over time between Unity versions. There could be a feature someday to recreate preview images with the current version of Unity to make them all uniform depending on interest.

SOME PREVIEWS ARE GREY AND DON'T SHOW ANYTHING

This can happen if the Asset Store tooling did not create a correct preview image while uploading an asset. An example is the *Danger Zone* asset from Unity released for 2021+.

CONSOLE SHOWS ERRORS DURING INDEXING

There are a couple of errors Unity creates that cannot be intercepted but can safely be ignored. These are:

Error reading JArray from JsonReader

- Will happen if conflicting Json converters are in the project. Json.net converters are a typical culprit. The recommendation is to remove the additional converters if not needed.

Could not determine image dimensions for 'filename': Not enough memory to complete operation [GDI+ status: OutOfMemory]

- Can happen when either the image dimensions are too big (5000px+), it is actually a different format (incorrectly named file) or the image file has a sub-format that Unity cannot read, e.g. special types of *png*.

“Cannot create FMOD::Sound instance for clip “” (FMOD error: Unsupported file or audio format.)”

- It typically means that an audio clip was saved with the wrong extension, e.g., a wave as an ogg by an asset publisher. The tool will try to auto-correct but Unity will show the error anyway. Use the **Open** button to start the native file player which will typically play the file correctly then.

“Assertion failed on expression: 'ps->array_size()'”

- In that case Unity encountered issues with a mesh to be imported. Ignore.

“TEXTURE HAS OUT OF RANGE WIDTH / HEIGHT”

- In that case Unity cannot read the texture dimensions and you will not be able to see the dimensions of the texture or filter for it. Otherwise, it does not have any effect.

“Mesh '': abnormal mesh bounds - most likely it has some invalid vertices (+/-inifinity or NANs) due to errors exporting.”

- In that case Unity cannot read the texture dimensions and you will not be able to see the dimensions of the texture or filter for it. Otherwise, it does not have any effect.

“ArgumentException: Getting control 4's position in a group with only 4 controls when doing repaint”

- This can happen if a lot of UI changes are going on while indexing is happening. Can safely be ignored and is without any effect.

“Curl error 28: Operation timed out after 30000 milliseconds with 0 bytes received”

- This can happen if Unity servers did not respond in time and the details for an asset could not be retrieved. This is solved by starting the Asset Store update process again.

“Invalid or expired API Token when contacting…”

- Happens if Unity was not in use for a longer period. The user login happens in the Unity hub and in that case the token for online access has expired. Solved by restarting Unity.

“Could not create asset from Assets/_AssetInventoryPreviewsTemp/1016771.png: File could not be read”

- Happens if Unity cannot create a preview for specific assets or if this is a malformed png. Can be ignored.

“TLS Allocator ALLOC_TEMP_THREAD, underlying allocator ALLOC_TEMP_THREAD has unfreed allocations, size 1261”

- Can occur after triggering a download. No negative side-effect seen so far. Can most likely be ignored and seems to be a Unity issue.

“You are trying to replace or create a Prefab from the instance '…' that references a missing script. This is not allowed. Please change the script or remove it from the GameObject.”

- Can occur during creating preview images when a prefab has script dependencies. Can be ignored since Asset Inventory fixes this situation automatically but the error cannot be suppressed.

“NormalMap settings dialog comes up”

- Can occur during creating preview images when a texture is misconfigured in a prefab. Can be ignored since this is typically invisible in previews.

SUPPORT

Web: https://www.wetzold.com/tools

Discord: https://discord.gg/uzeHzEMM4B

Roadmap: https://trello.com/b/SOSDzX3s/asset-inventory